

Genomic null calibration: first 100 per population

2026-10-07 | Frozen snapshot: **600 / 6,000** completed simulations (10.0%). Target: 1,000 per population. Live runner at report generation: running; 8 workers; 0 reported failures.

Decoded TMRCA: all sampled haplotype pairs

Four requested numbers per population. Entries are percentages of pairs with TMRCA below the specified age; they are descriptive empirical 95th/99th percentiles across retained null focal sites.

Population	Nulls	10 kya: top 5%	10 kya: top 1%	50 kya: top 5%	50 kya: top 1%
AFR	100	1.11%	1.45%	8.27%	10.07%
AMR	100	2.28%	2.38%	26.91%	32.74%
EAS	100	5.18%	6.36%	33.03%	42.28%
EUR	100	3.98%	5.64%	29.95%	43.97%
MID	100	2.69%	3.17%	28.74%	32.71%
SAS	100	3.61%	4.34%	26.00%	30.01%

Empirical p-values: ties included, no +1 adjustment

p = number of null scores greater than or equal to the observation / n. With n = 100, p <= 0.05 requires strictly exceeding the 95th-percentile boundary; p <= 0.01 requires strictly exceeding the 99th-percentile boundary. Equality does not reject.

The p-value grid has 0.01 steps. The 99th percentile is the second-largest null score, so the 1% tail is still imprecisely estimated. An observation above every null has empirical p = 0, which does not establish zero population tail probability.

Pooled CDFs are shown without demographic confidence bands. Every represented history currently has one completed region; ten are planned per history. This report freezes the prespecified cohort; later completed replicates are excluded.

True TMRCA and allele-pair availability

For the decoded all-pairs statistic, reject at $p \leq 0.05$ only when the target score is **strictly greater** than the top-5% value on the previous page. The true-TMRCA boundaries below use the same rule.

True TMRCA: all sampled haplotype pairs

Population	Nulls	10 kya: top 5%	10 kya: top 1%	50 kya: top 5%	50 kya: top 1%
AFR	100	0.49%	0.51%	5.06%	5.60%
AMR	100	0.63%	0.71%	14.77%	17.16%
EAS	100	0.99%	1.06%	25.38%	29.10%
EUR	100	1.33%	1.41%	23.96%	29.39%
MID	100	0.69%	0.74%	17.50%	21.06%
SAS	100	0.84%	0.94%	23.62%	27.66%

ALT/ALT availability and saturation at 50 kya

Population	Available / n	Missing	p at 100%: true	p at 100%: decoded
AFR	75 / 100	25	0.47	0.29
AMR	89 / 100	11	0.82	0.26
EAS	92 / 100	8	0.83	0.32
EUR	95 / 100	5	0.85	0.27
MID	85 / 100	15	0.72	0.28
SAS	90 / 100	10	0.83	0.32

At a target score of 100%, p is the fraction of null scores also equal to 100%. This table counts ties in the numerator. If every null equals the target, $p = 1$.

Raw ALT/ALT fractions: empirical cutoffs

Percentages of carrier pairs; no carrier-mass weighting. The denominator remains all 100 null simulations. Undefined null carrier-pair scores are retained below finite support; undefined targets are no-calls. Available-only percentiles and tied counts are separately labeled in the full cutoff CSV.

Decoded TMRCA: ALT/ALT pairs

Population	Nulls	10 kya: top 5%	10 kya: top 1%	50 kya: top 5%	50 kya: top 1%
AFR	100	100.00%	100.00%	100.00%	100.00%
AMR	100	57.14%	100.00%	100.00%	100.00%
EAS	100	66.67%	100.00%	100.00%	100.00%
EUR	100	55.56%	100.00%	100.00%	100.00%
MID	100	75.00%	100.00%	100.00%	100.00%
SAS	100	100.00%	100.00%	100.00%	100.00%

Truth TMRCA: ALT/ALT pairs

Population	Nulls	10 kya: top 5%	10 kya: top 1%	50 kya: top 5%	50 kya: top 1%
AFR	100	66.67%	100.00%	100.00%	100.00%
AMR	100	46.67%	100.00%	100.00%	100.00%
EAS	100	50.00%	100.00%	100.00%	100.00%
EUR	100	56.00%	100.00%	100.00%	100.00%
MID	100	50.00%	100.00%	100.00%	100.00%
SAS	100	60.00%	100.00%	100.00%	100.00%

Model, allele frequencies and limitations

Neutral archaic-derived alleles, segregating immediately after 2% introgression at 50 kya and observed in the present sample (fixation retained). Neanderthal $N_e = 3,600$; split = 500 kya; 25 years/generation. 200 sampled diploids; 10,000 fixed sampled haplotype pairs; no rescaling.

Random autosomal regions use the local deCODE GRCh38 map in simulation. Gamma-SMC uses the cropped region mean recombination rate, accurate $\exp(10)$, and fixed backward alignment. Mutation rate = 1.29×10^{-8} . Assembly gaps and missing map coverage are masked; the HMMix strict mask is not used.

True versus decoded and ALT/ALT results

The following pages include allele-frequency distributions, separate true/decoded CDFs, and all-pairs/ALT-ALT comparisons. ALT/ALT scores use raw carrier-pair fractions. Null no-calls remain below finite score support in the CDF and percentile denominator. See the CSVs for available counts, tied fractions and p at each cutoff.

This campaign contains calibration nulls only. It does not supply new selected-treatment power or independent neutral-target FPR estimates.

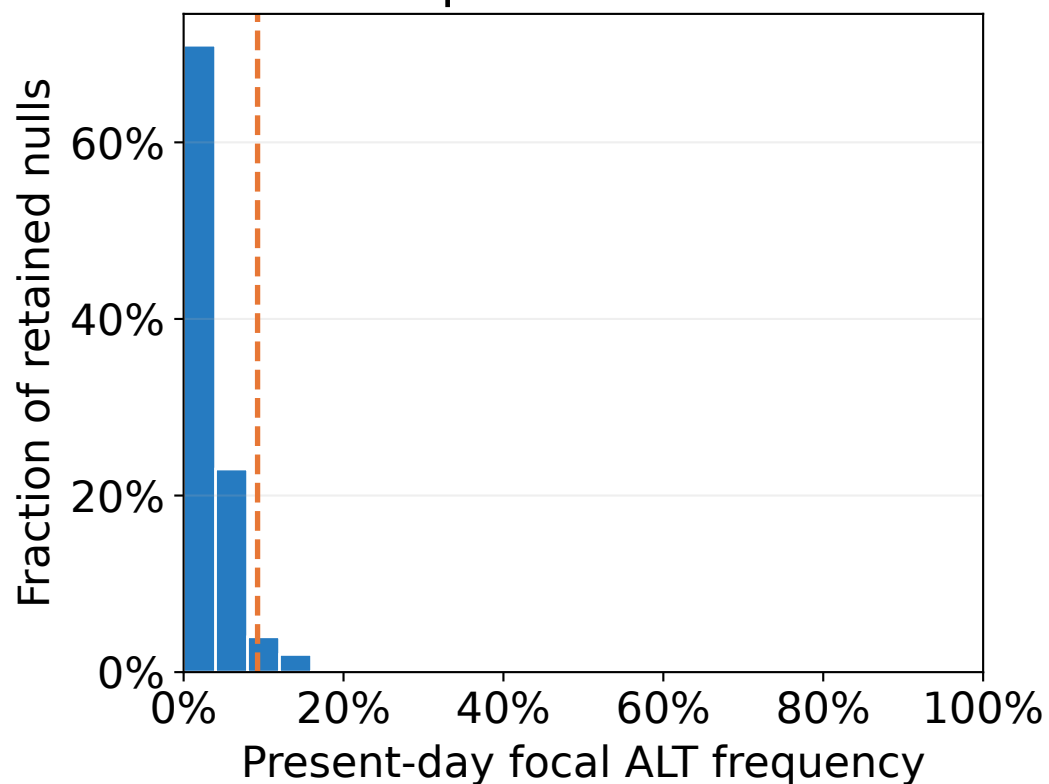
ALT/ALT at 50 kya is saturated: every population has a 100% upper-tail boundary for both true and decoded TMRCA. Even a target score of 100% cannot reject at $p \leq 0.05$ with these nulls.

Focal ALT frequencies among retained nulls

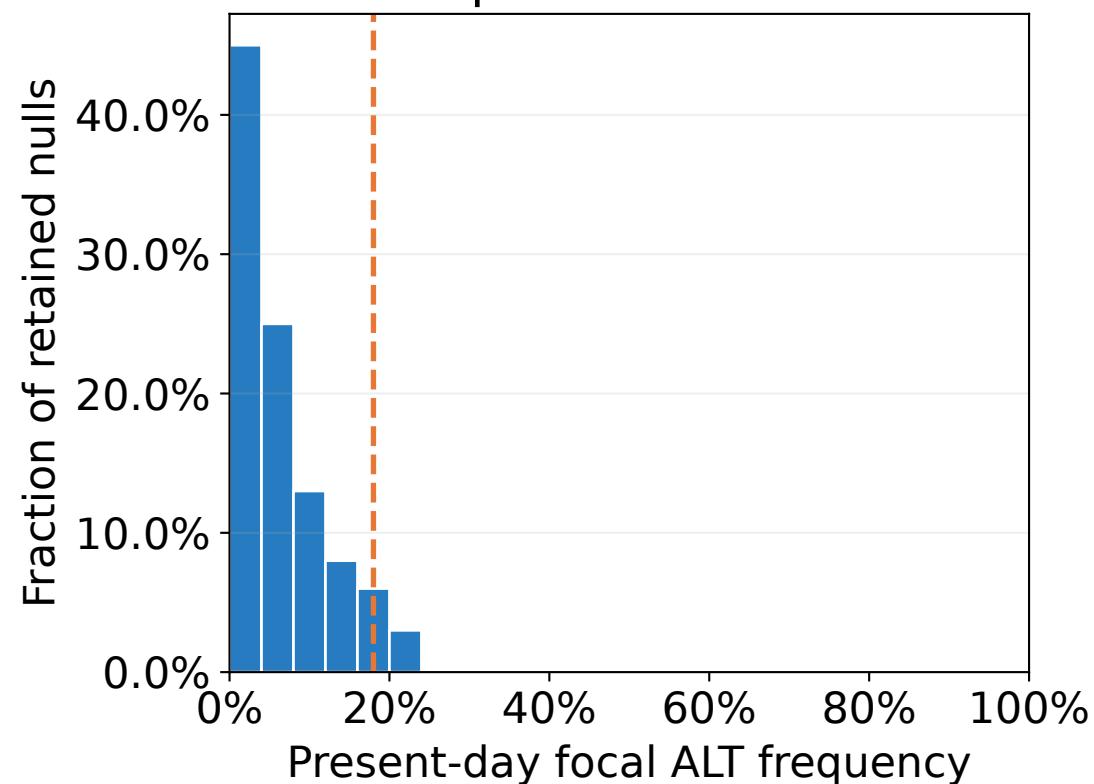
Population	Median AF	95th-percentile AF	Fixed alleles	Median attempts
AFR	2.0%	9.2%	0	7
AMR	4.8%	18.0%	0	11.5
EAS	6.0%	22.5%	0	18.5
EUR	7.9%	28.0%	0	13.5
MID	5.2%	20.8%	0	12
SAS	6.2%	26.0%	0	14

Empirical 95th percentile

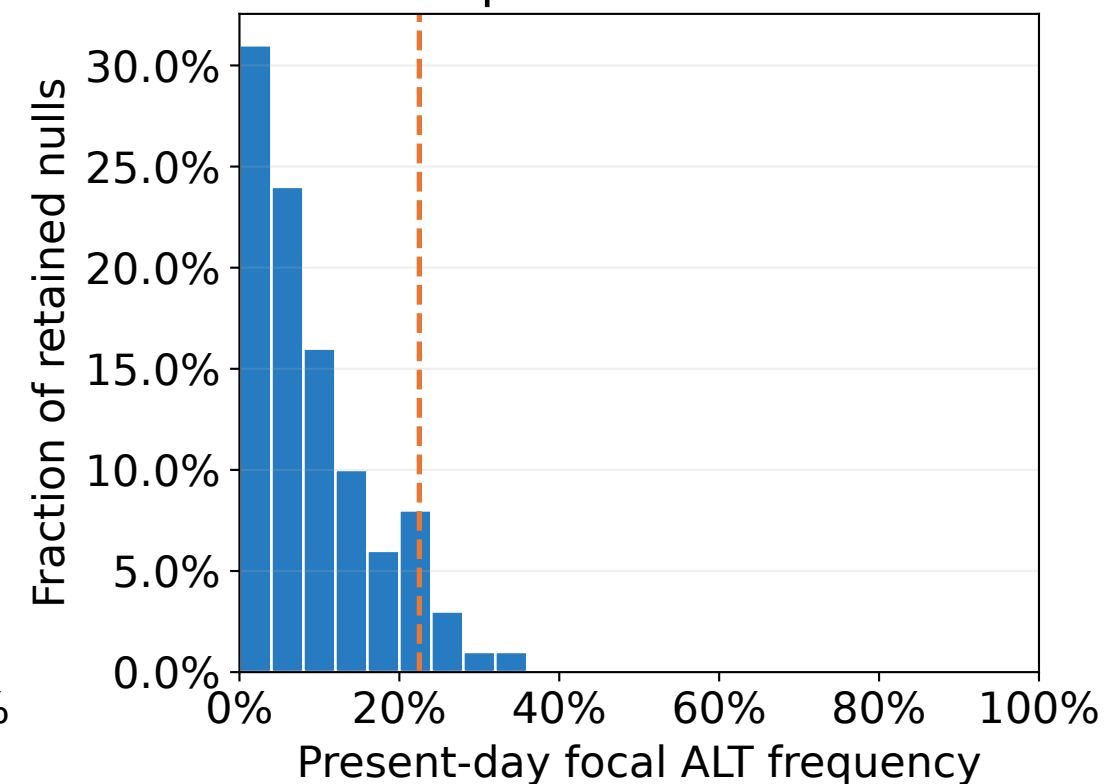
AFR | $n = 100$
95th percentile: 9.2%



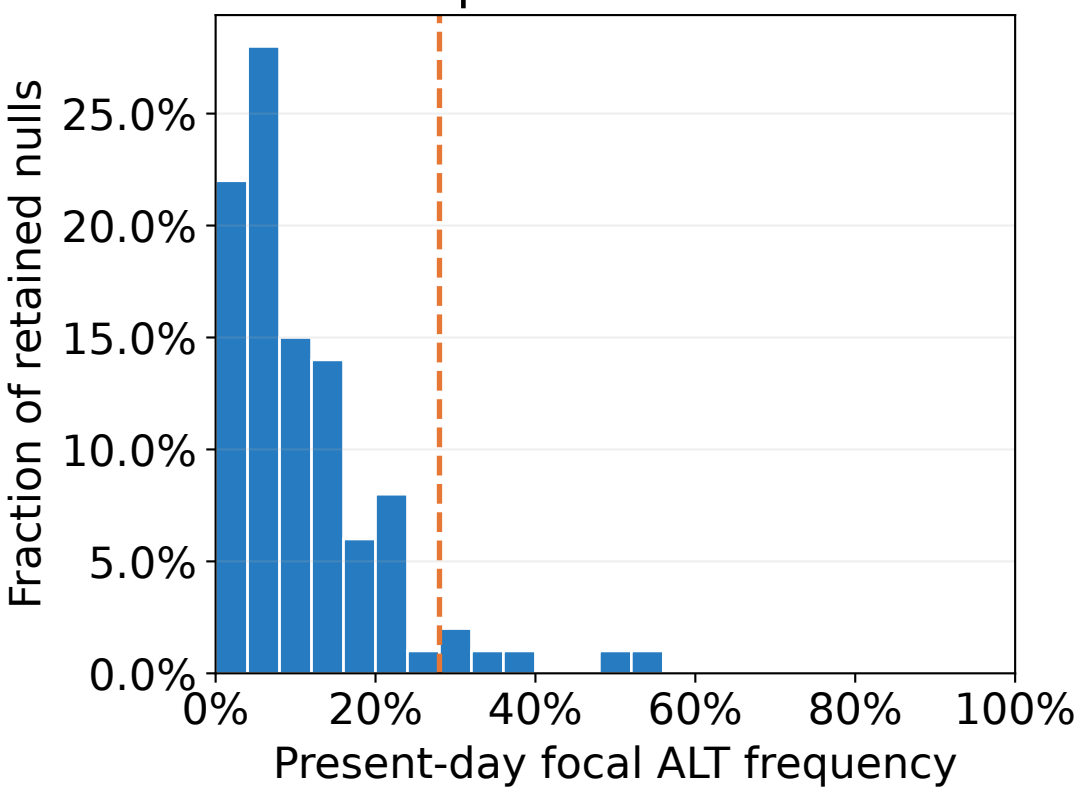
AMR | $n = 100$
95th percentile: 18.0%



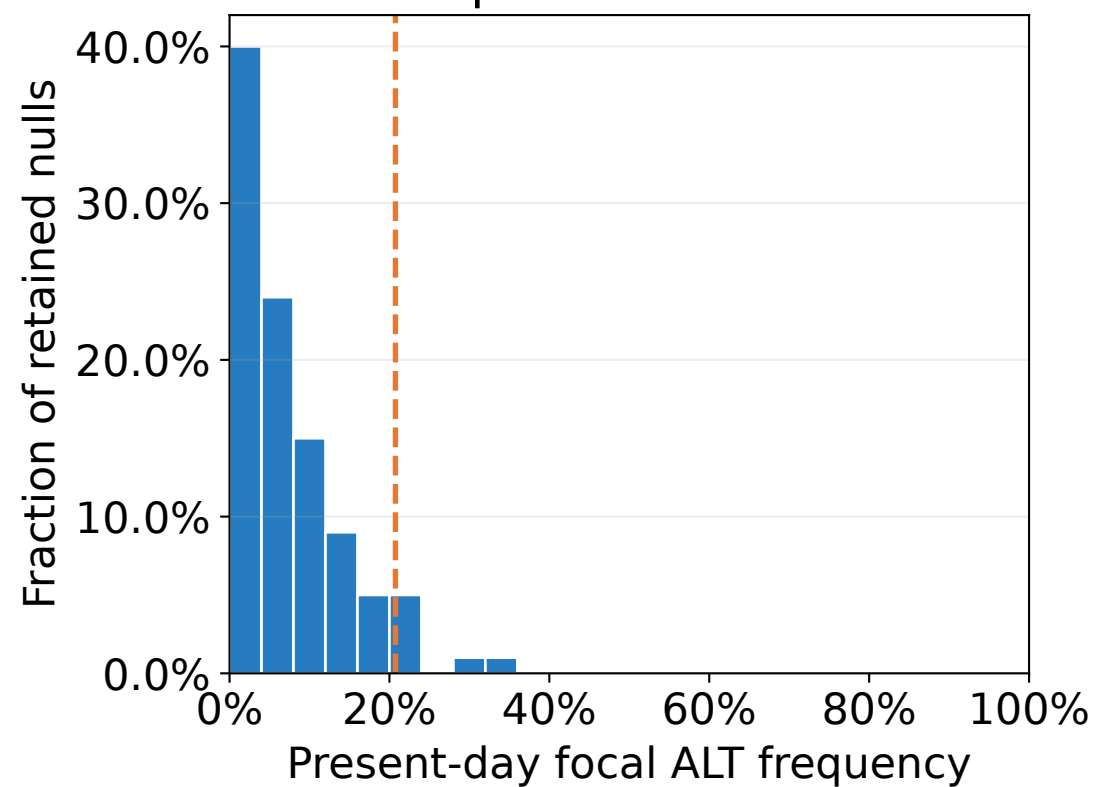
EAS | $n = 100$
95th percentile: 22.5%



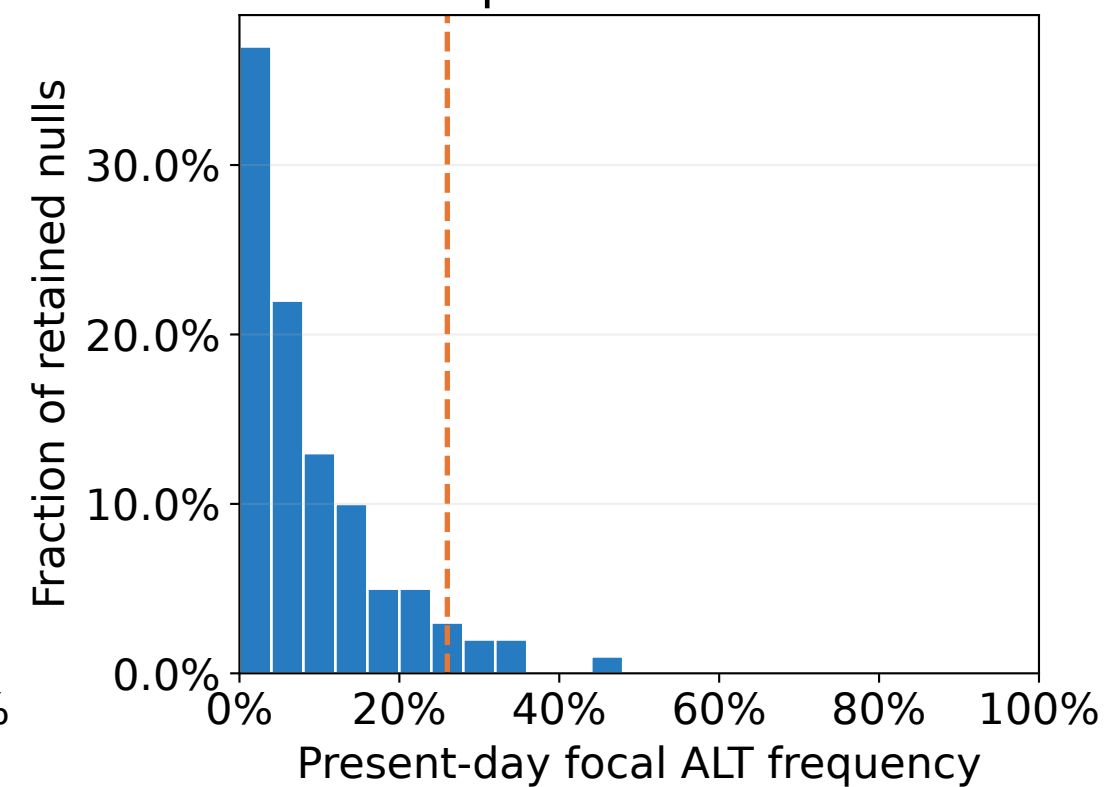
EUR | $n = 100$
95th percentile: 28.0%



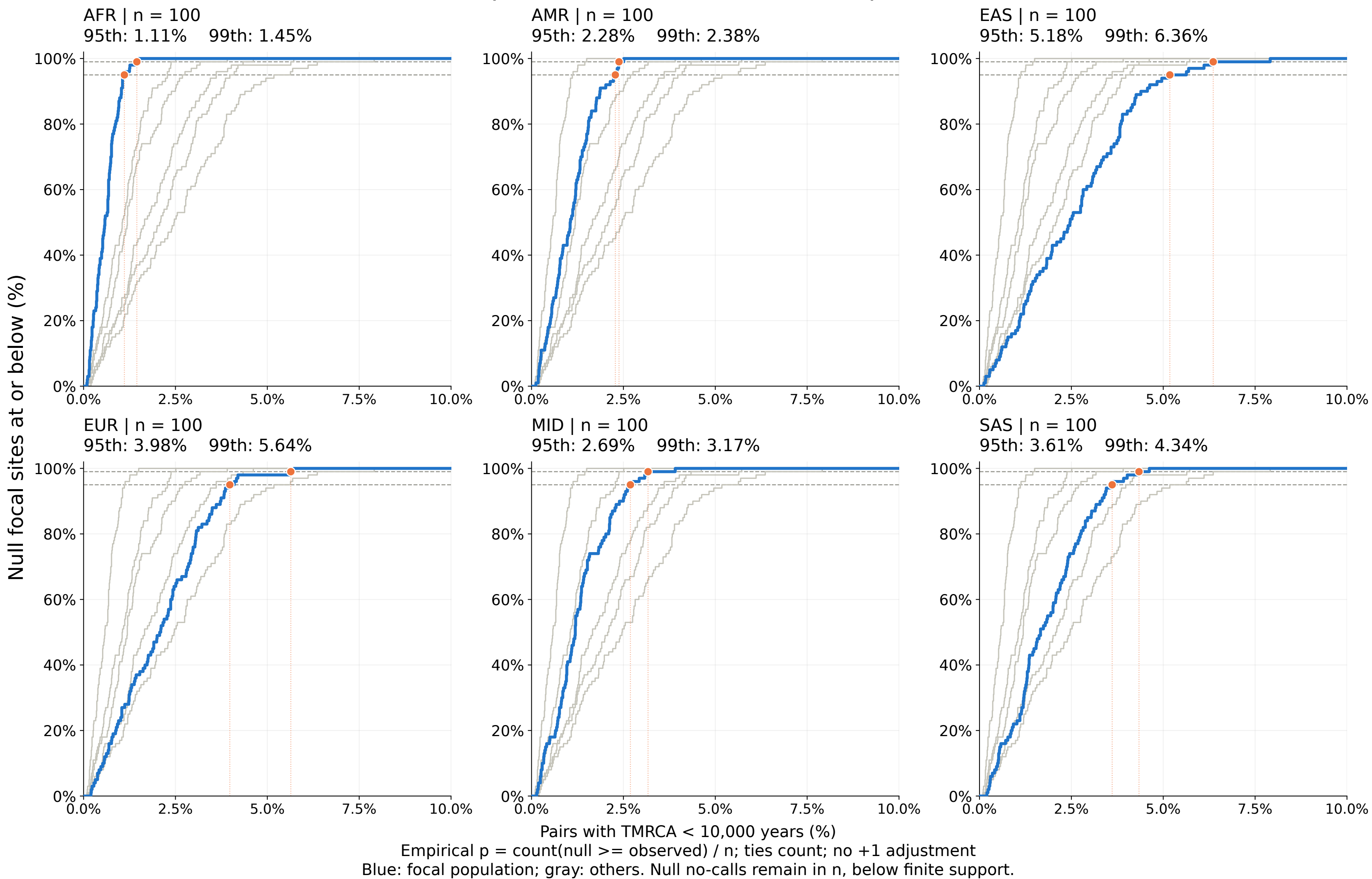
MID | $n = 100$
95th percentile: 20.8%



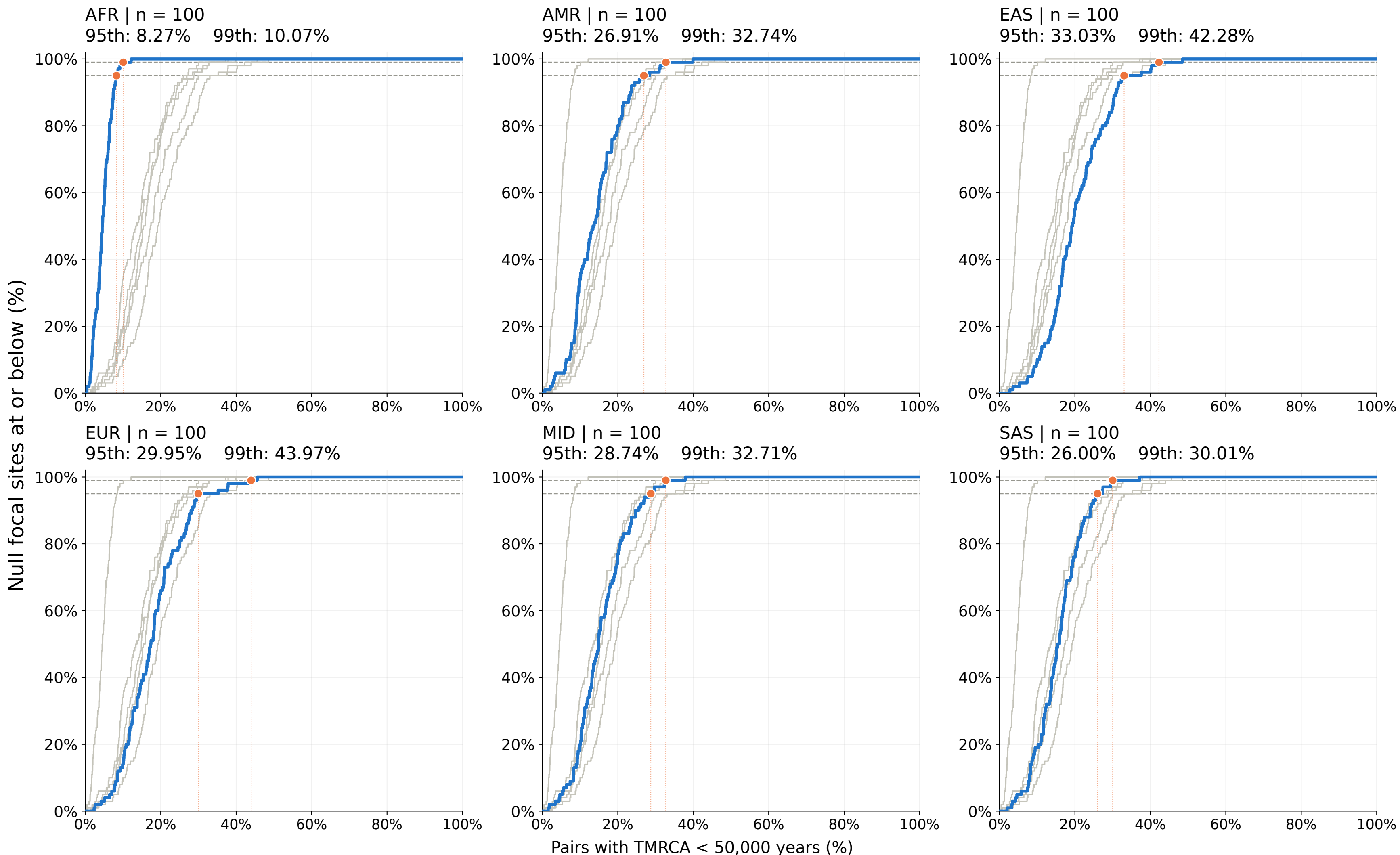
SAS | $n = 100$
95th percentile: 26.0%



Decoded TMRCA | All sampled haplotype pairs | $T < 10,000$ years



Decoded TMRCA | All sampled pairs | T < 50,000 years

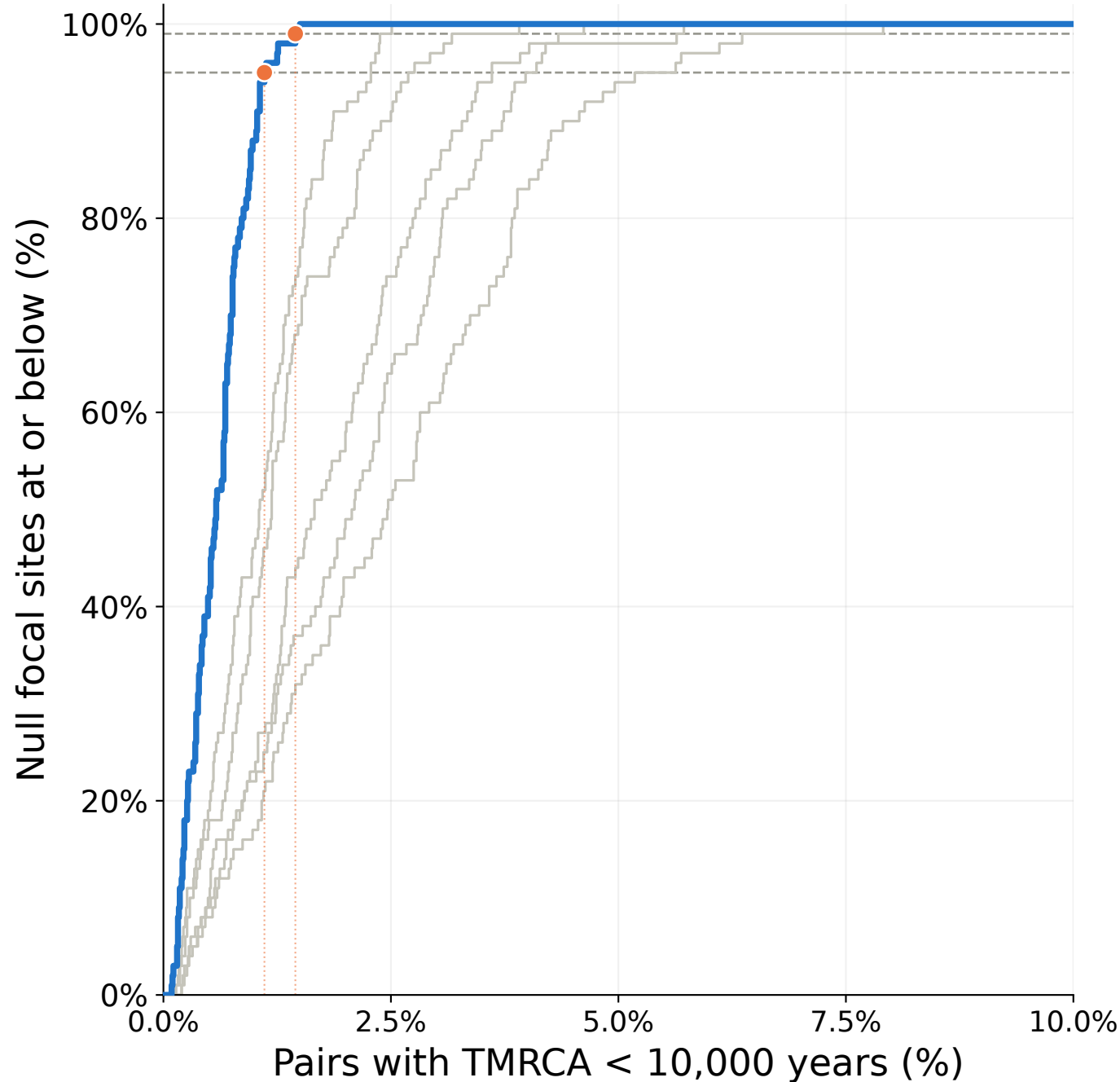


Empirical p = count(null >= observed) / n; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Decoded TMRCA | All sampled haplotype pairs

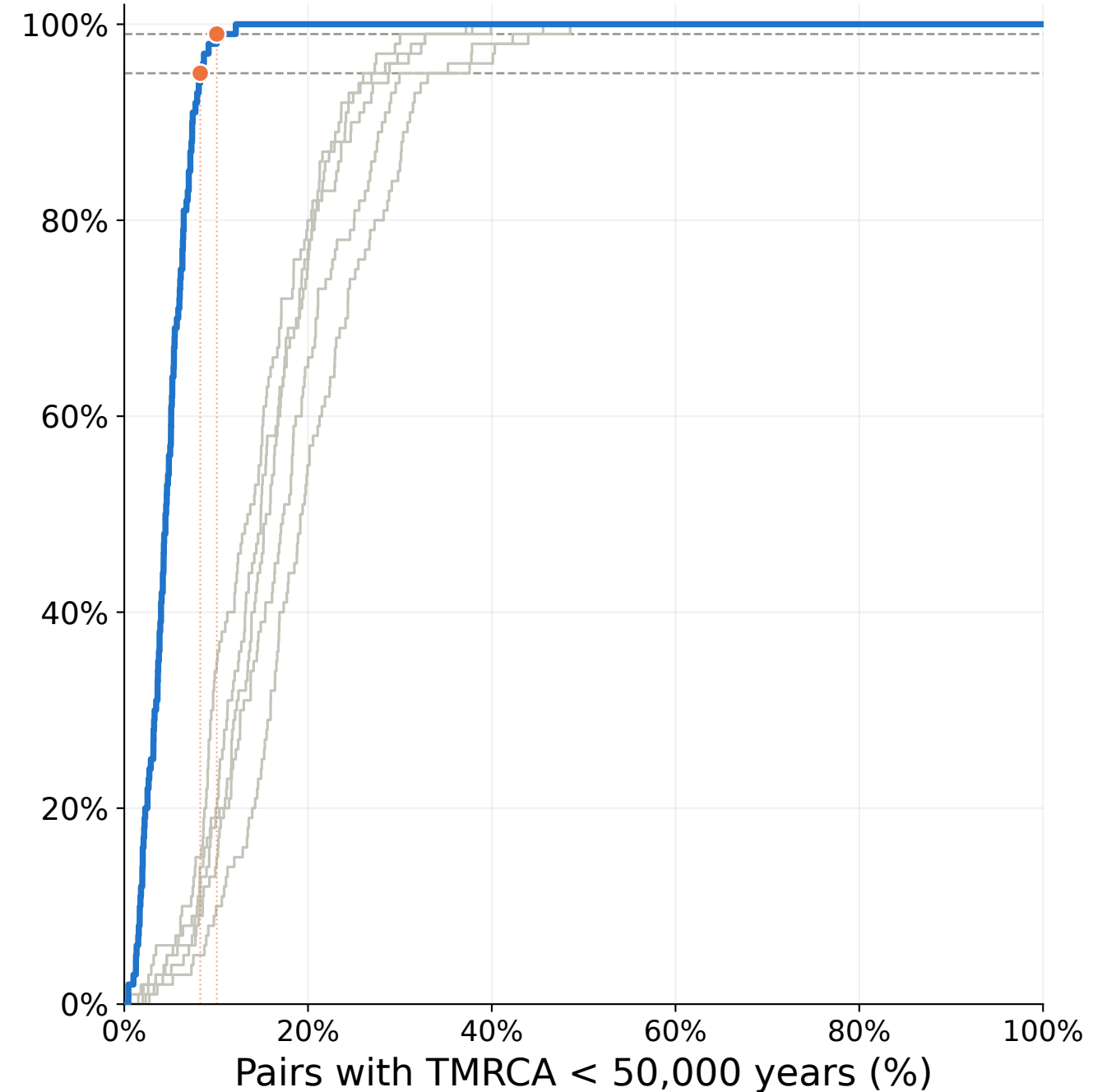
AFR | n = 100

95th: 1.11% 99th: 1.45%



AFR | n = 100

95th: 8.27% 99th: 10.07%



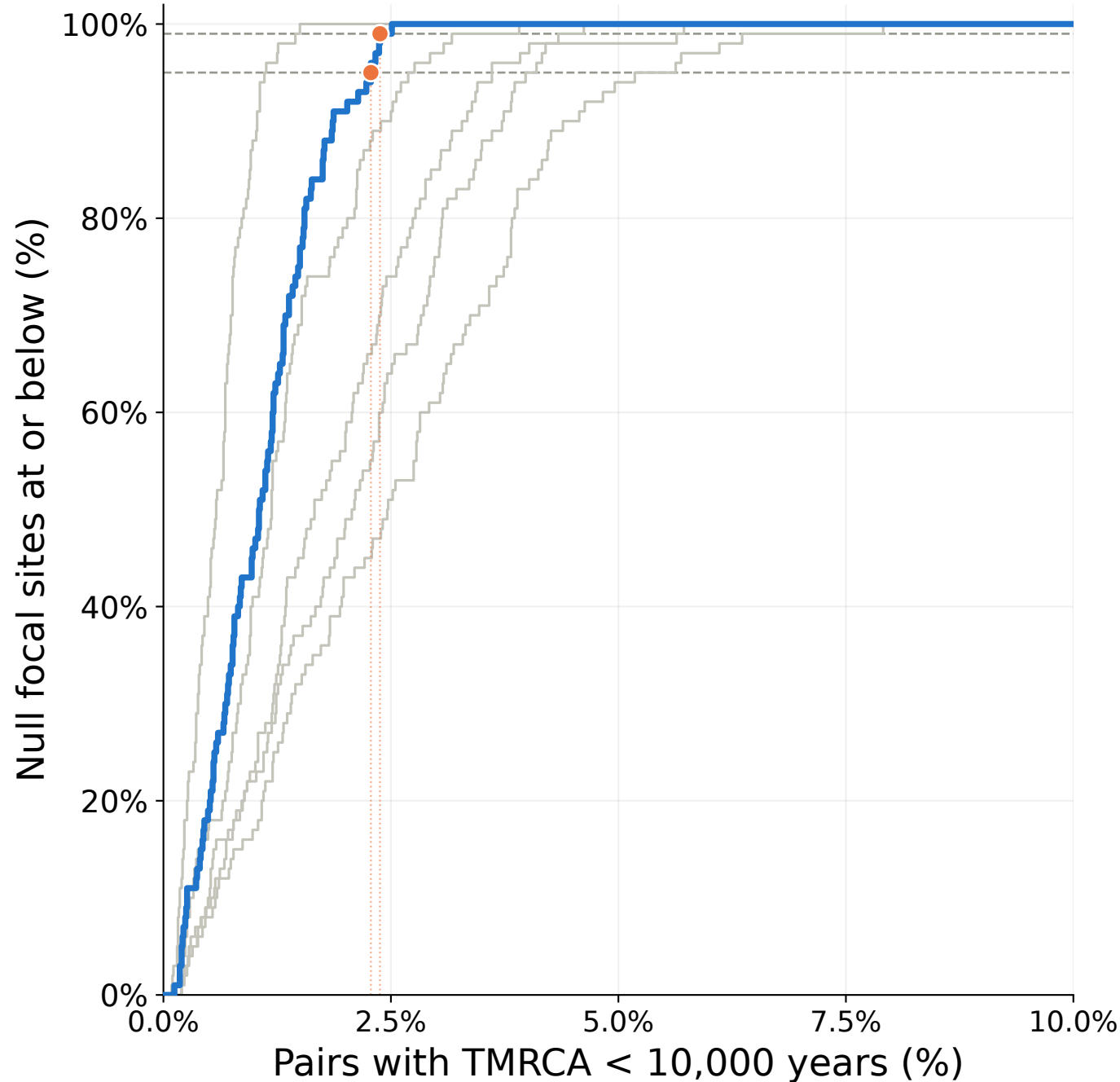
Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment

Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Decoded TMRCA | All sampled haplotype pairs

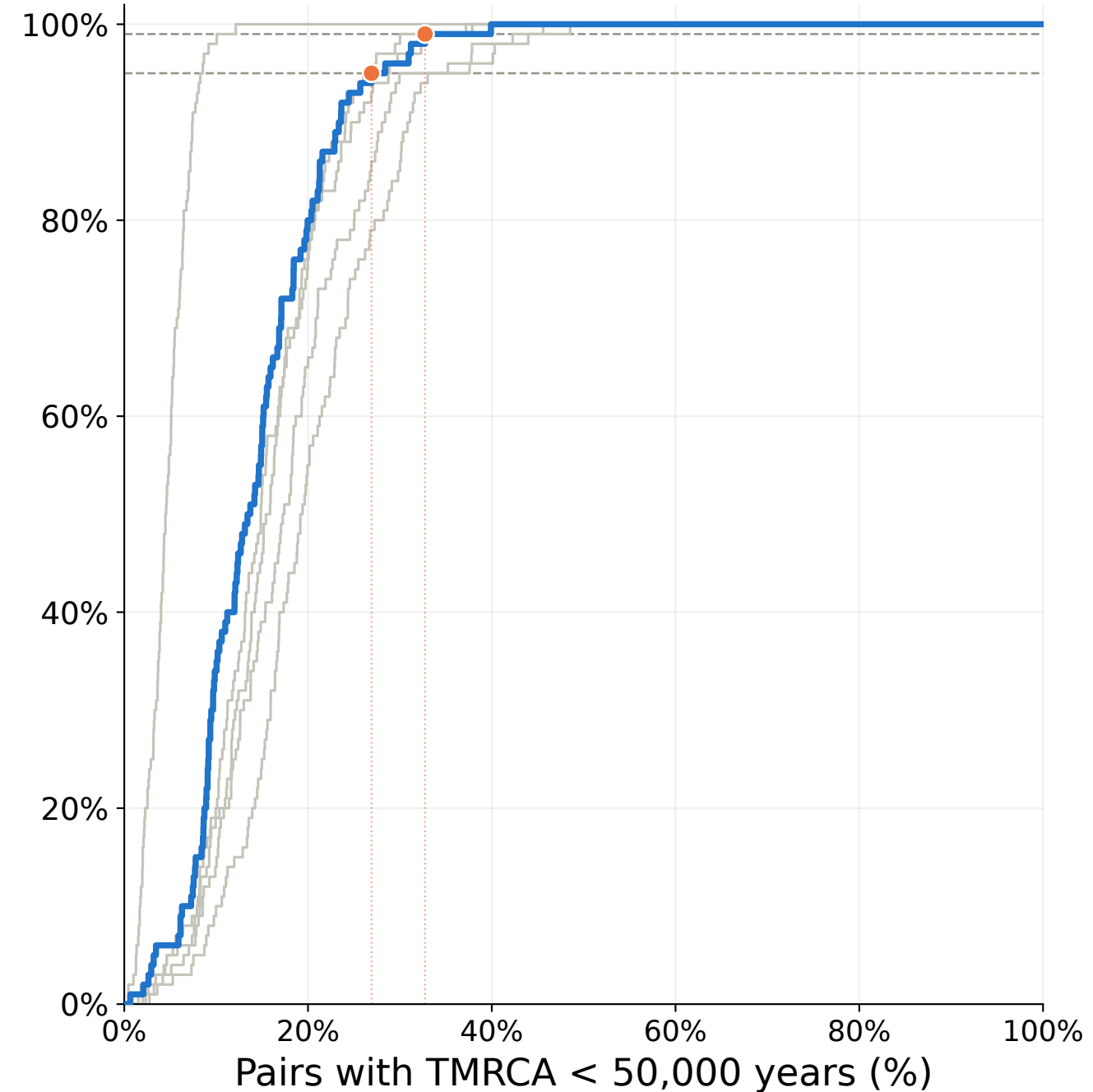
AMR | n = 100

95th: 2.28% 99th: 2.38%



AMR | n = 100

95th: 26.91% 99th: 32.74%



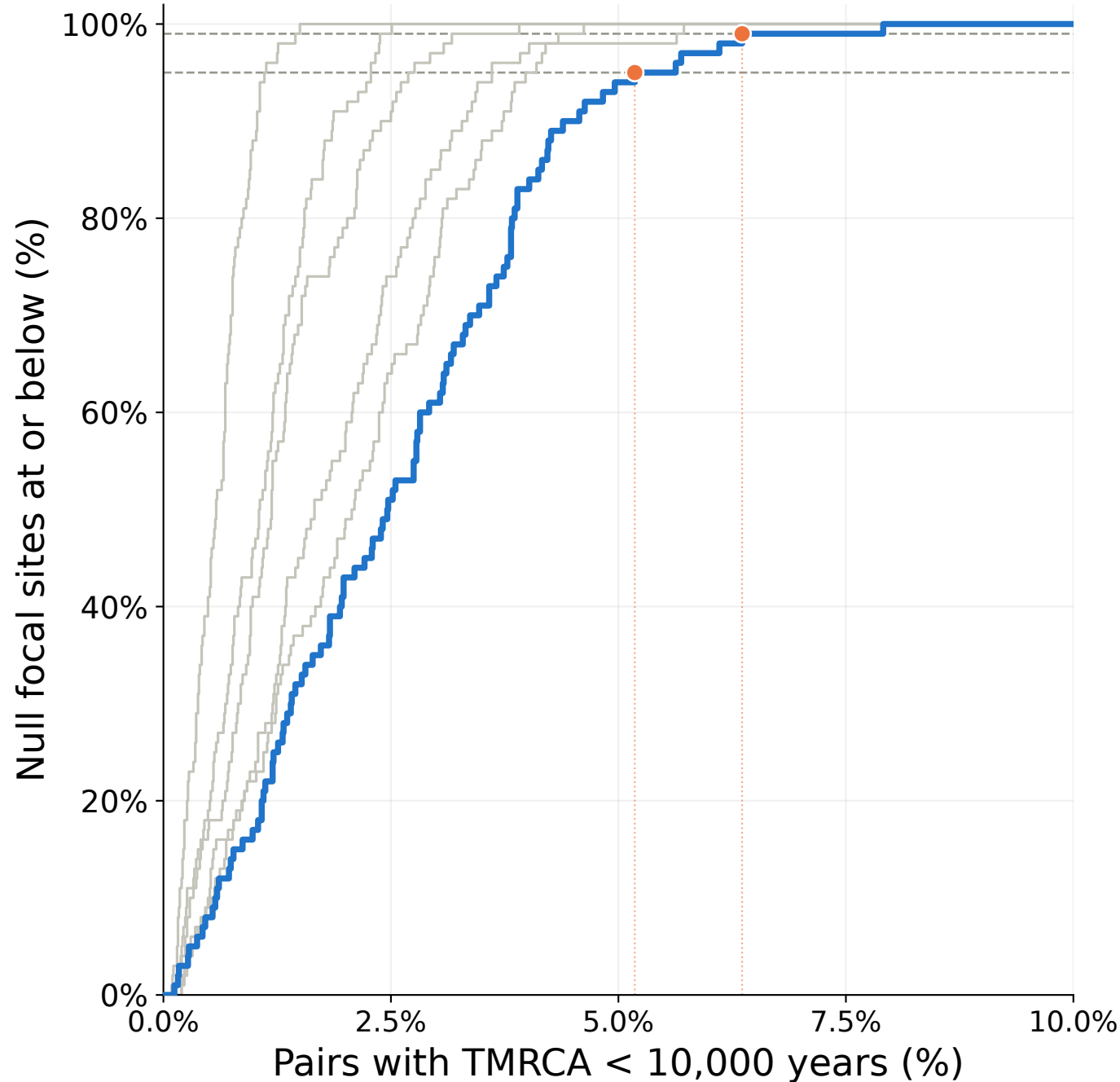
Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment

Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Decoded TMRCA | All sampled haplotype pairs

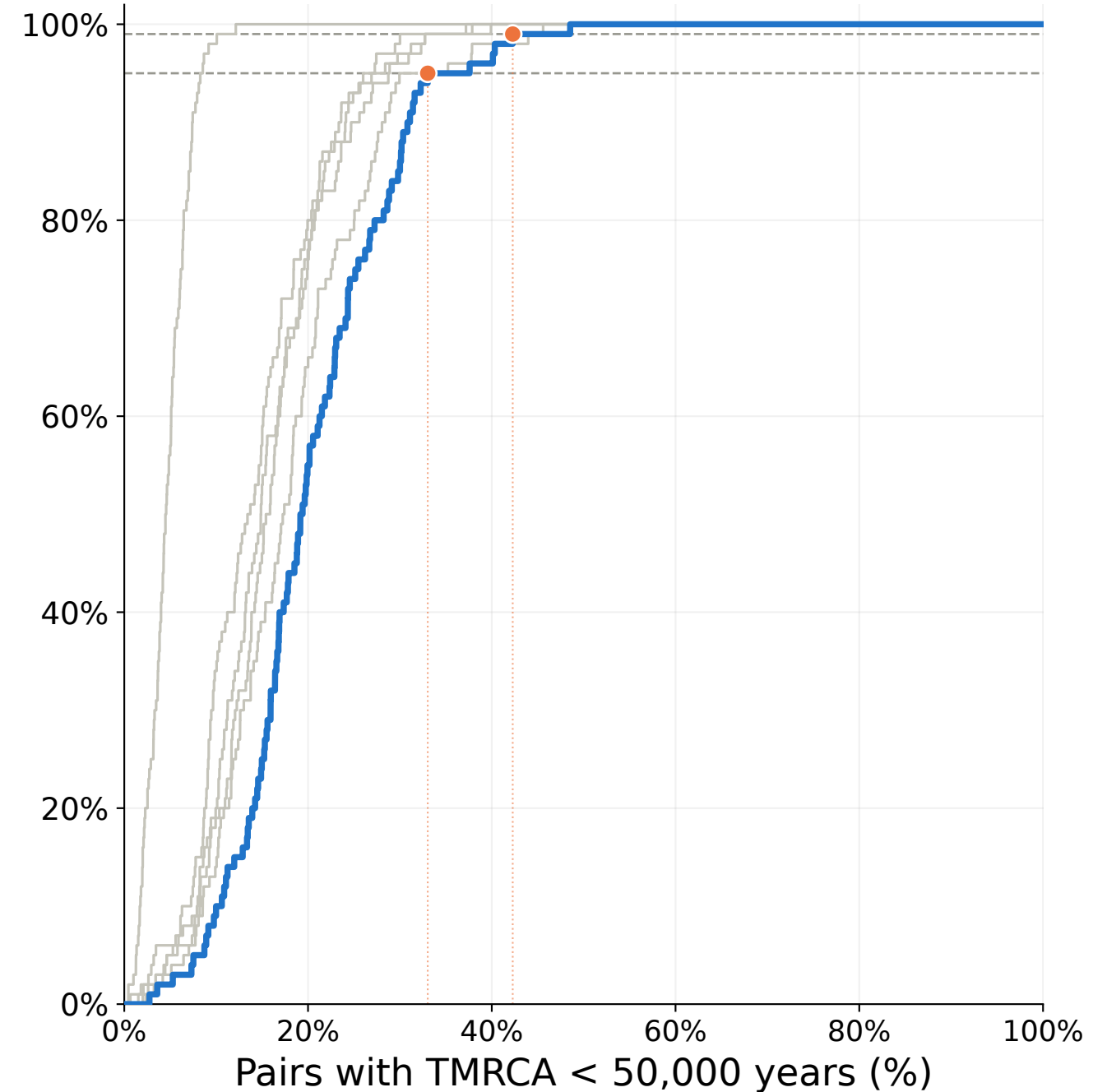
EAS | n = 100

95th: 5.18% 99th: 6.36%



EAS | n = 100

95th: 33.03% 99th: 42.28%

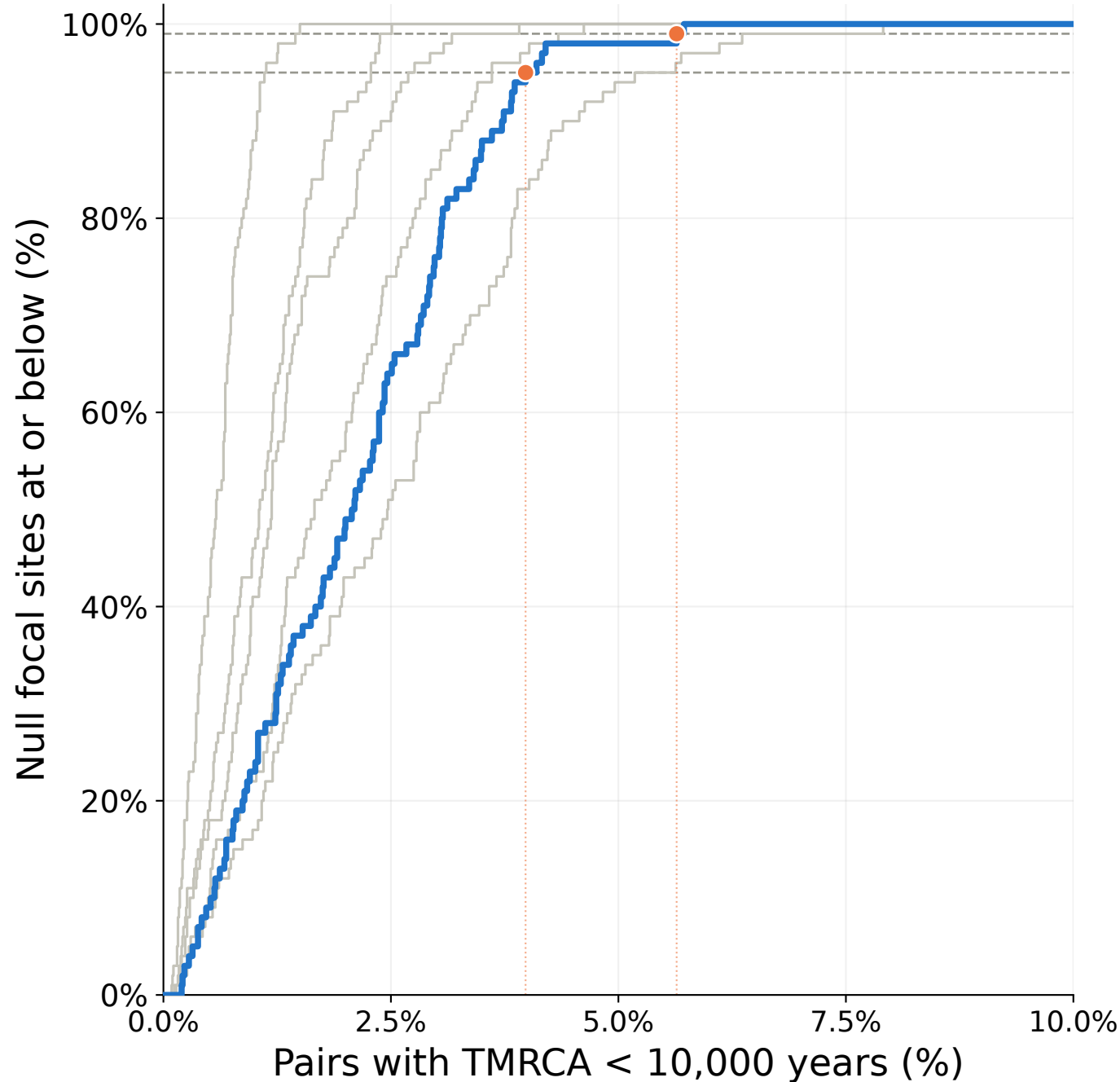


Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n , below finite support.

Decoded TMRCA | All sampled haplotype pairs

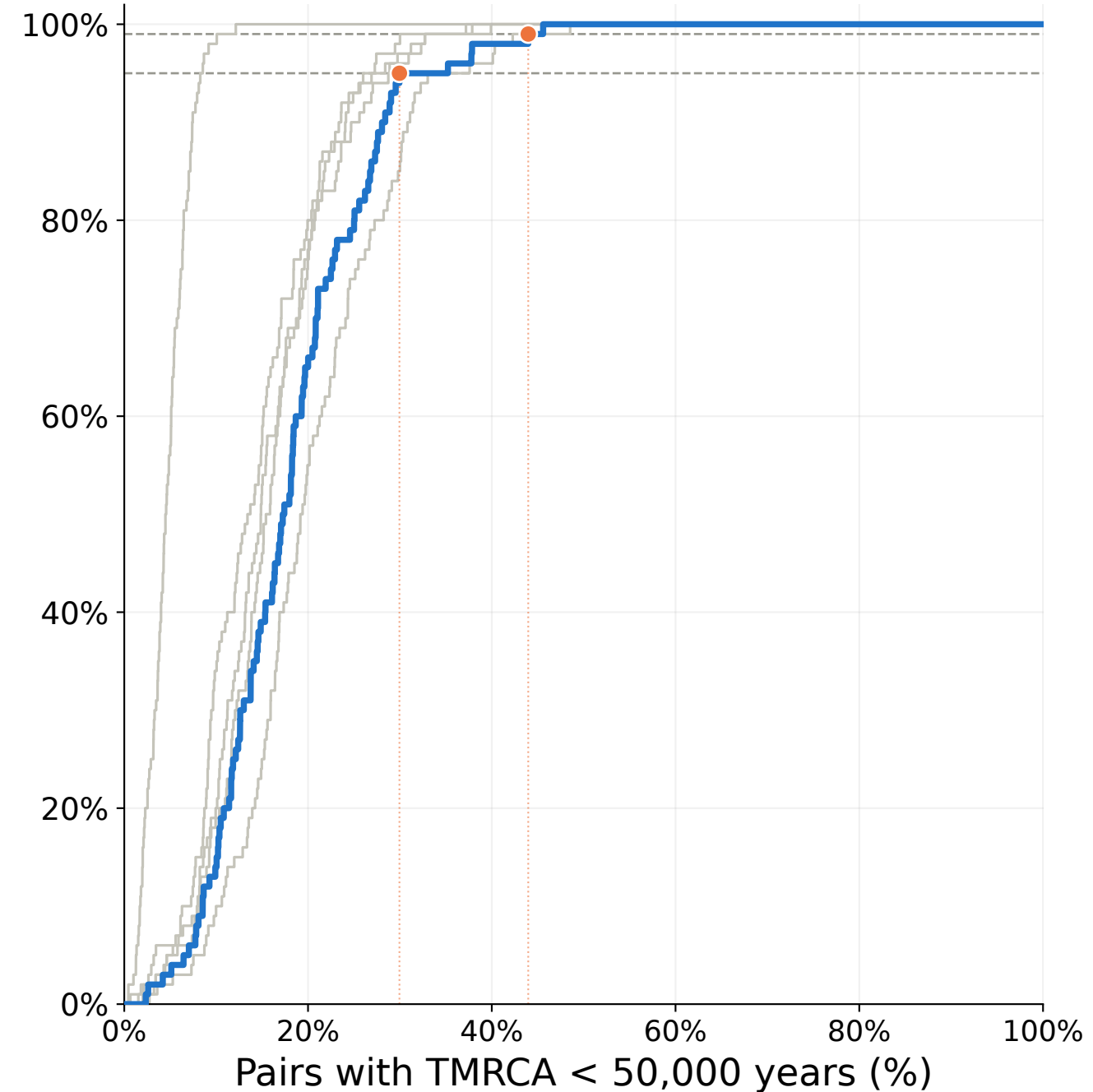
EUR | n = 100

95th: 3.98% 99th: 5.64%



EUR | n = 100

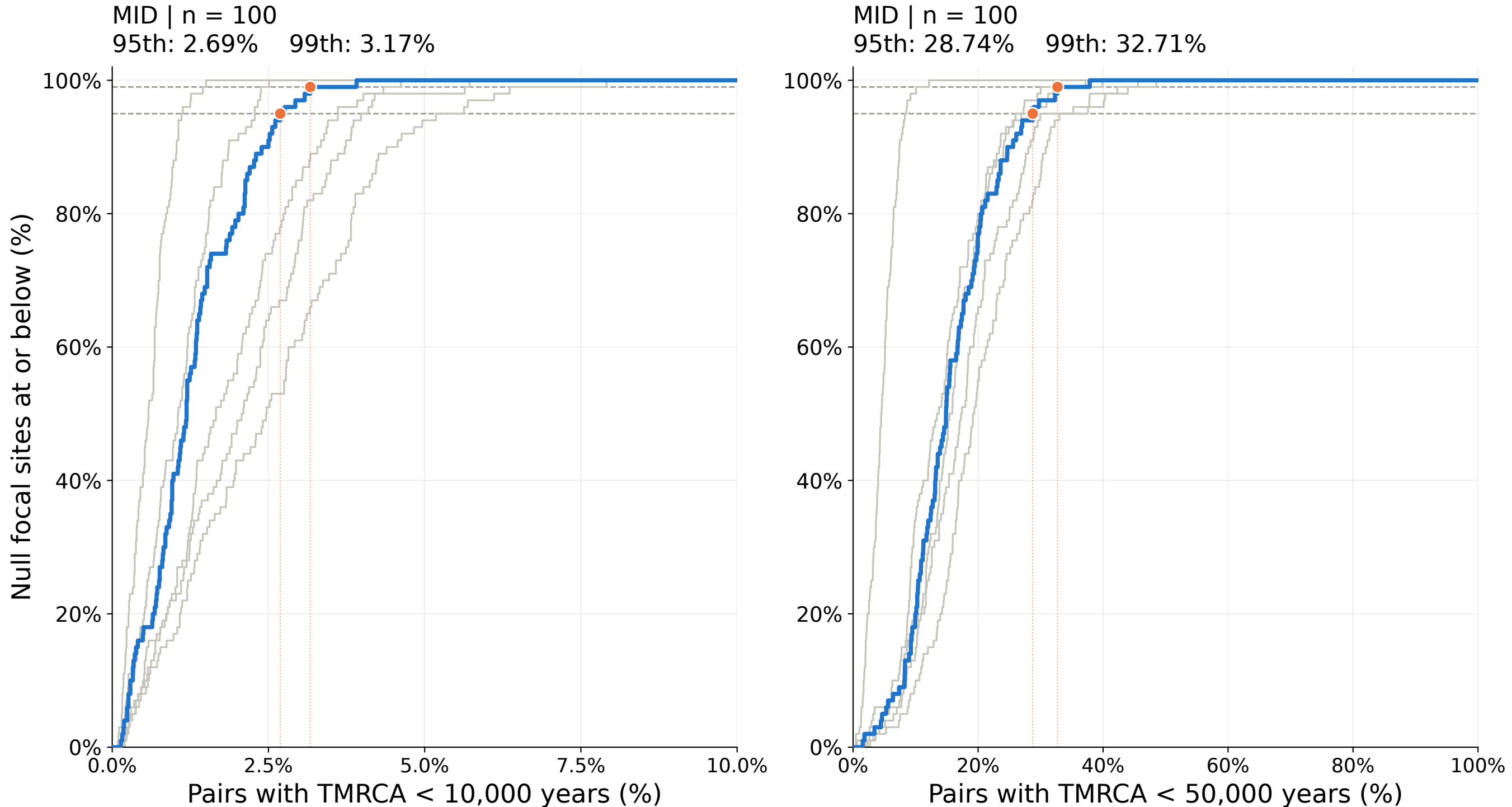
95th: 29.95% 99th: 43.97%



Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment

Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Decoded TMRCA | All sampled haplotype pairs

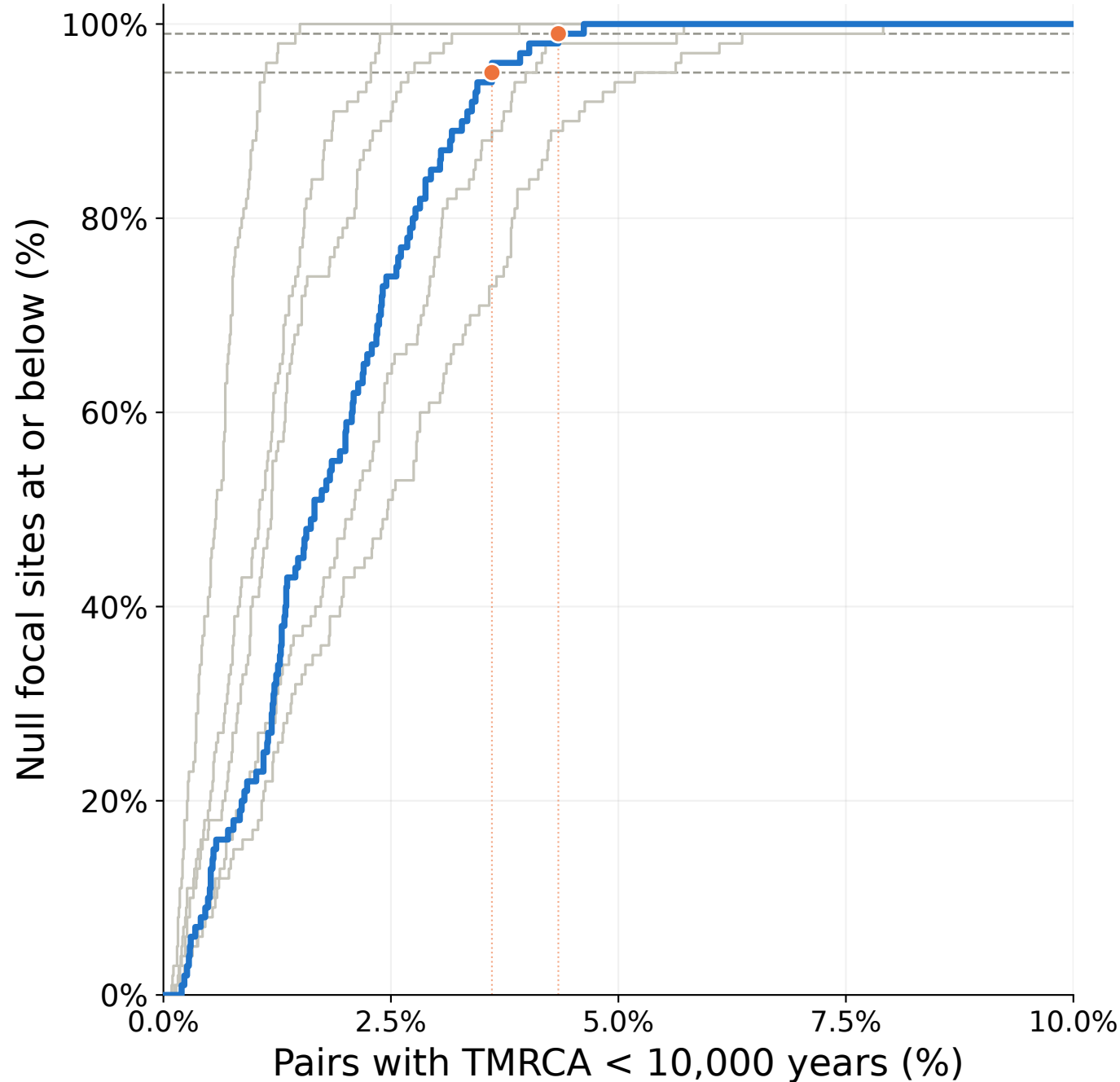


Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n , below finite support.

Decoded TMRCA | All sampled haplotype pairs

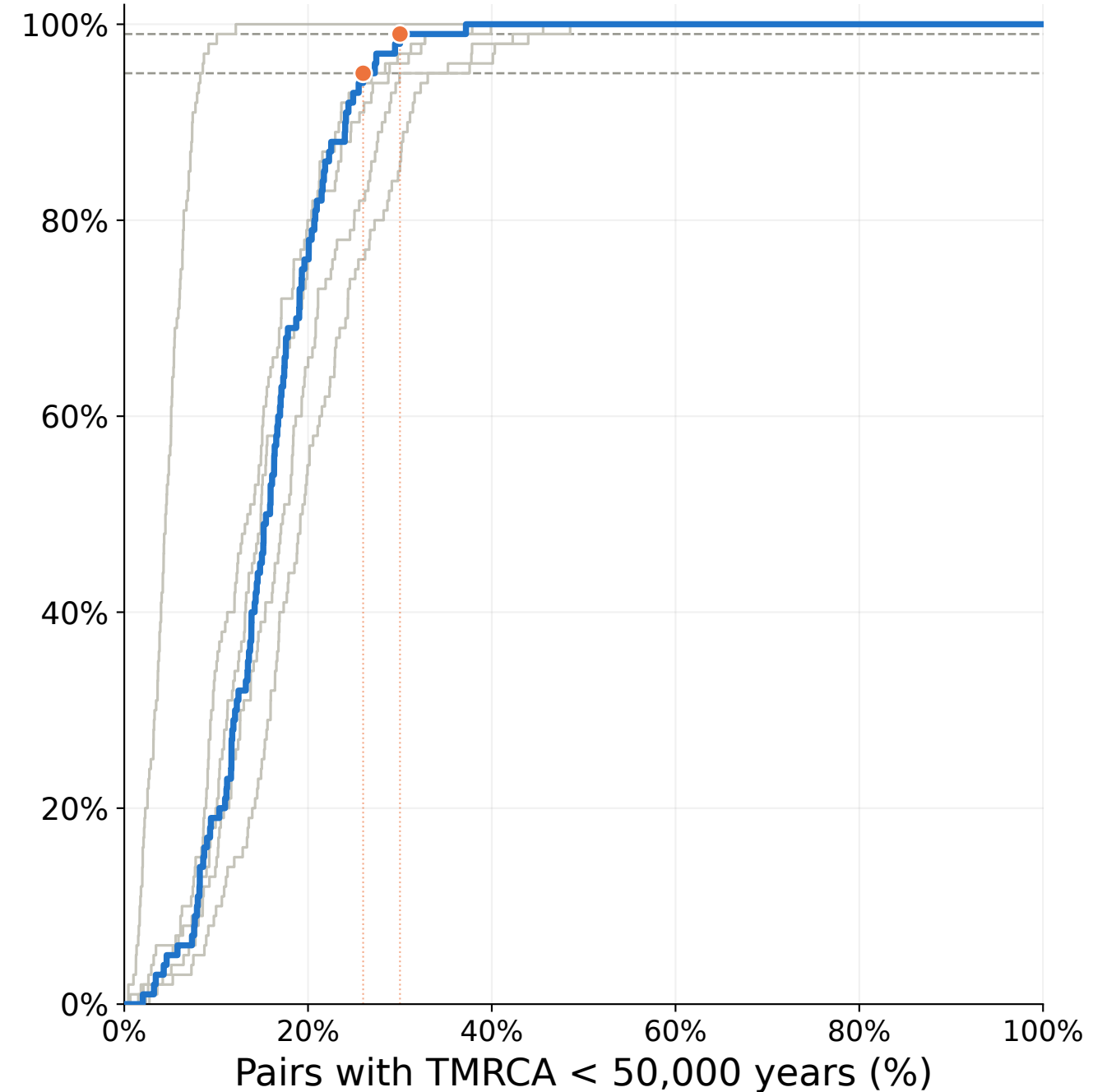
SAS | n = 100

95th: 3.61% 99th: 4.34%



SAS | n = 100

95th: 26.00% 99th: 30.01%

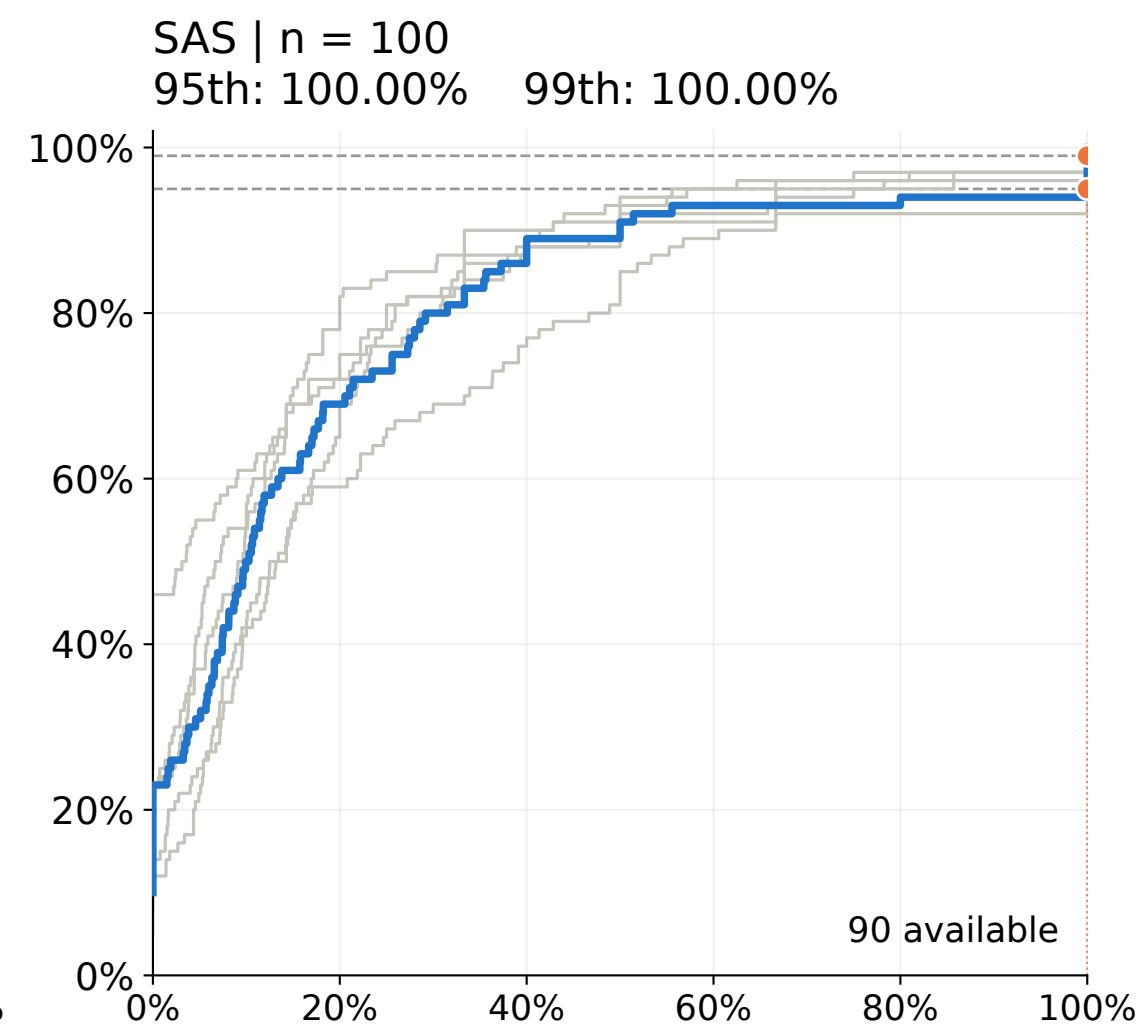
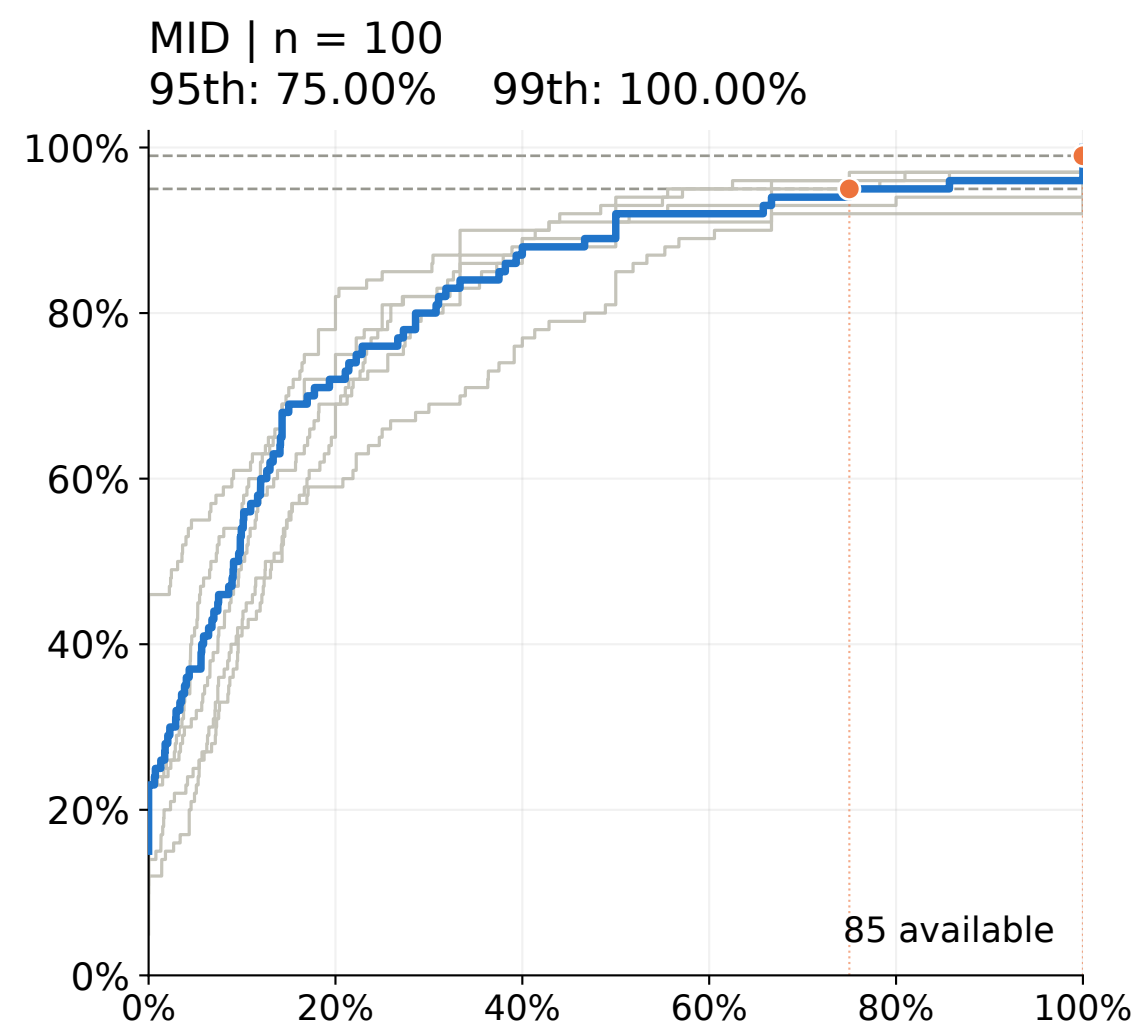
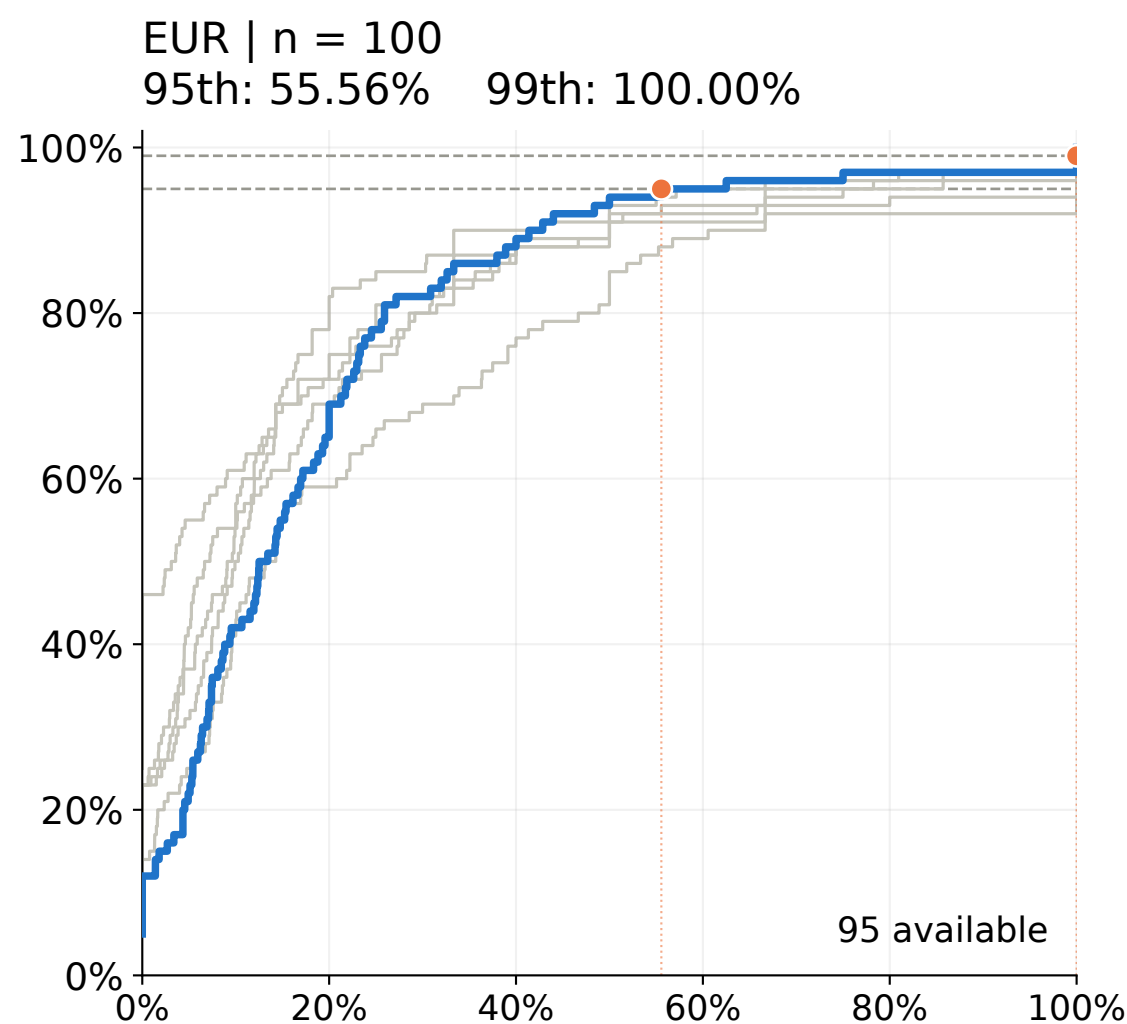
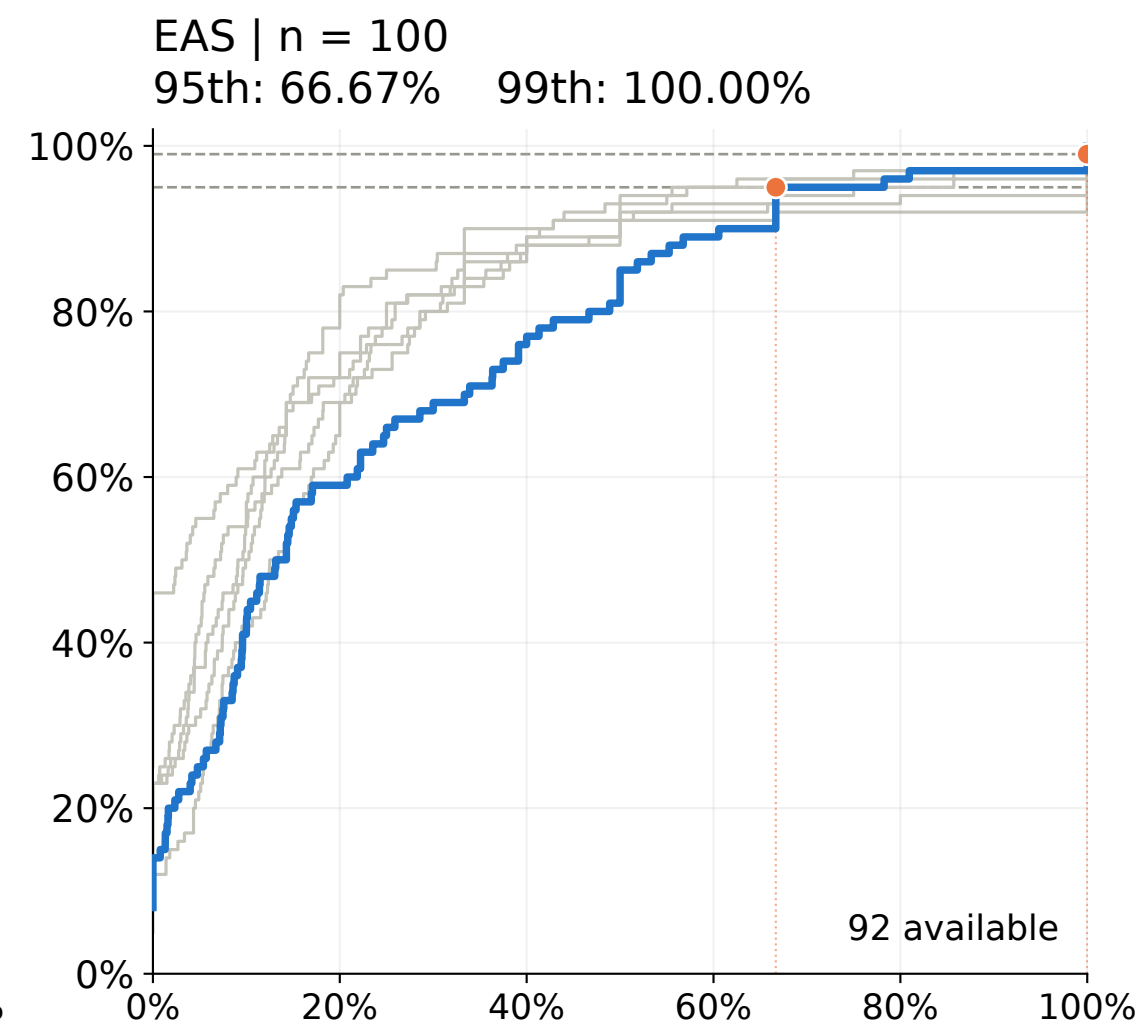
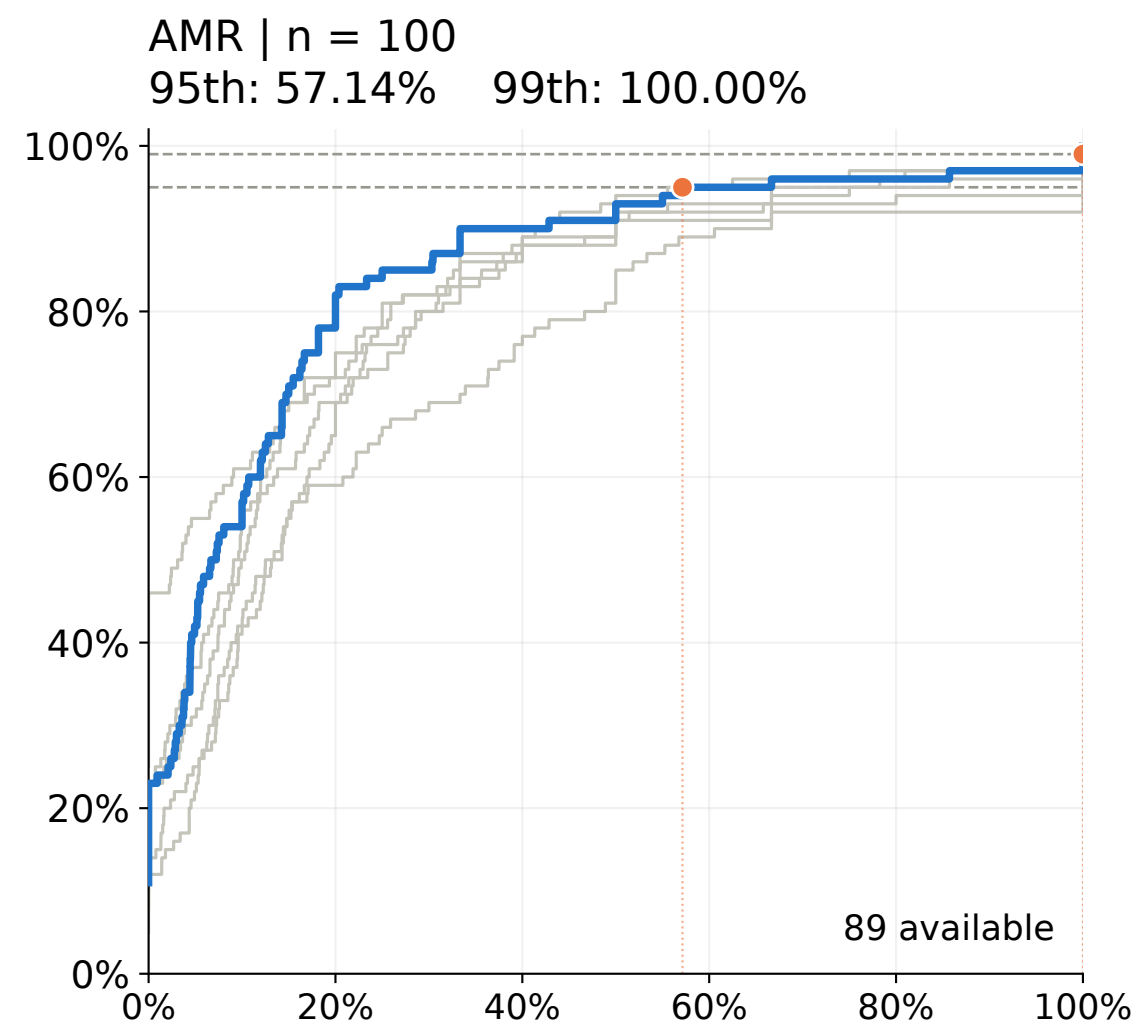
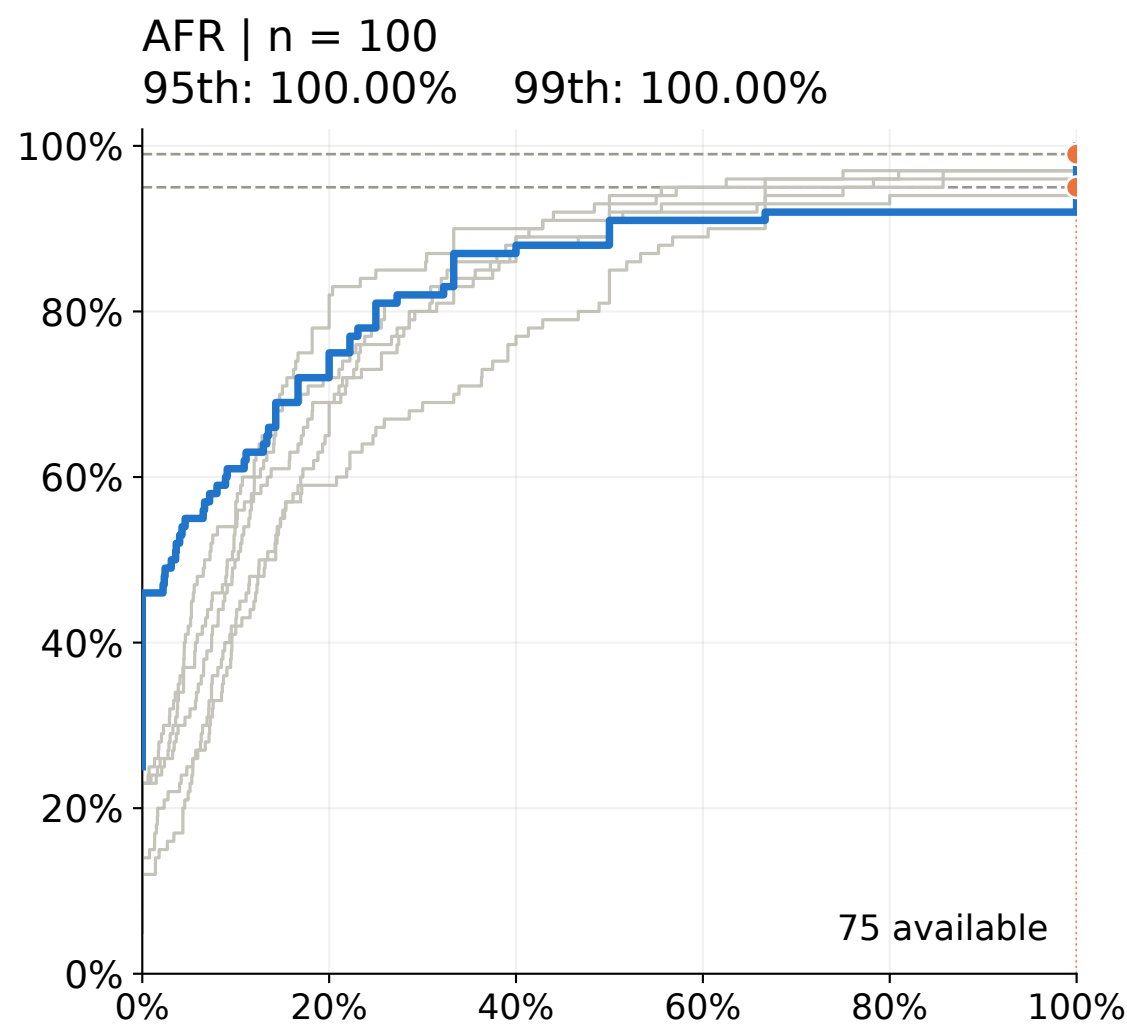


Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment

Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Decoded TMRCA | ALT/ALT pairs | T < 10,000 years

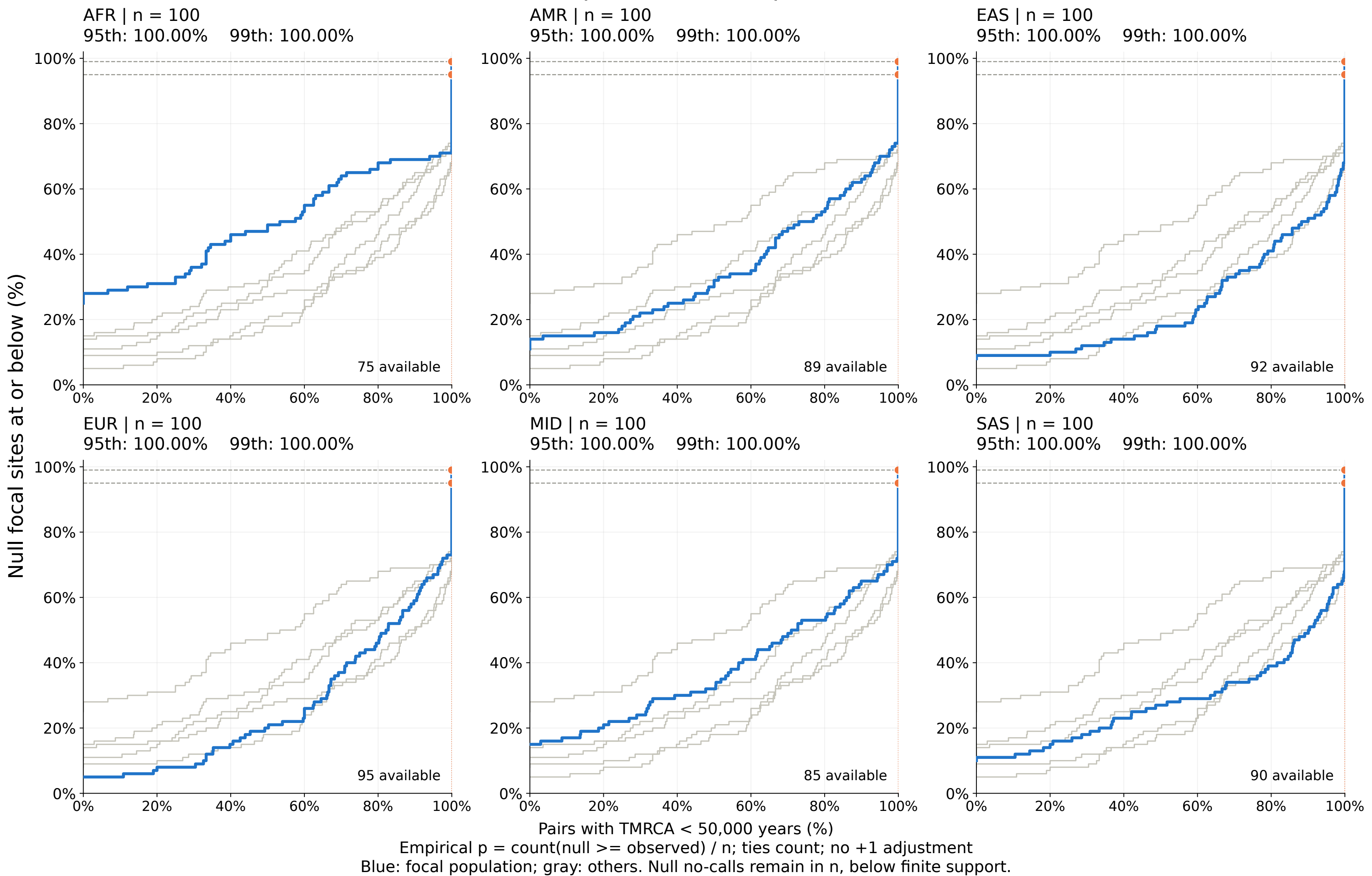
Null focal sites at or below (%)



Pairs with TMRCA < 10,000 years (%)

Empirical p = count(null >= observed) / n; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

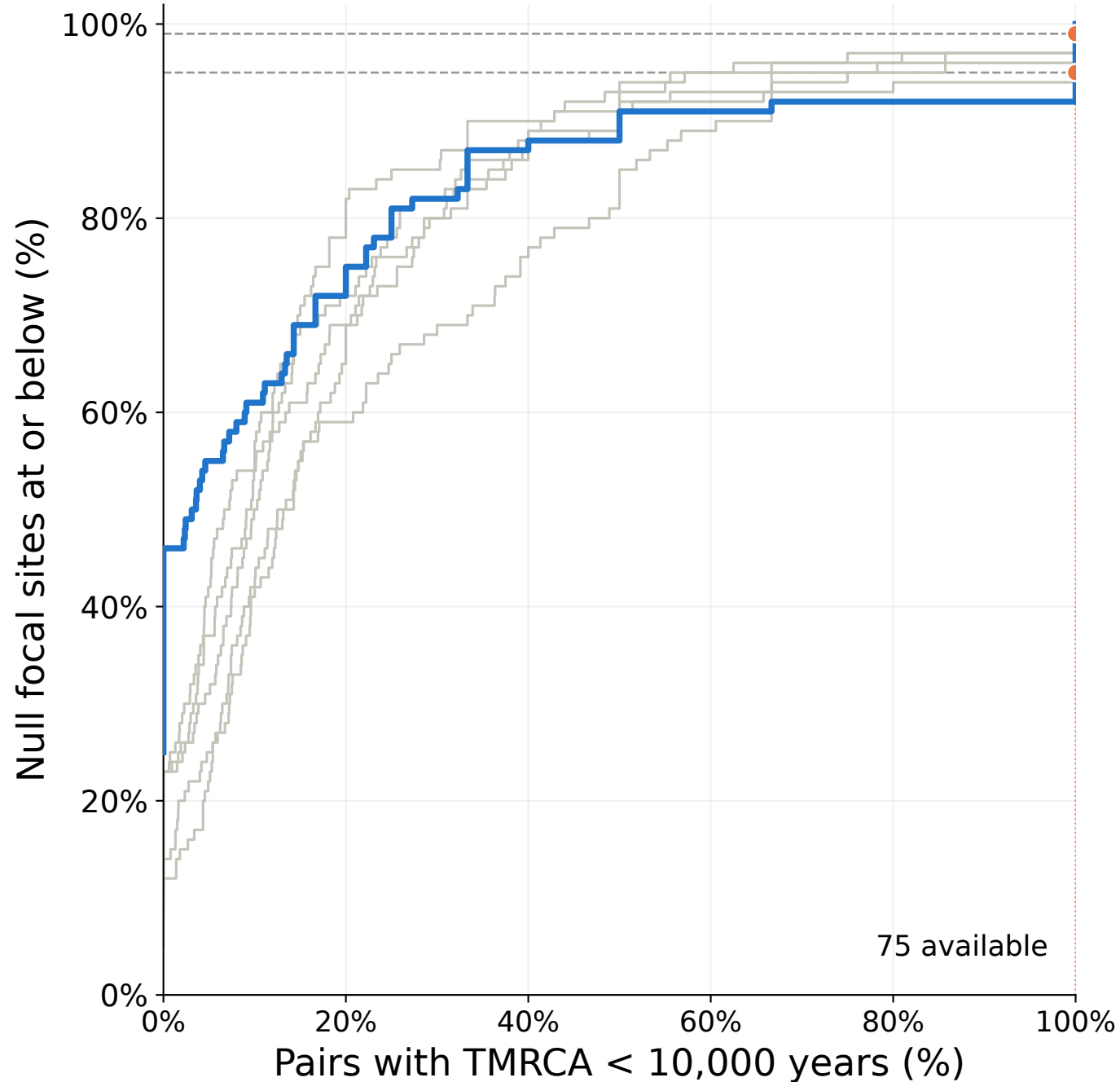
Decoded TMRCA | ALT/ALT pairs | T < 50,000 years



Decoded TMRCA | ALT/ALT pairs

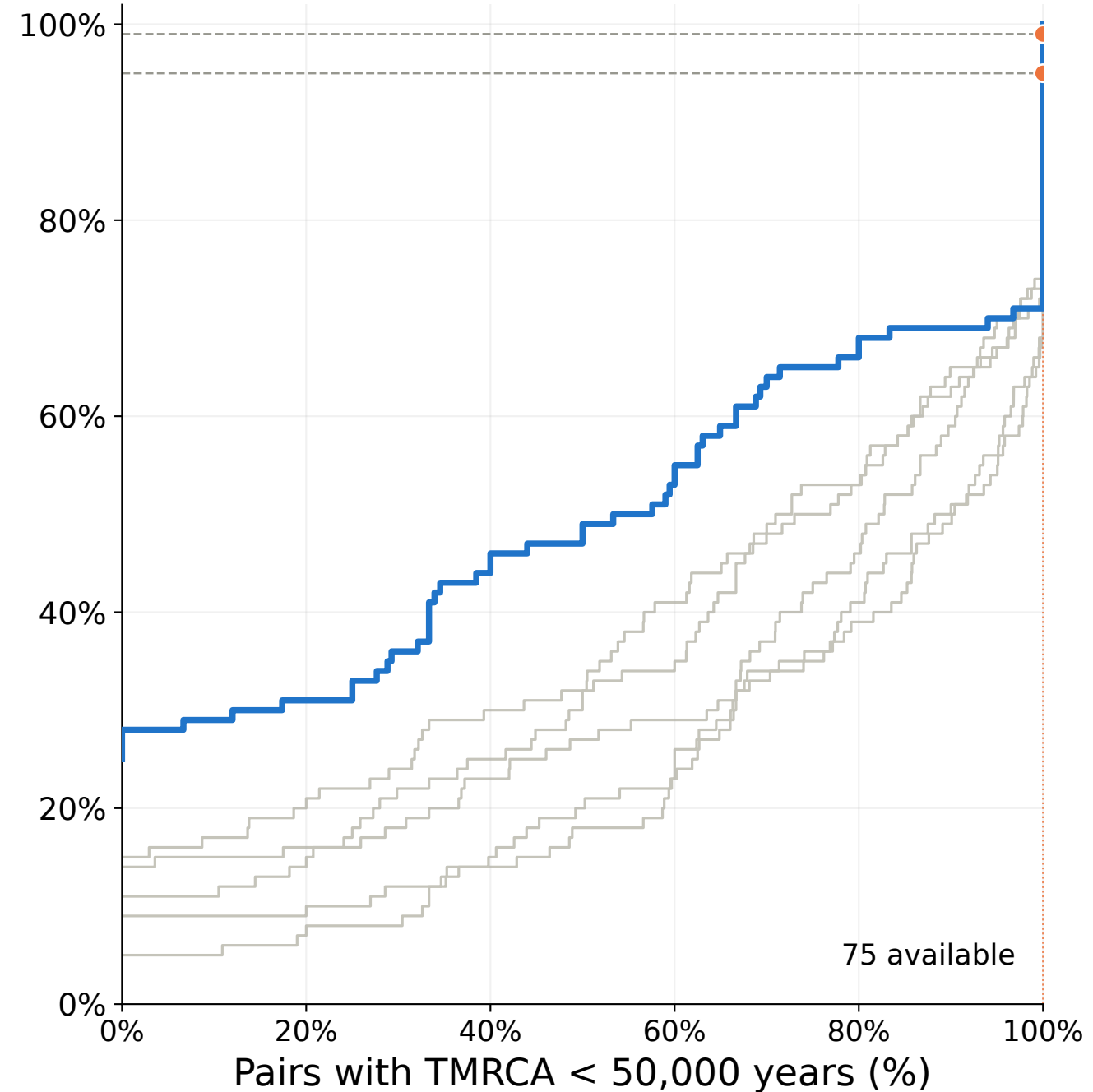
AFR | n = 100

95th: 100.00% 99th: 100.00%



AFR | n = 100

95th: 100.00% 99th: 100.00%

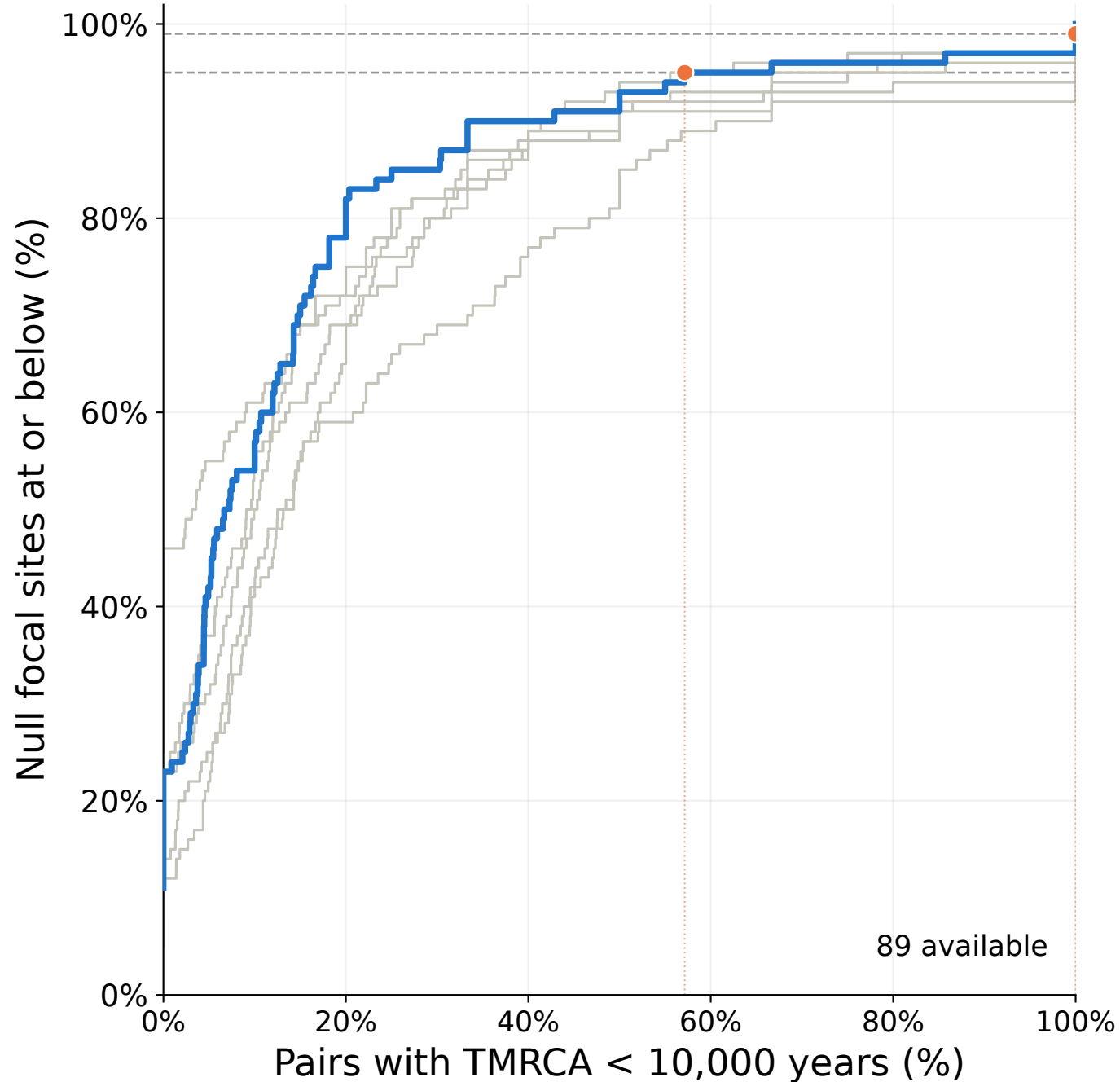


Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Decoded TMRCA | ALT/ALT pairs

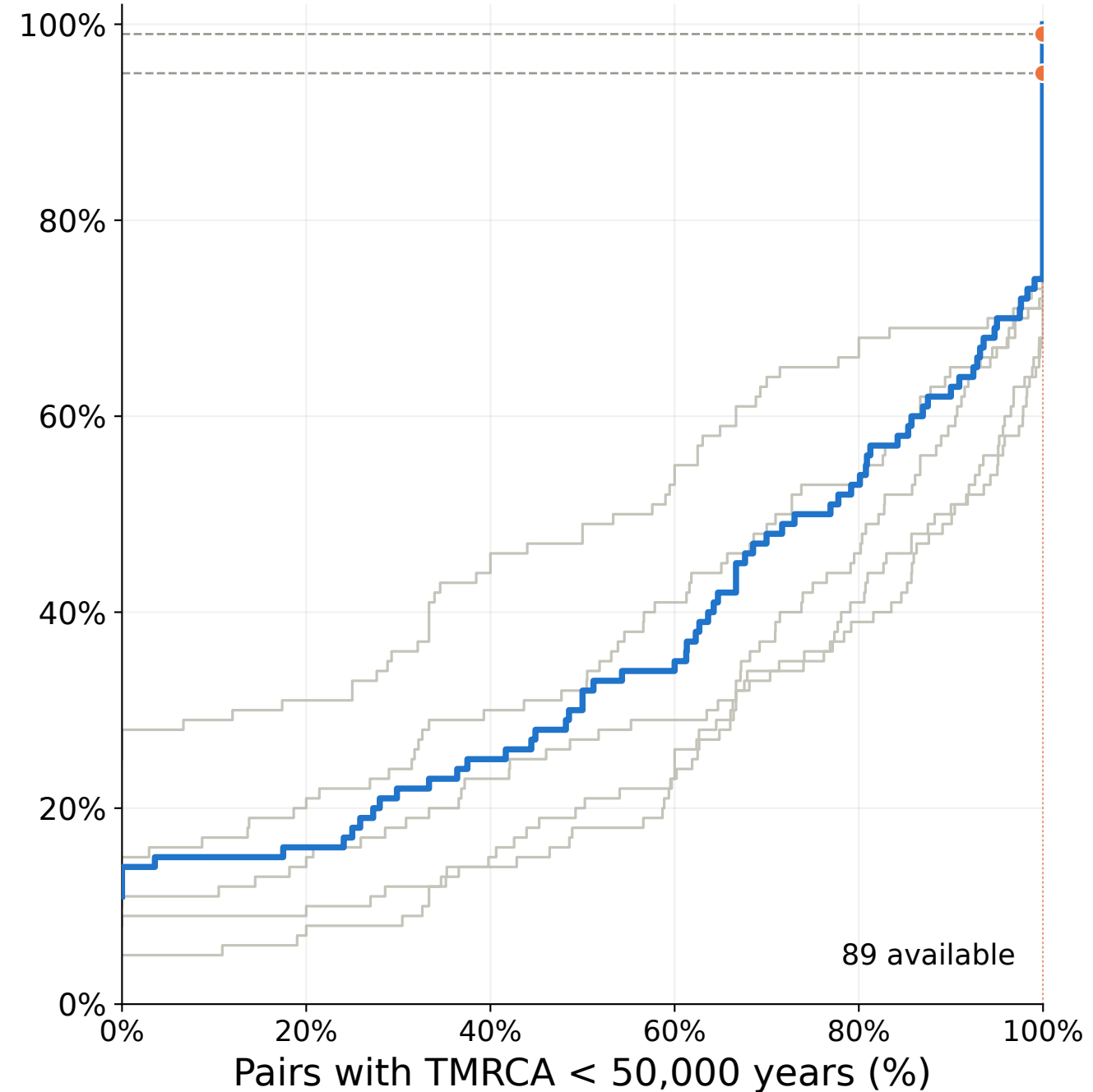
AMR | n = 100

95th: 57.14% 99th: 100.00%



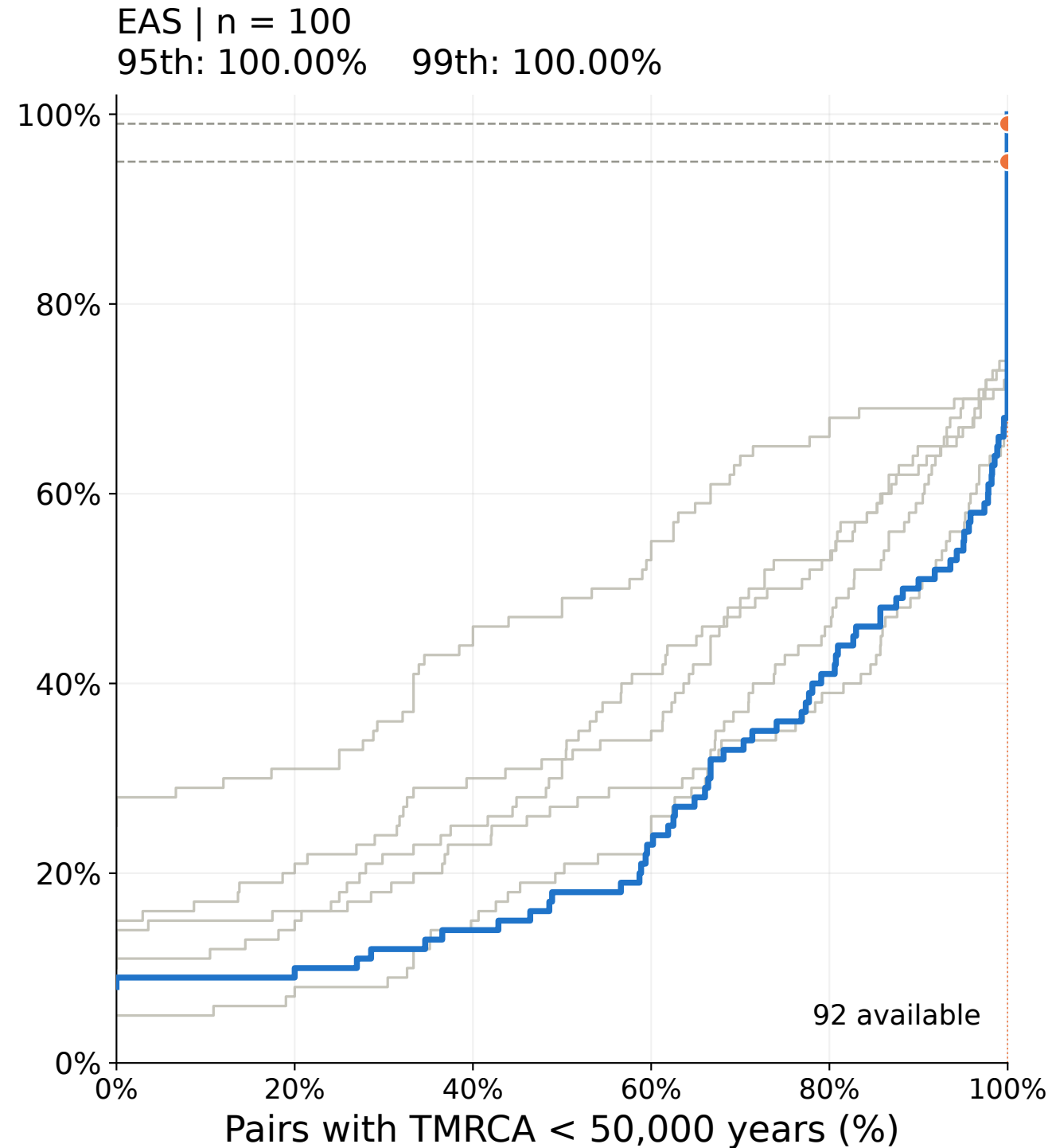
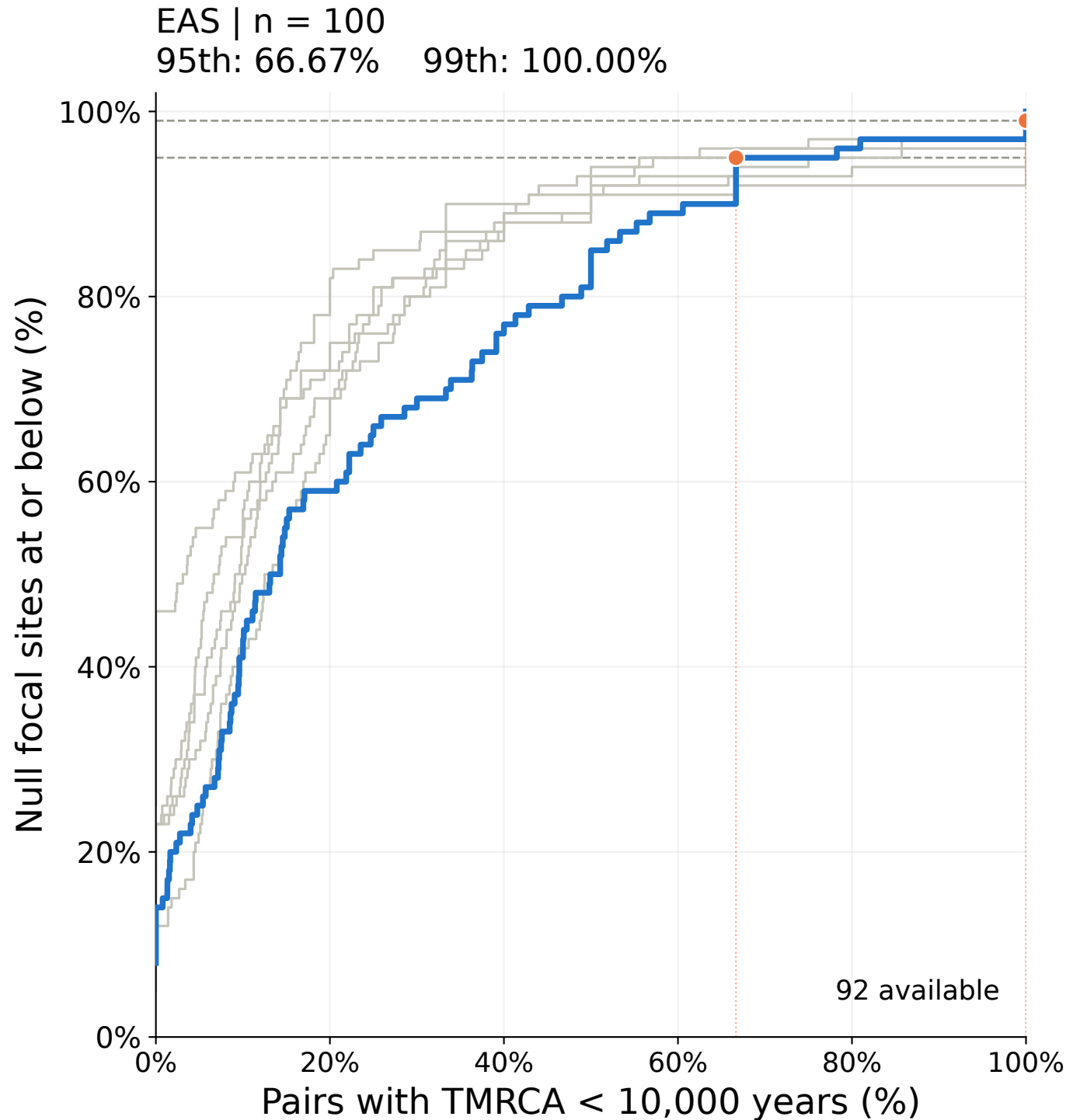
AMR | n = 100

95th: 100.00% 99th: 100.00%



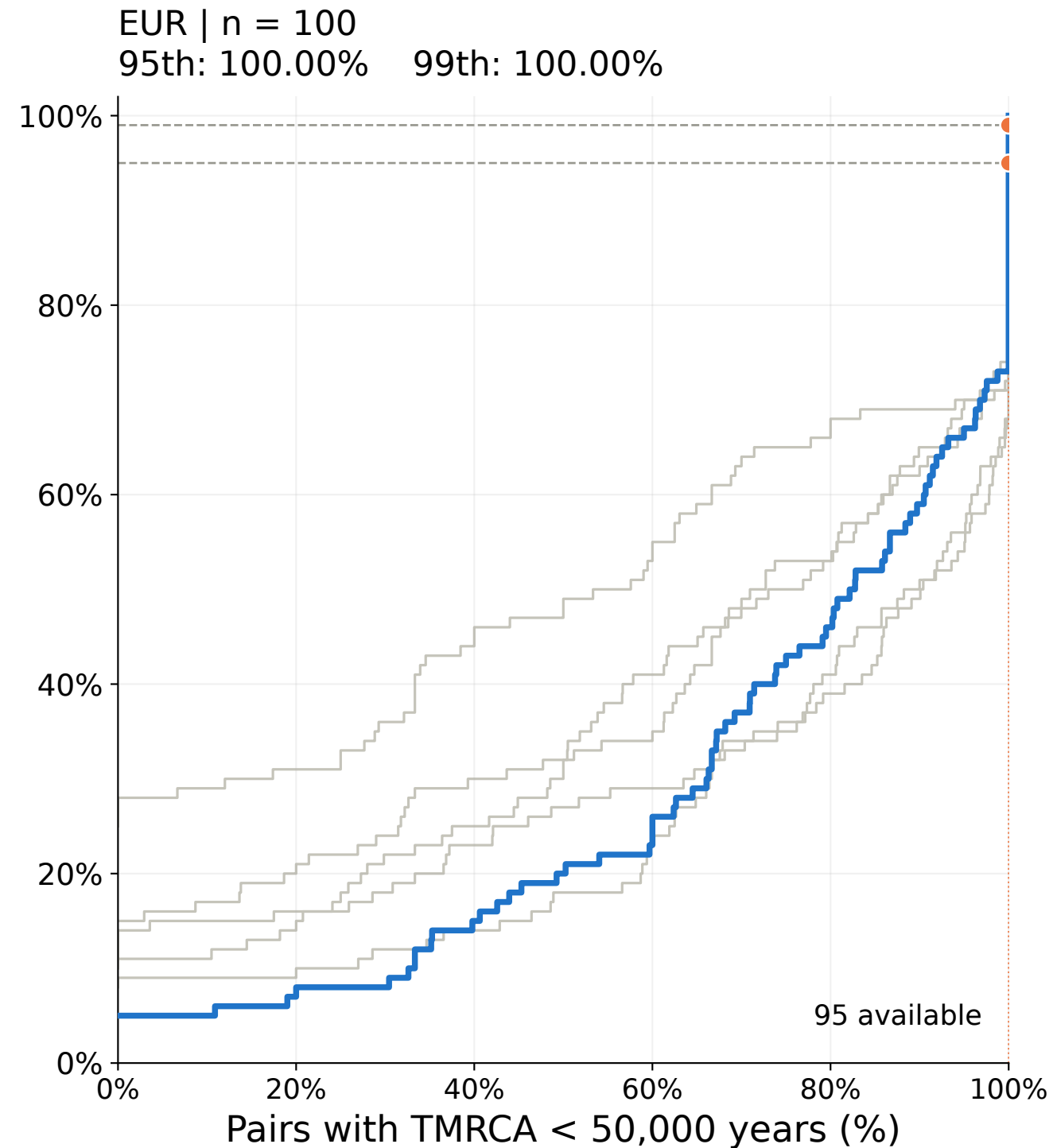
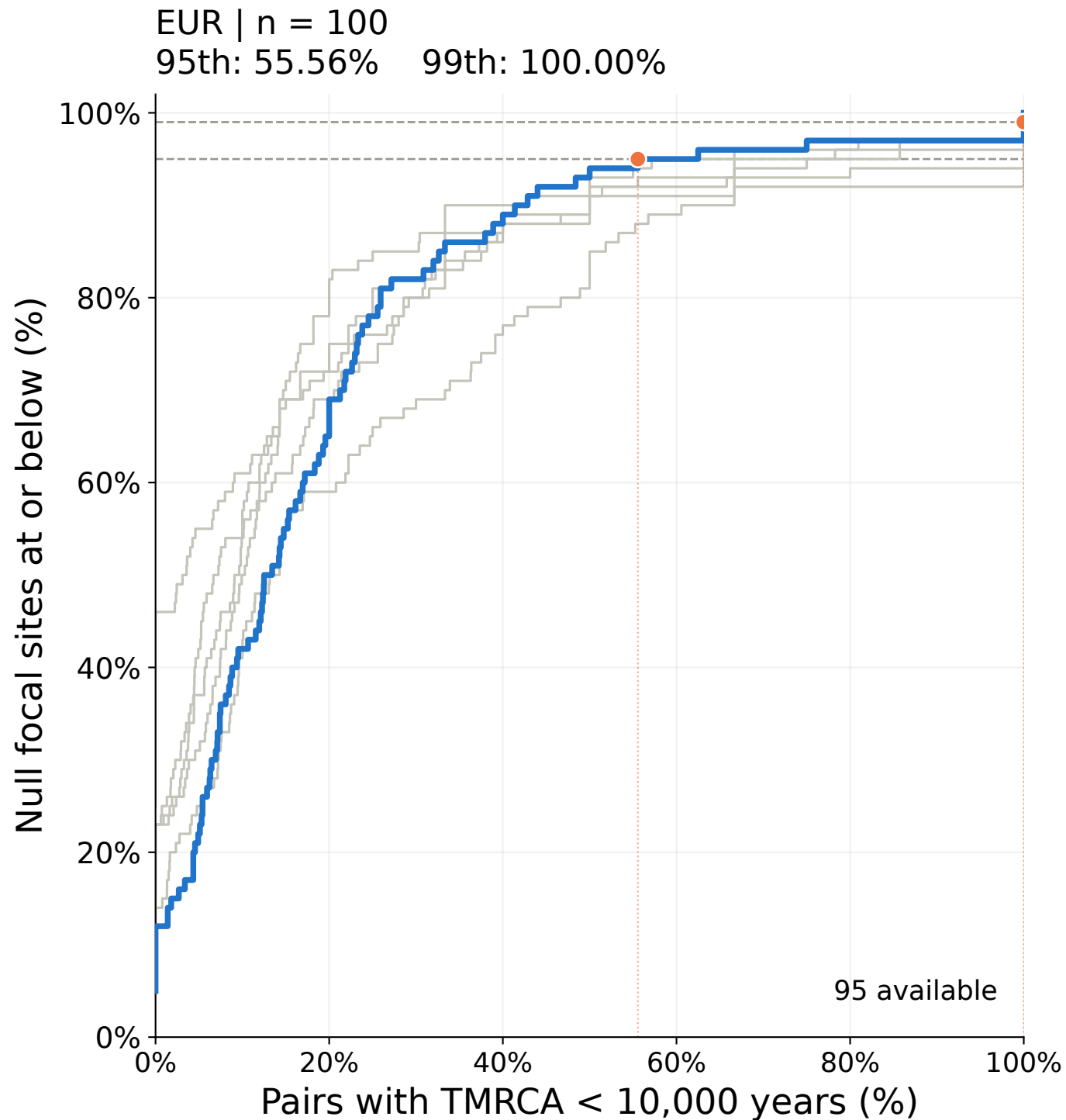
Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Decoded TMRCA | ALT/ALT pairs



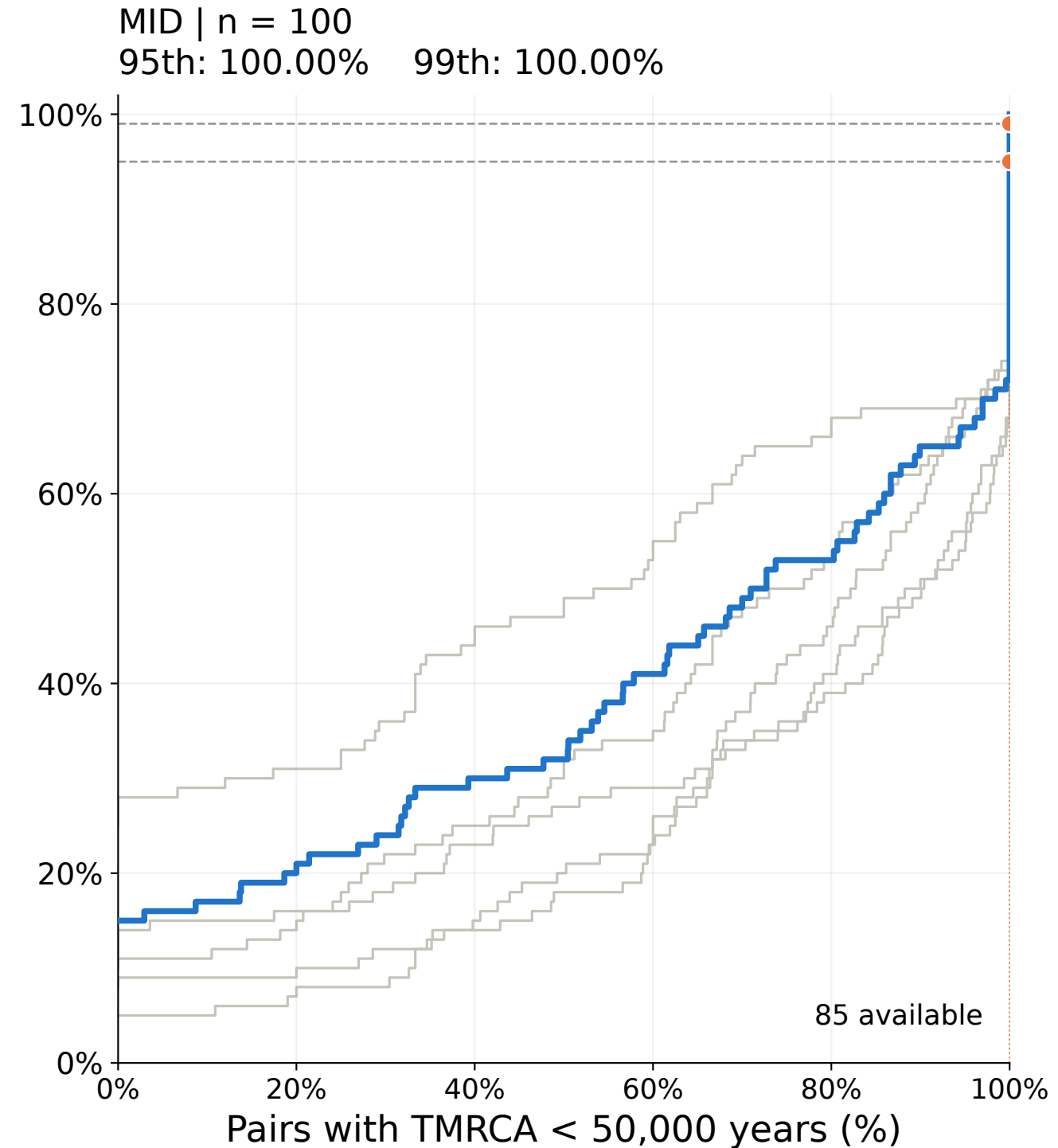
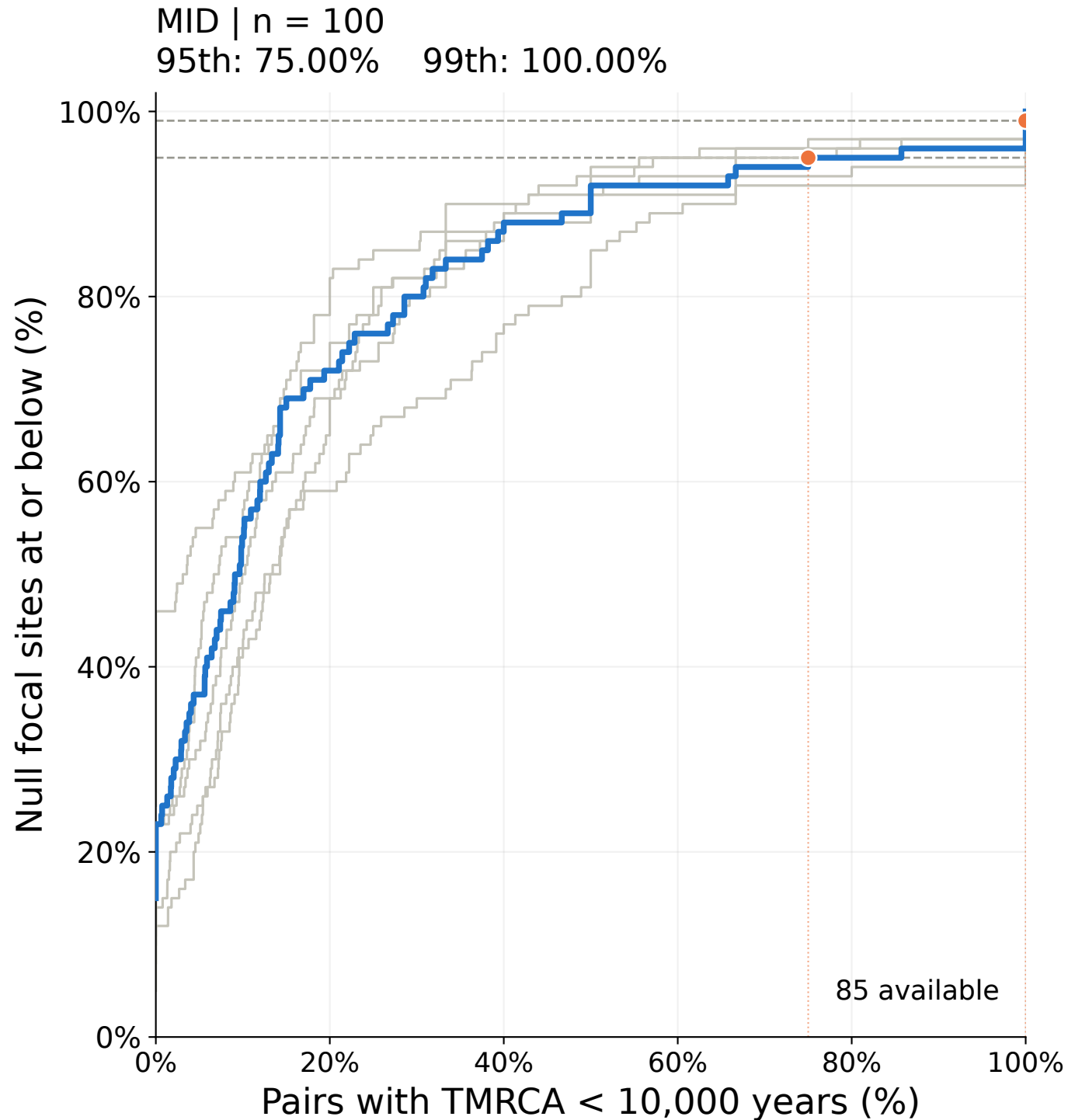
Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n , below finite support.

Decoded TMRCA | ALT/ALT pairs



Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n , below finite support.

Decoded TMRCA | ALT/ALT pairs

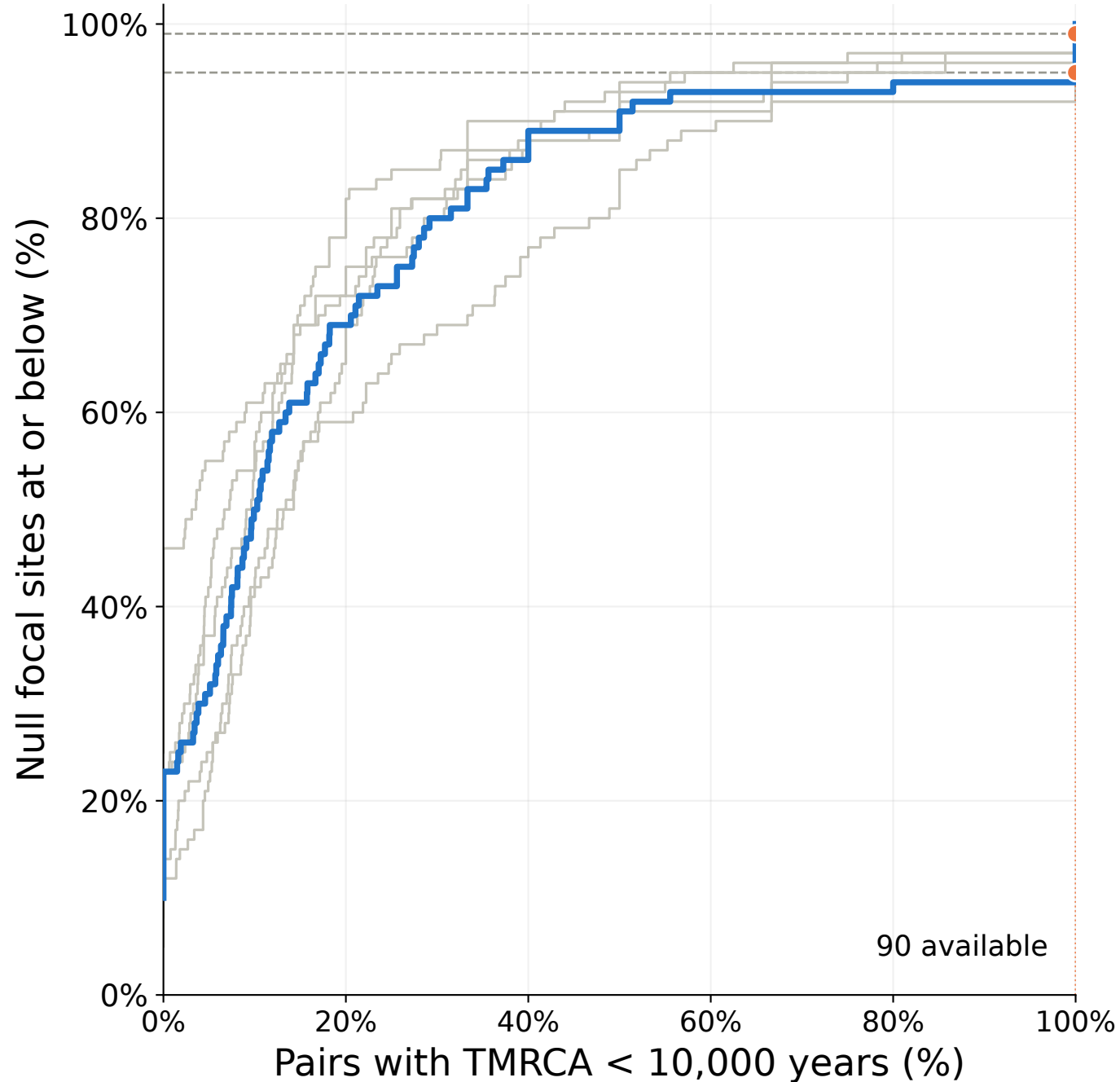


Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n , below finite support.

Decoded TMRCA | ALT/ALT pairs

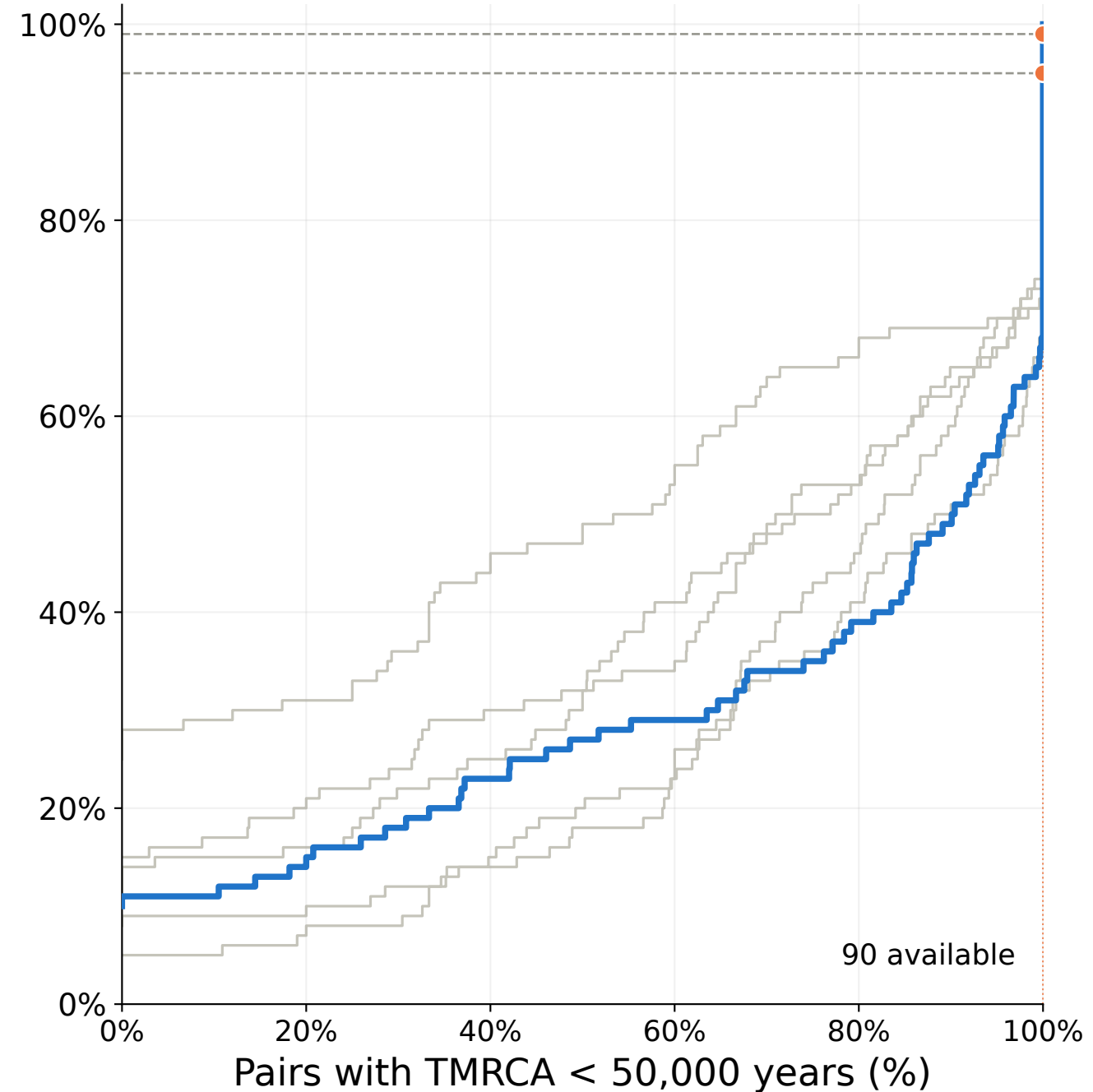
SAS | n = 100

95th: 100.00% 99th: 100.00%



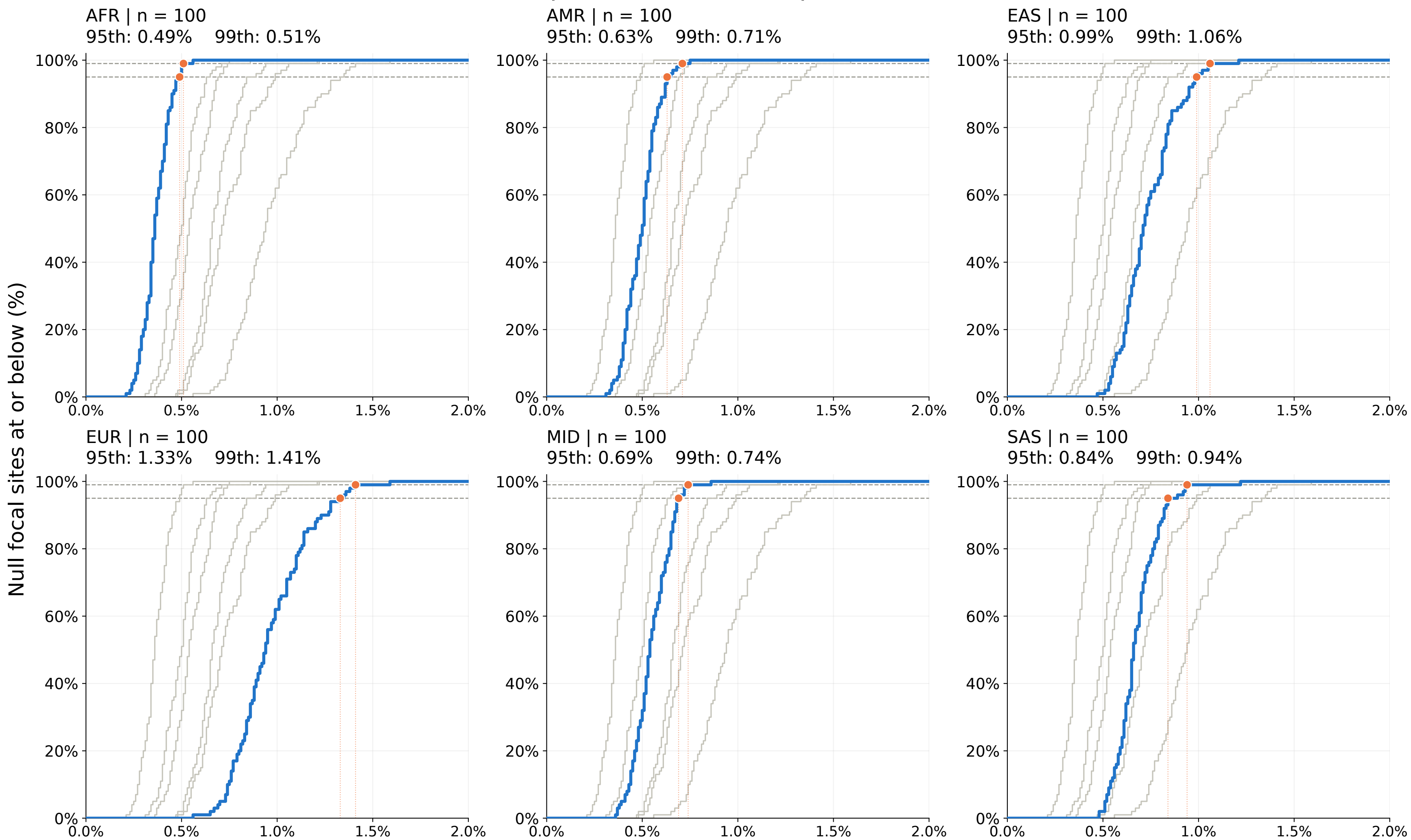
SAS | n = 100

95th: 100.00% 99th: 100.00%



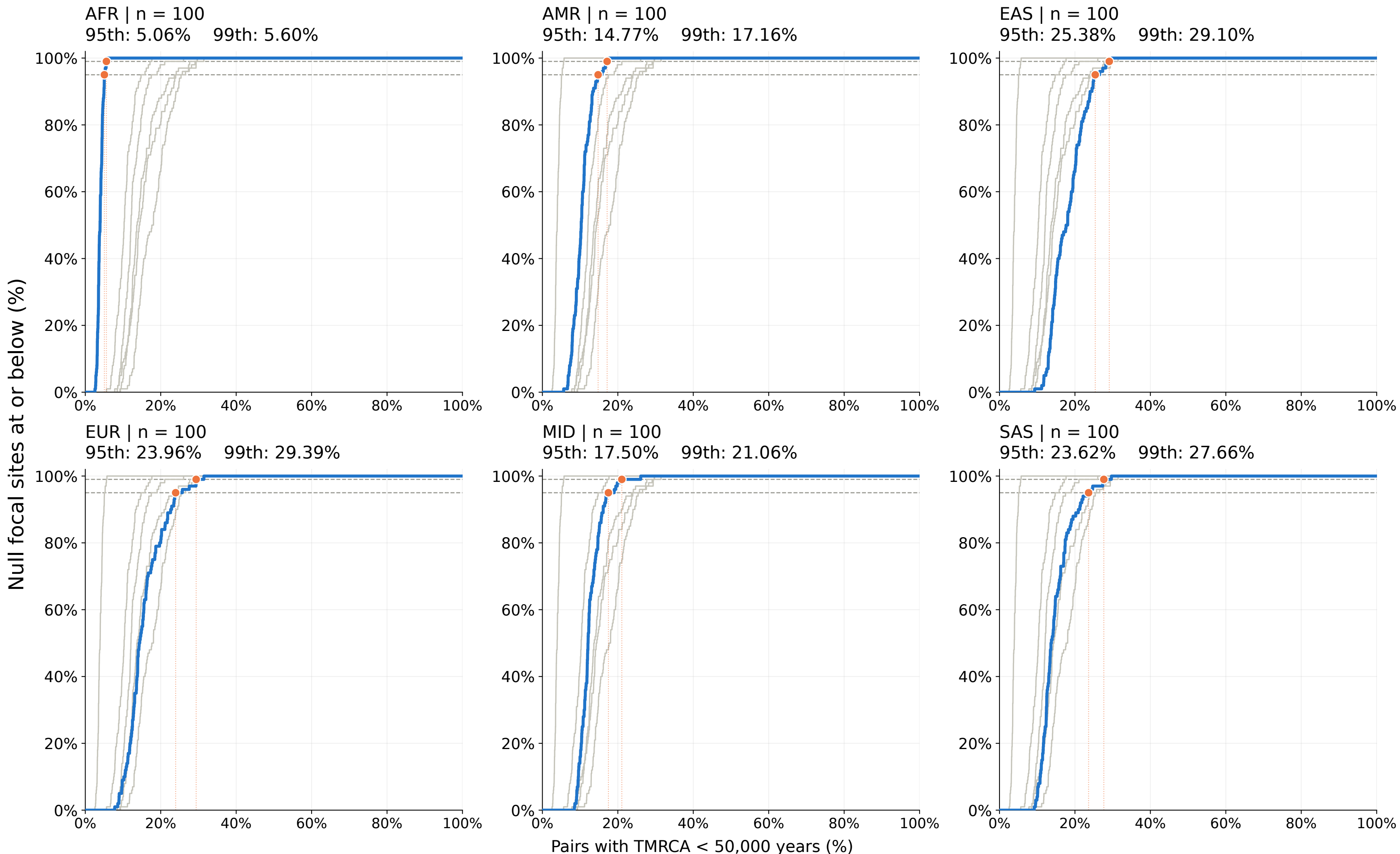
Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Truth TMRCA | All sampled pairs | T < 10,000 years



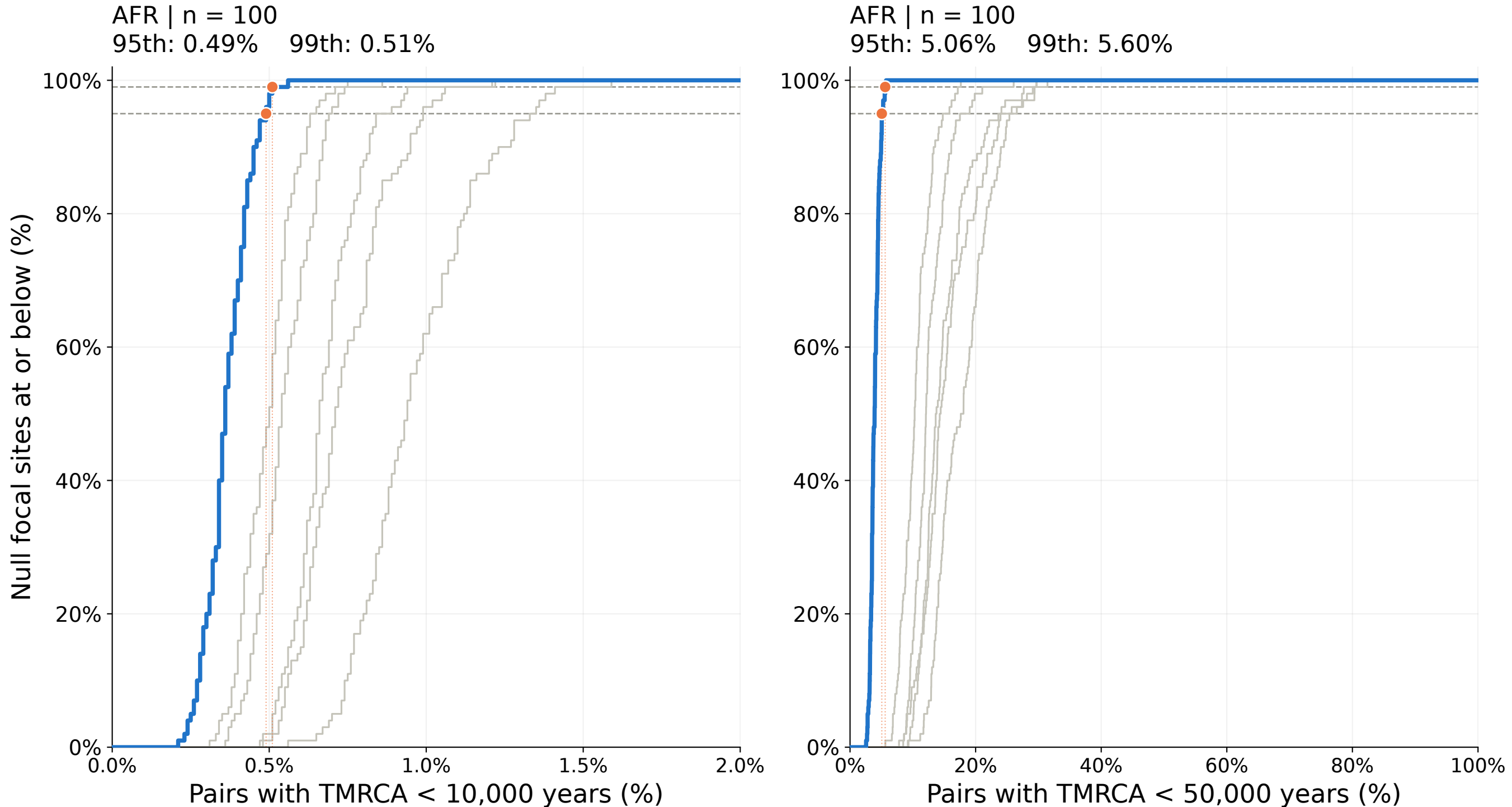
Empirical p = count(null >= observed) / n; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Truth TMRCA | All sampled pairs | T < 50,000 years



Empirical p = count(null >= observed) / n; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Truth TMRCA | All sampled pairs

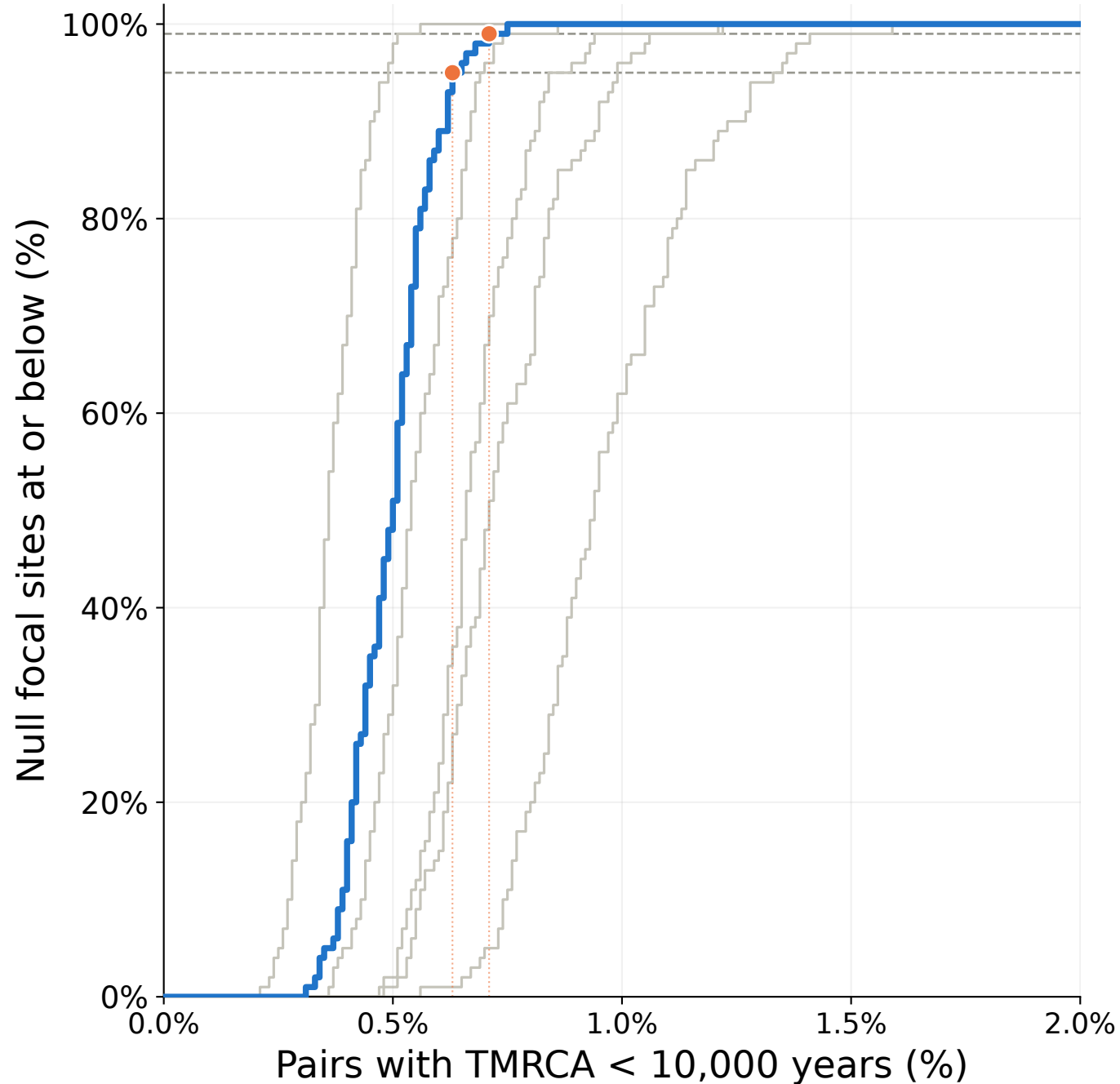


Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Truth TMRCA | All sampled pairs

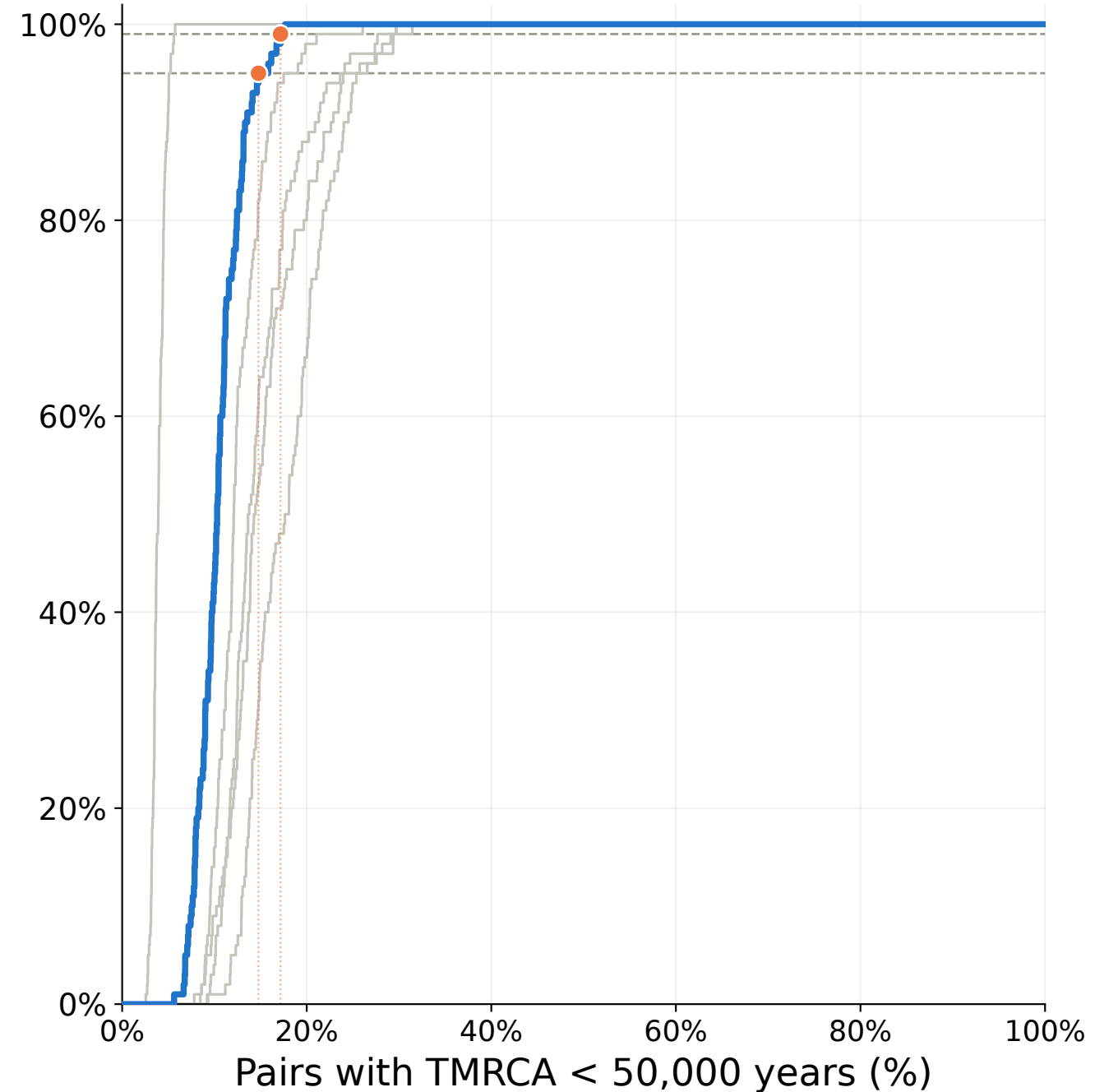
AMR | n = 100

95th: 0.63% 99th: 0.71%



AMR | n = 100

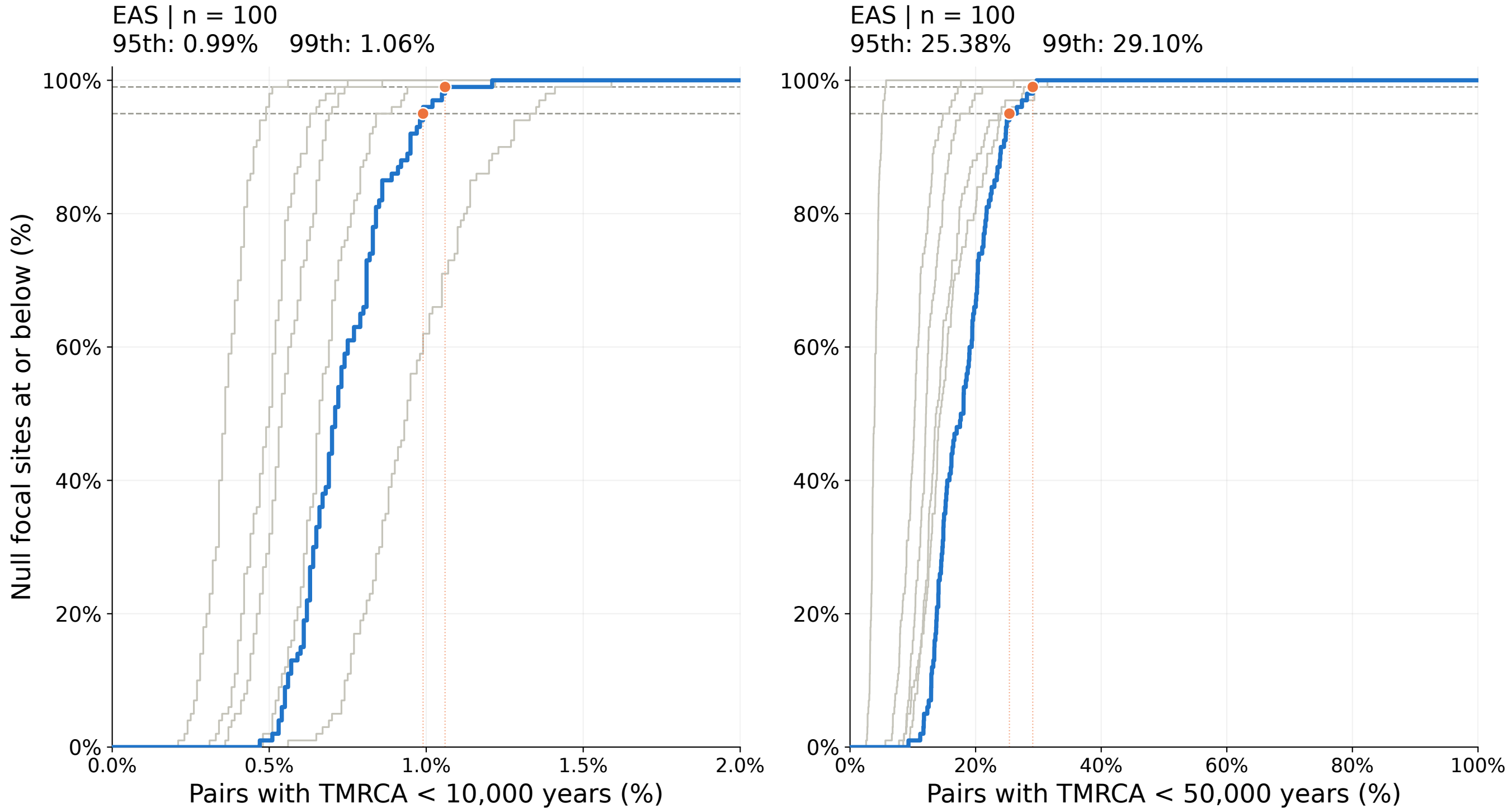
95th: 14.77% 99th: 17.16%



Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment

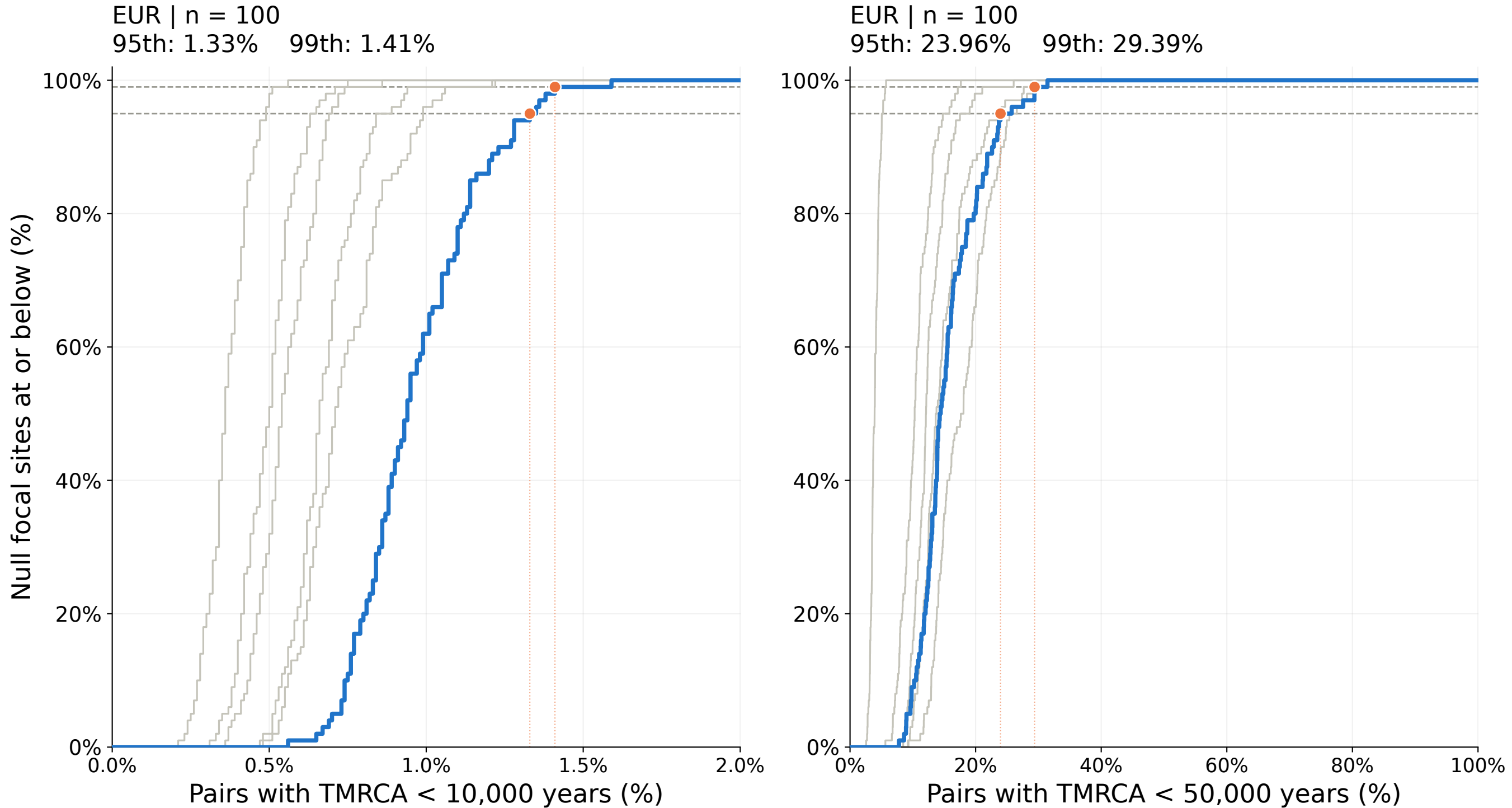
Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Truth TMRCA | All sampled pairs



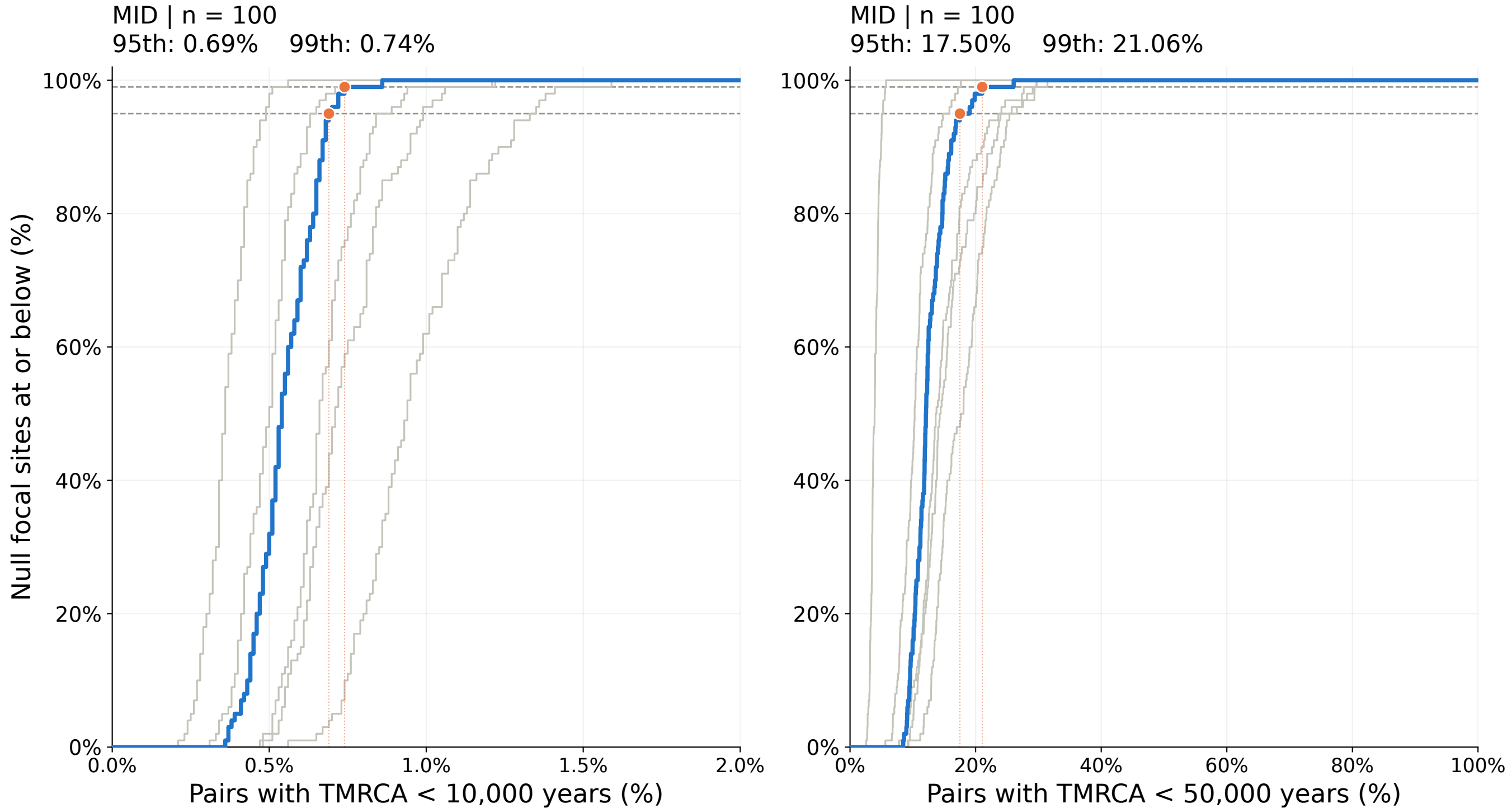
Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
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Truth TMRCA | All sampled pairs



Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
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Truth TMRCA | All sampled pairs

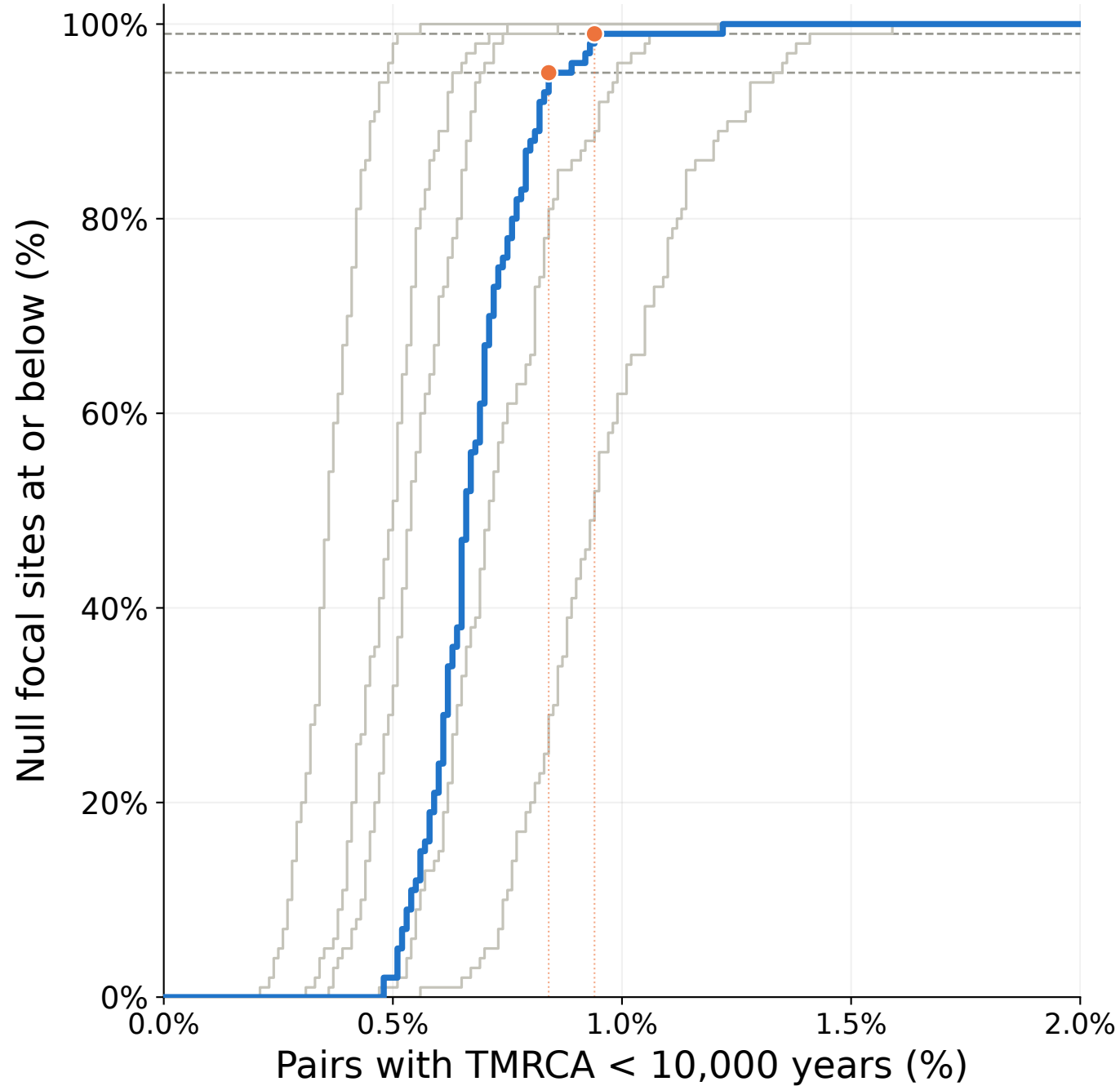


Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Truth TMRCA | All sampled pairs

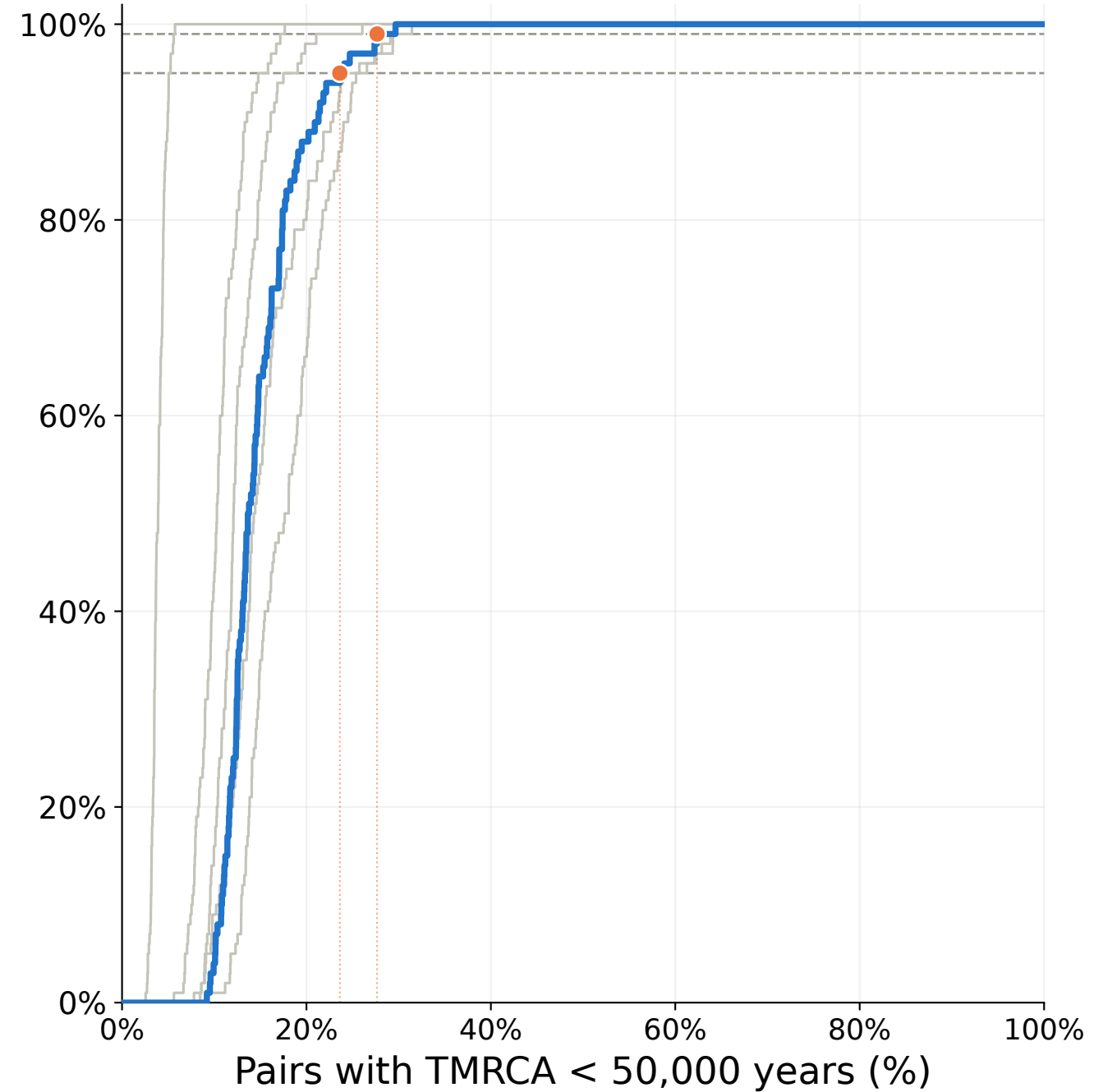
SAS | n = 100

95th: 0.84% 99th: 0.94%



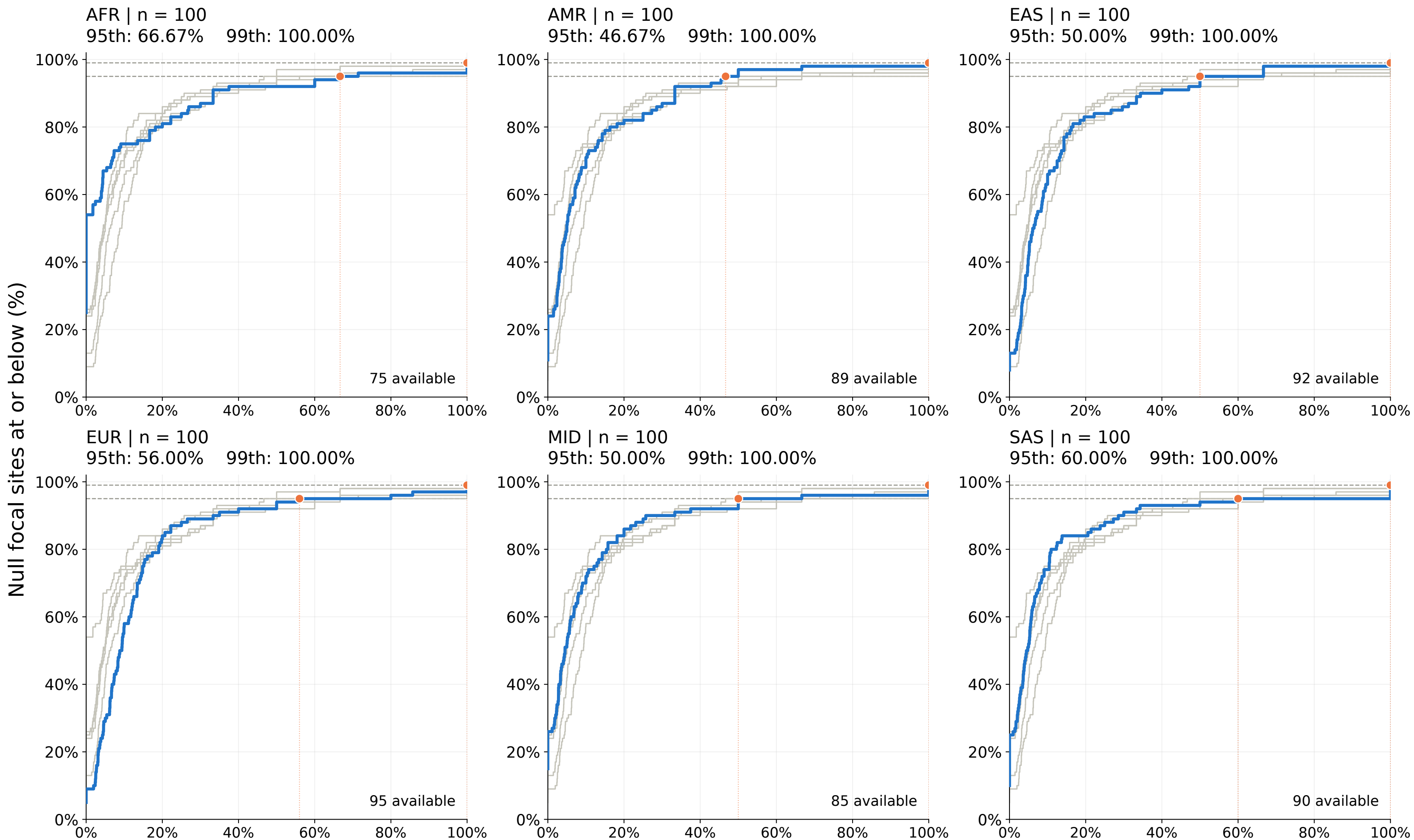
SAS | n = 100

95th: 23.62% 99th: 27.66%



Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n , below finite support.

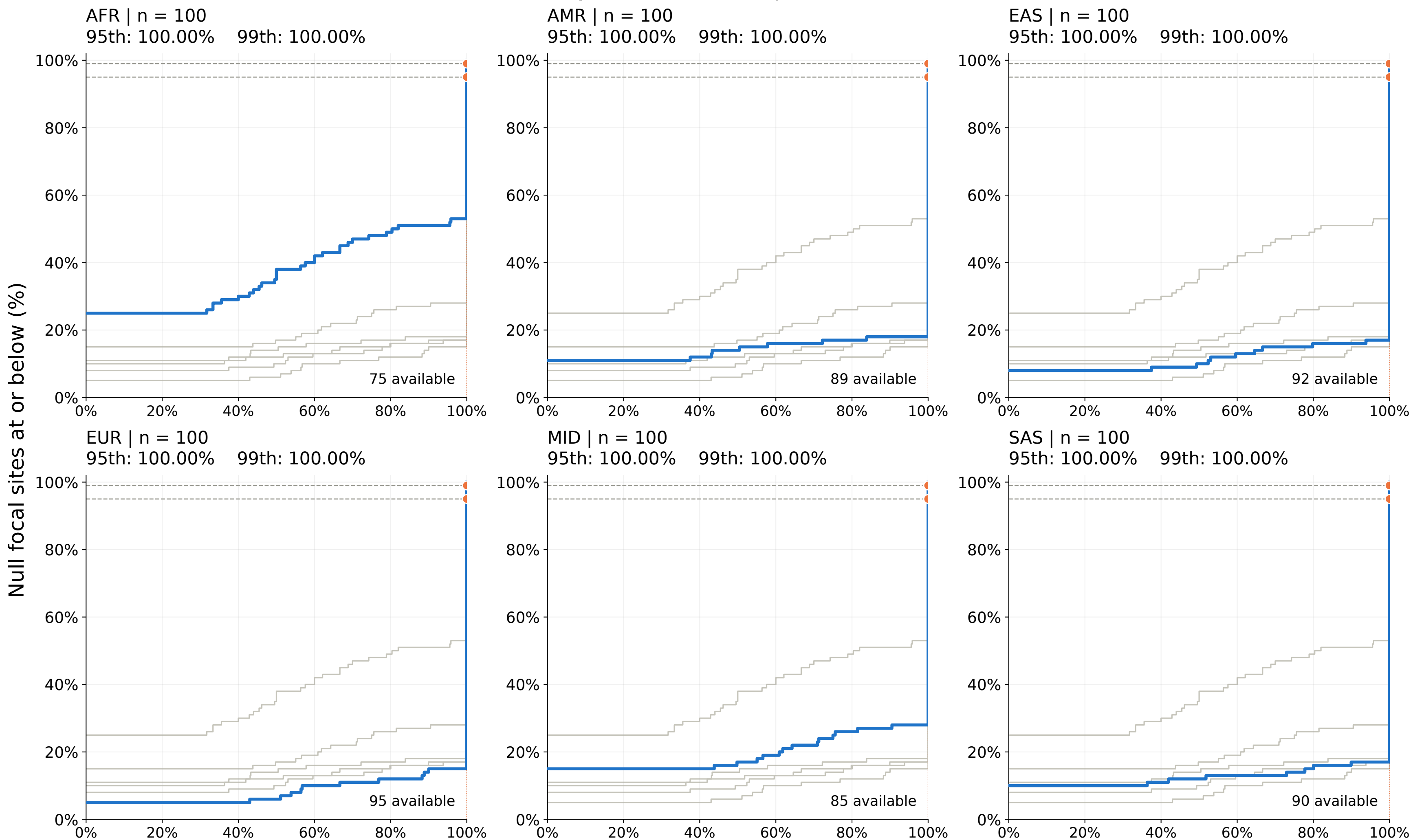
Truth TMRCA | ALT/ALT pairs | T < 10,000 years



Pairs with TMRCA < 10,000 years (%)

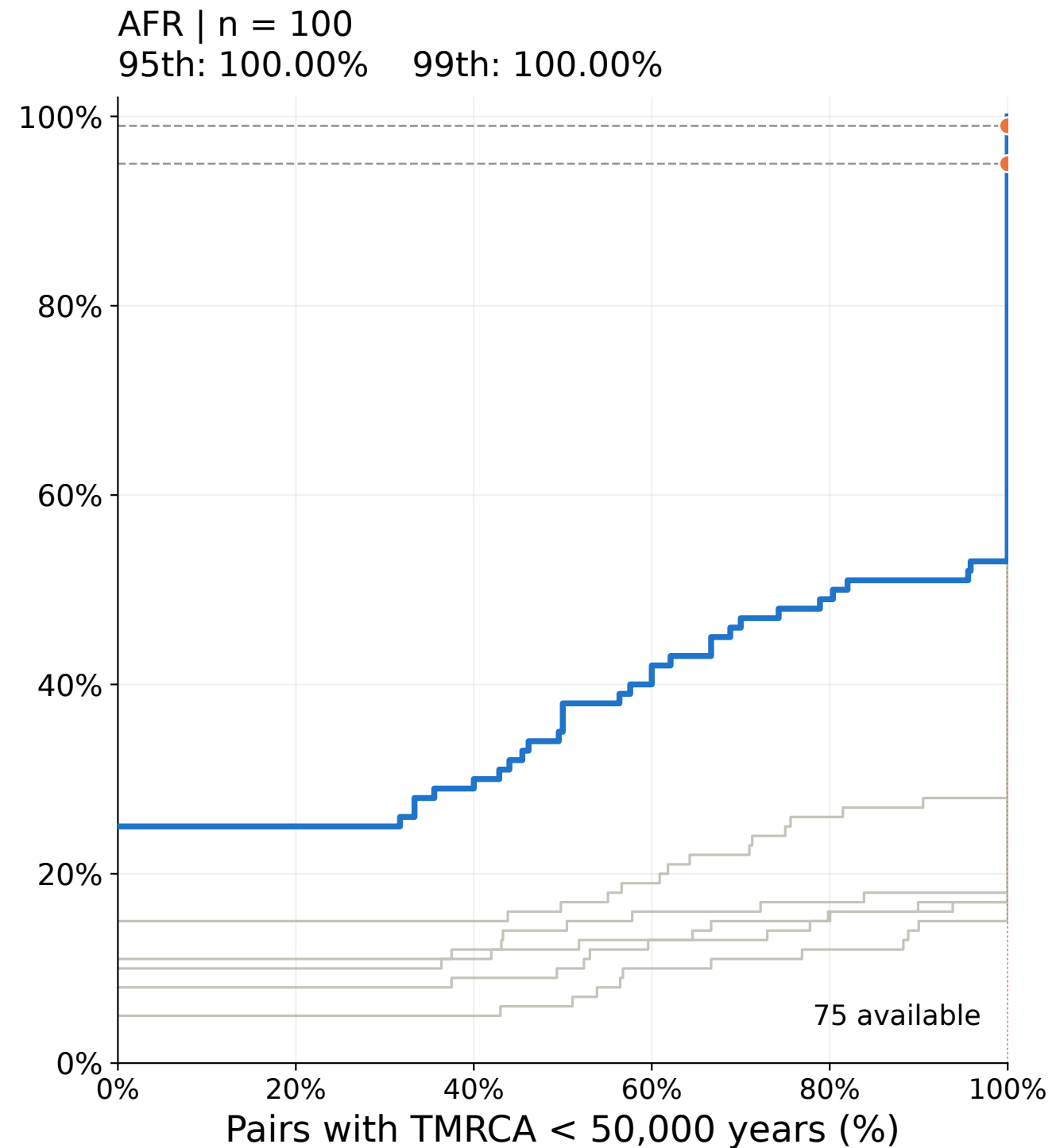
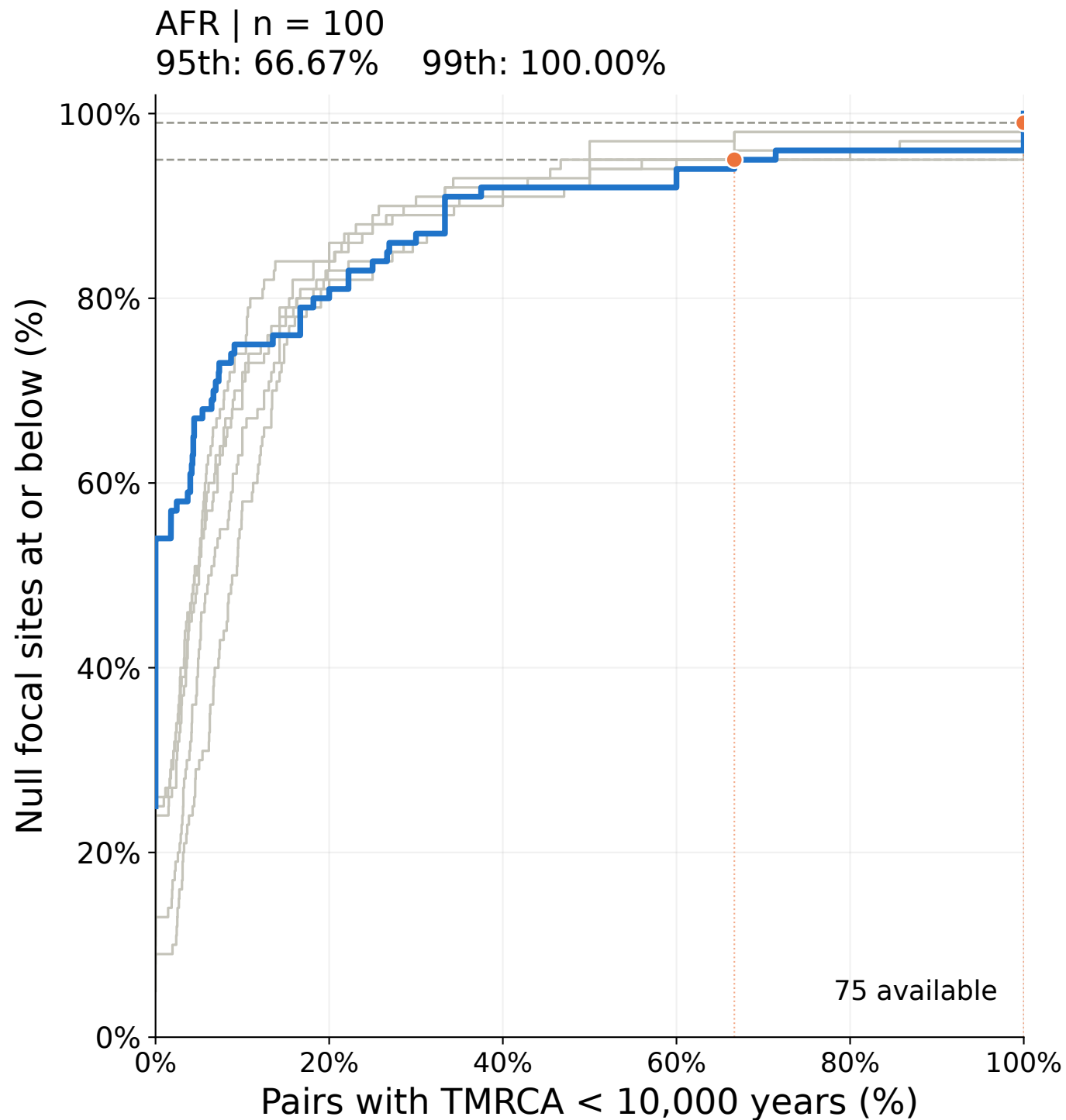
Empirical p = count(null >= observed) / n; ties count; no +1 adjustment
 Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Truth TMRCA | ALT/ALT pairs | T < 50,000 years



Empirical p = count(null >= observed) / n; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Truth TMRCA | ALT/ALT pairs

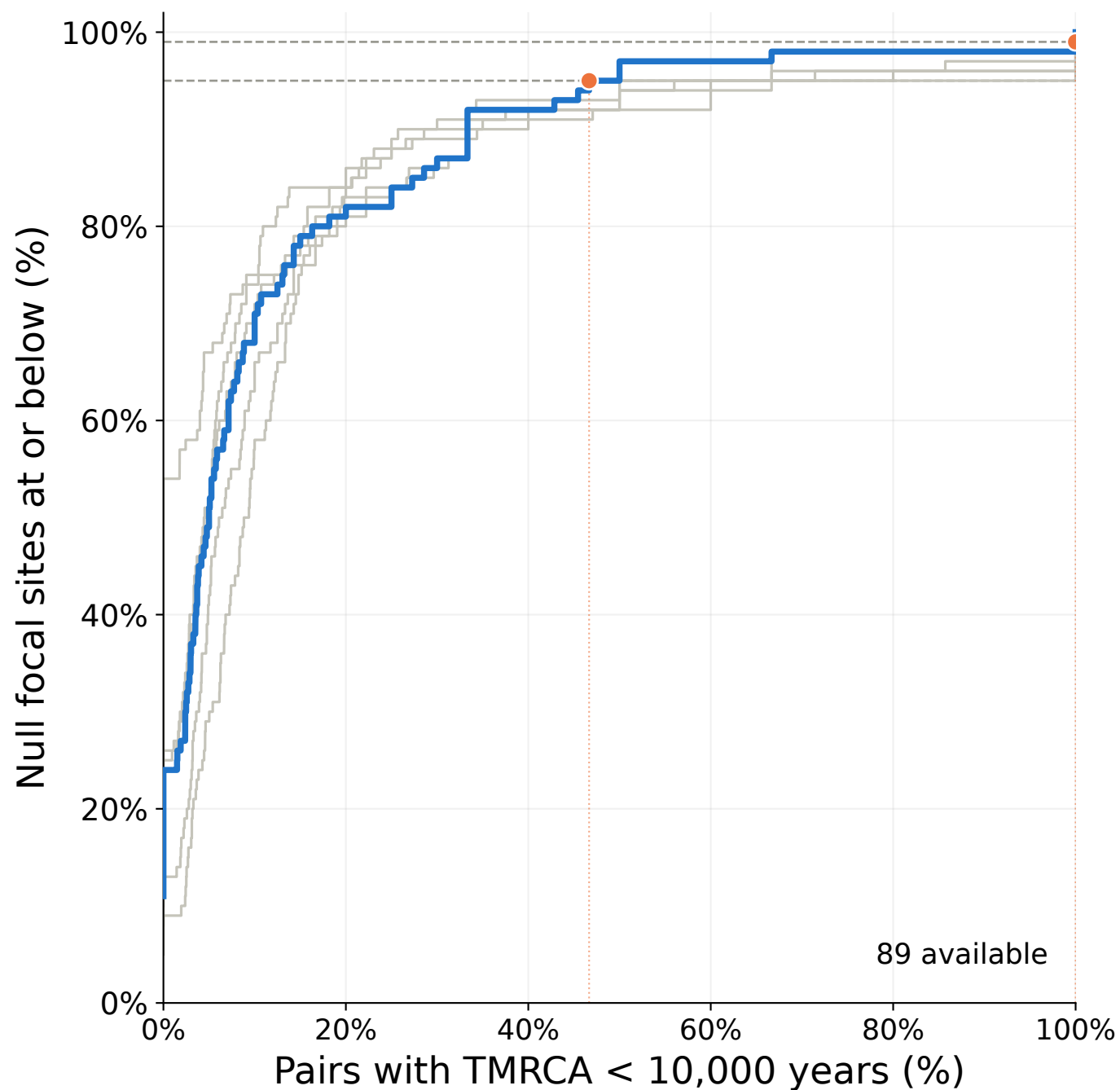


Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n , below finite support.

Truth TMRCA | ALT/ALT pairs

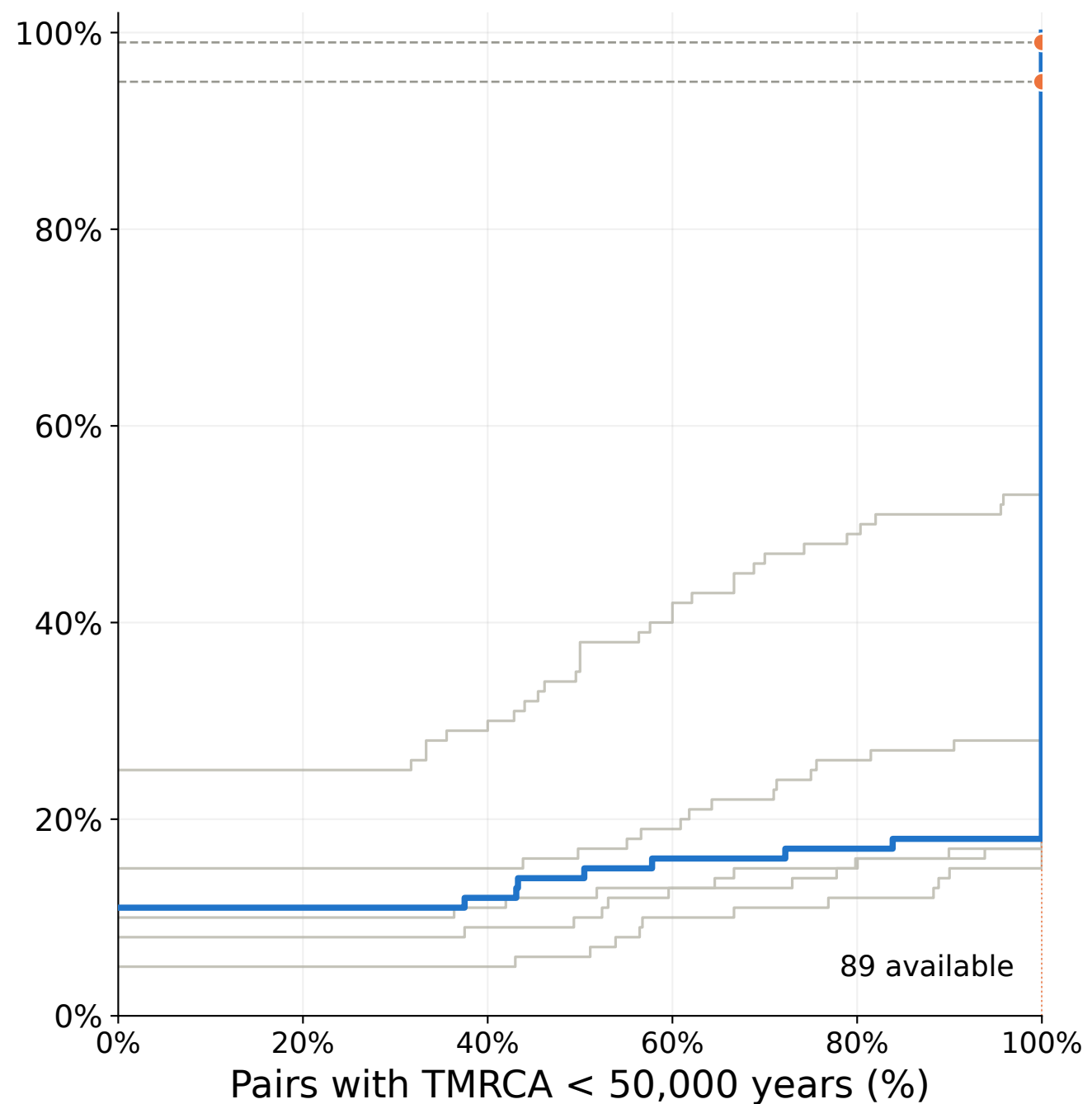
AMR | n = 100

95th: 46.67% 99th: 100.00%



AMR | n = 100

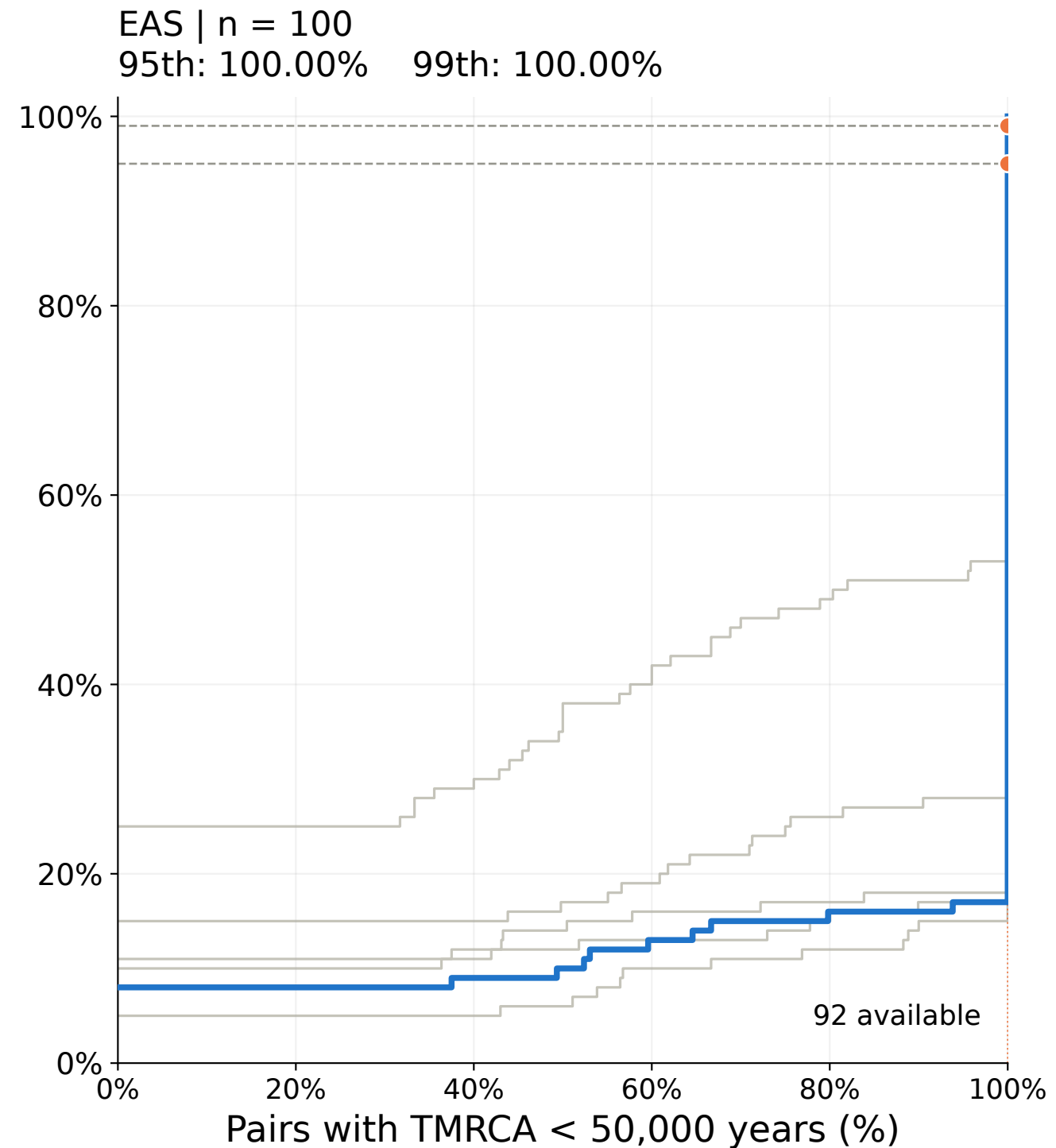
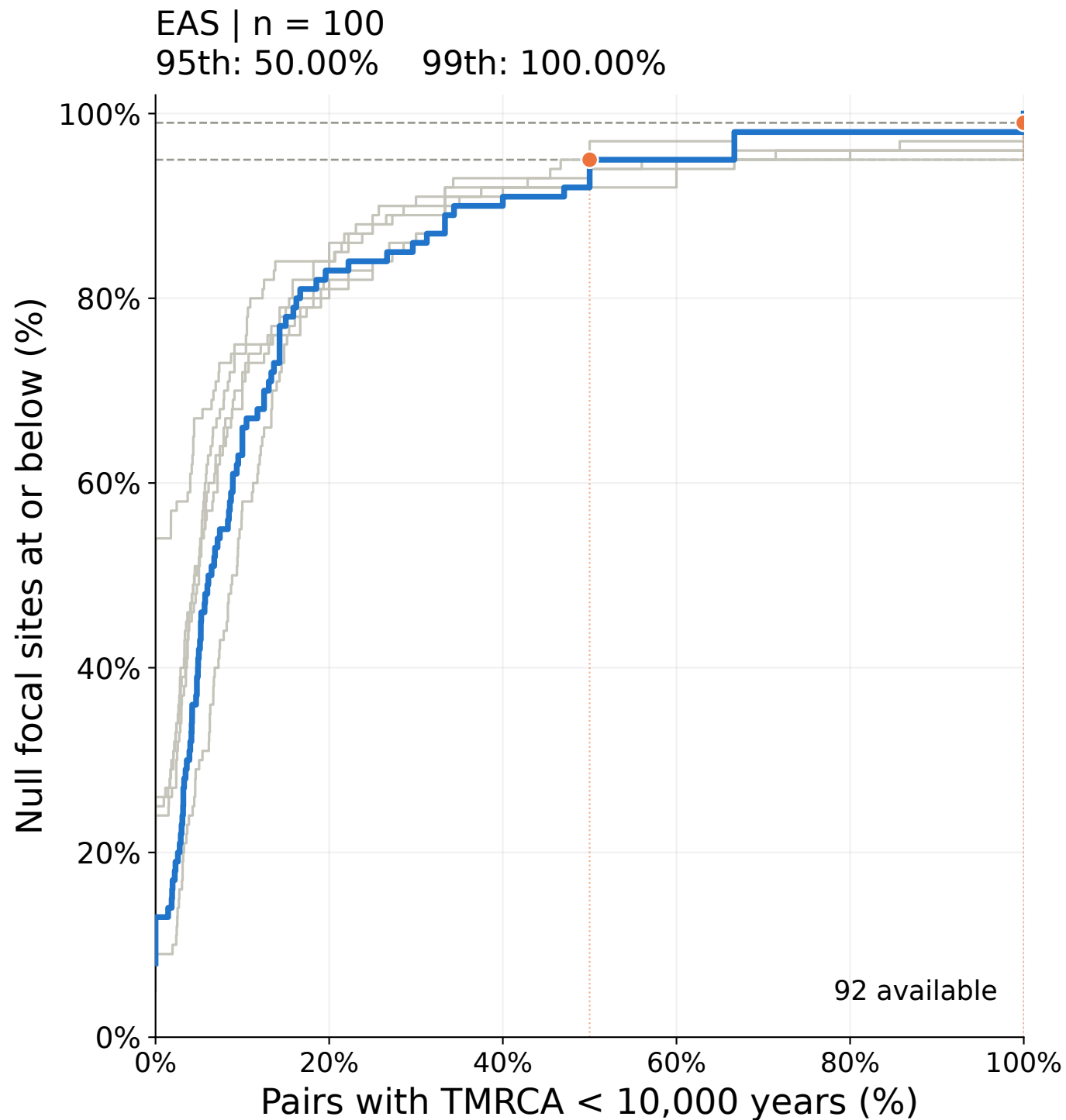
95th: 100.00% 99th: 100.00%



Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment

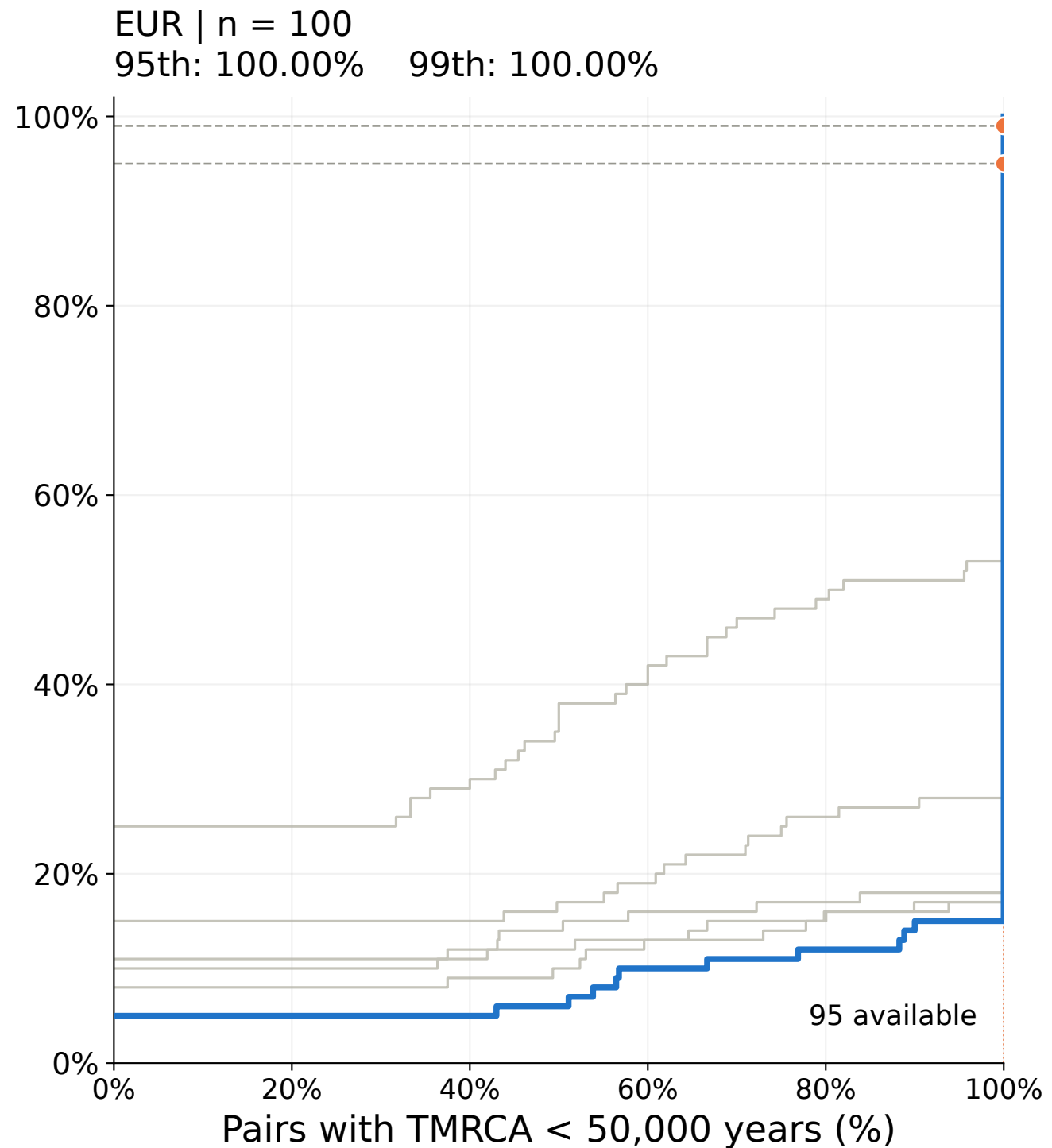
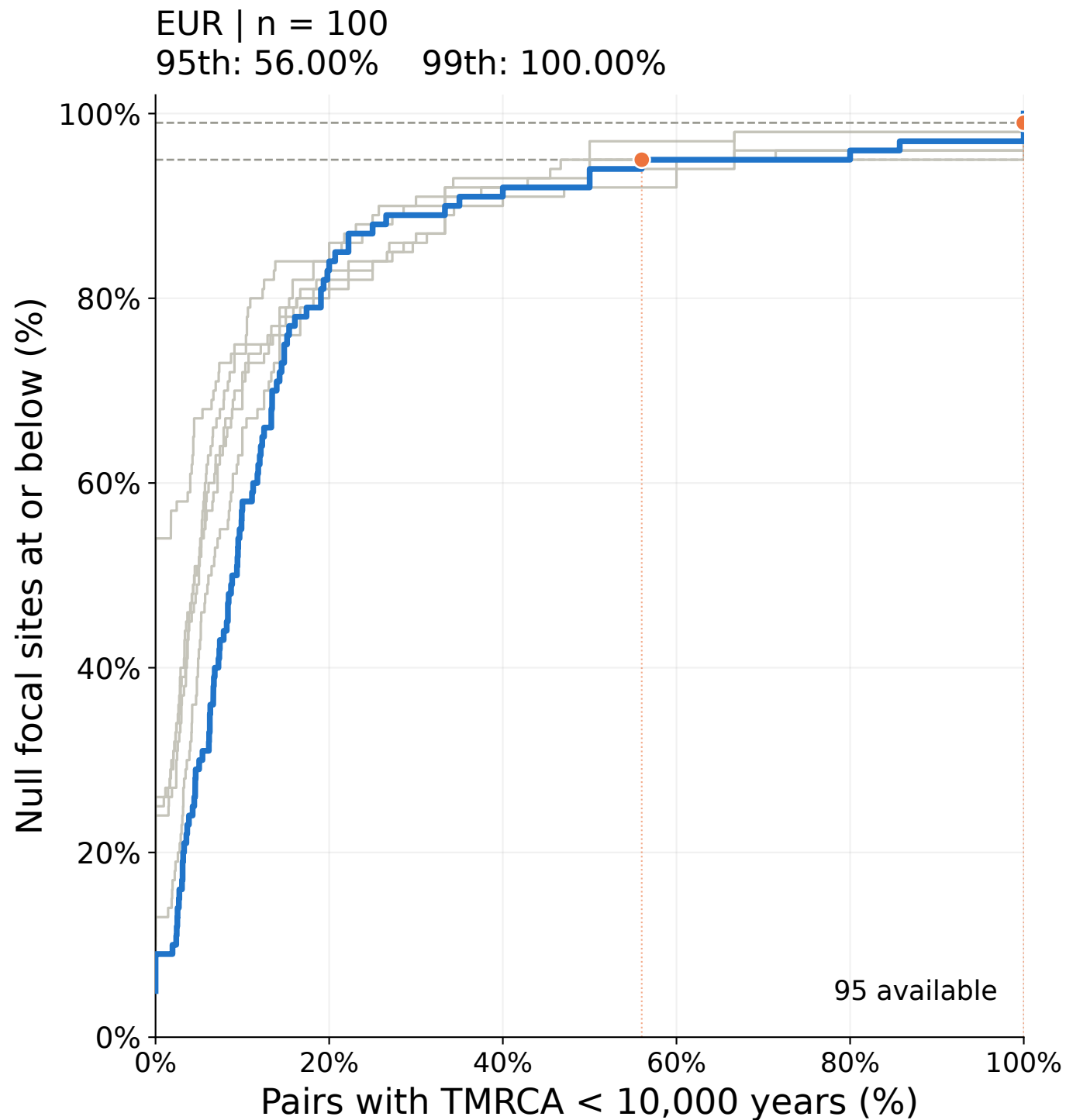
Blue: focal population; gray: others. Null no-calls remain in n, below finite support.

Truth TMRCA | ALT/ALT pairs



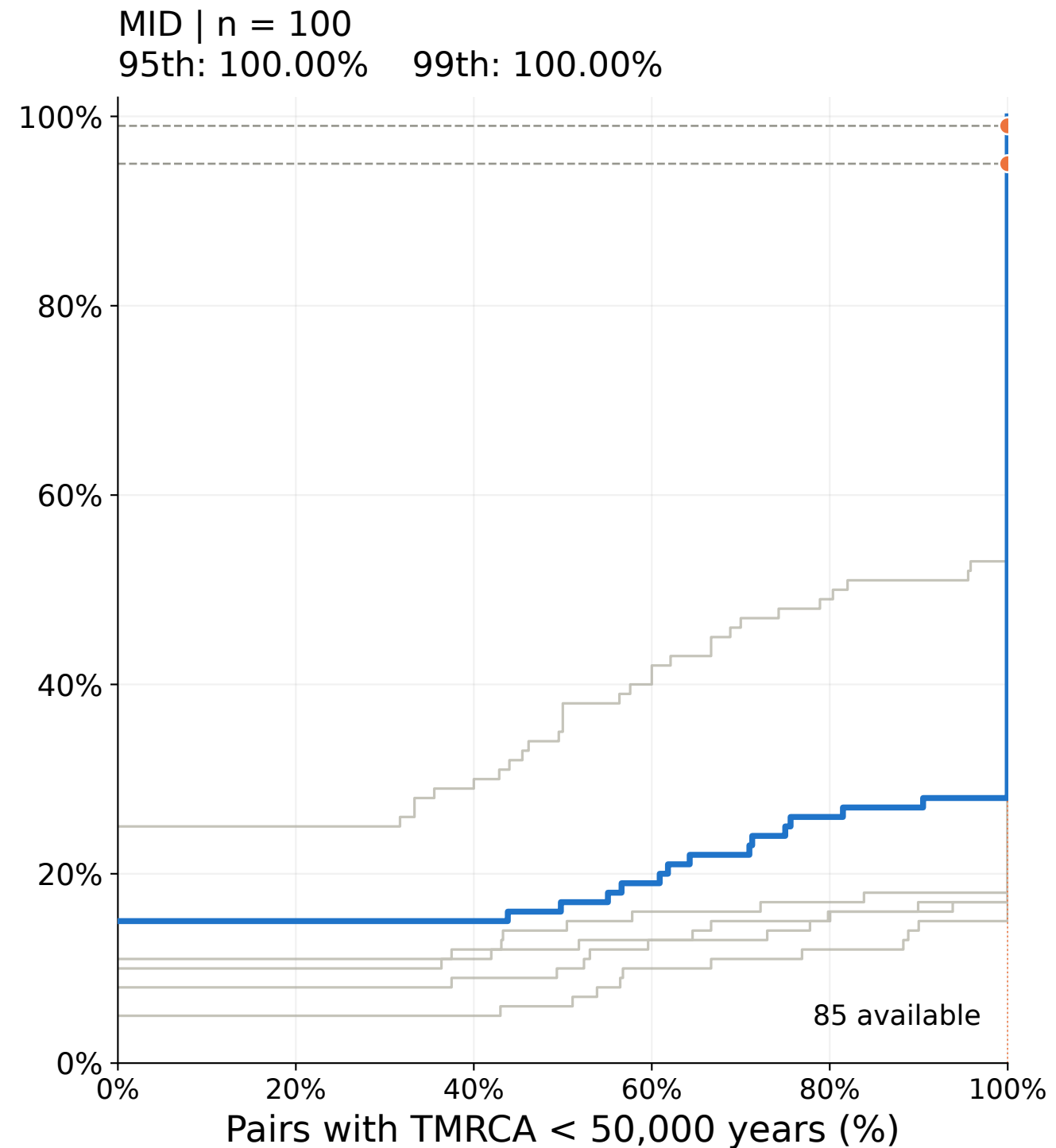
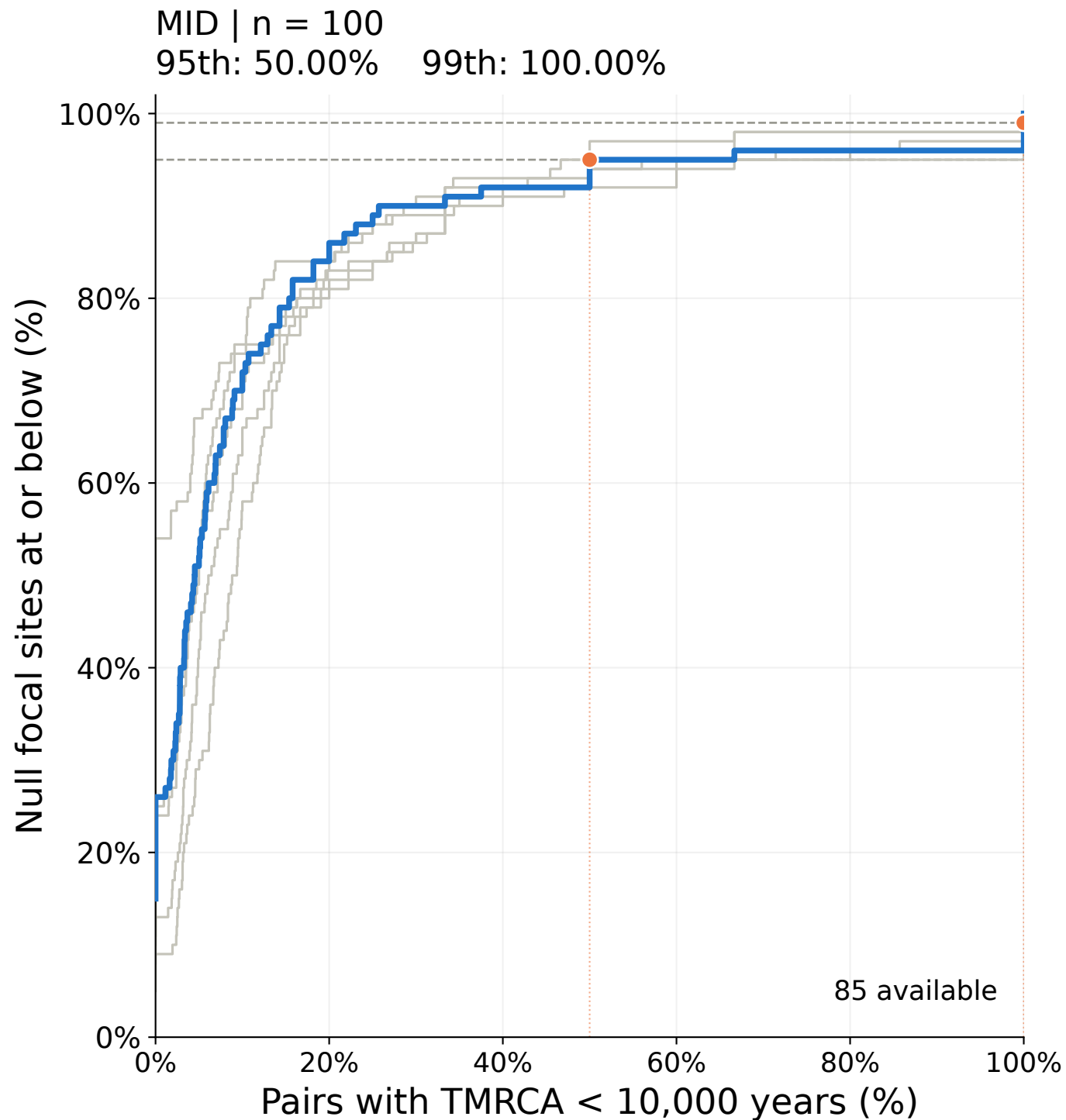
Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n , below finite support.

Truth TMRCA | ALT/ALT pairs



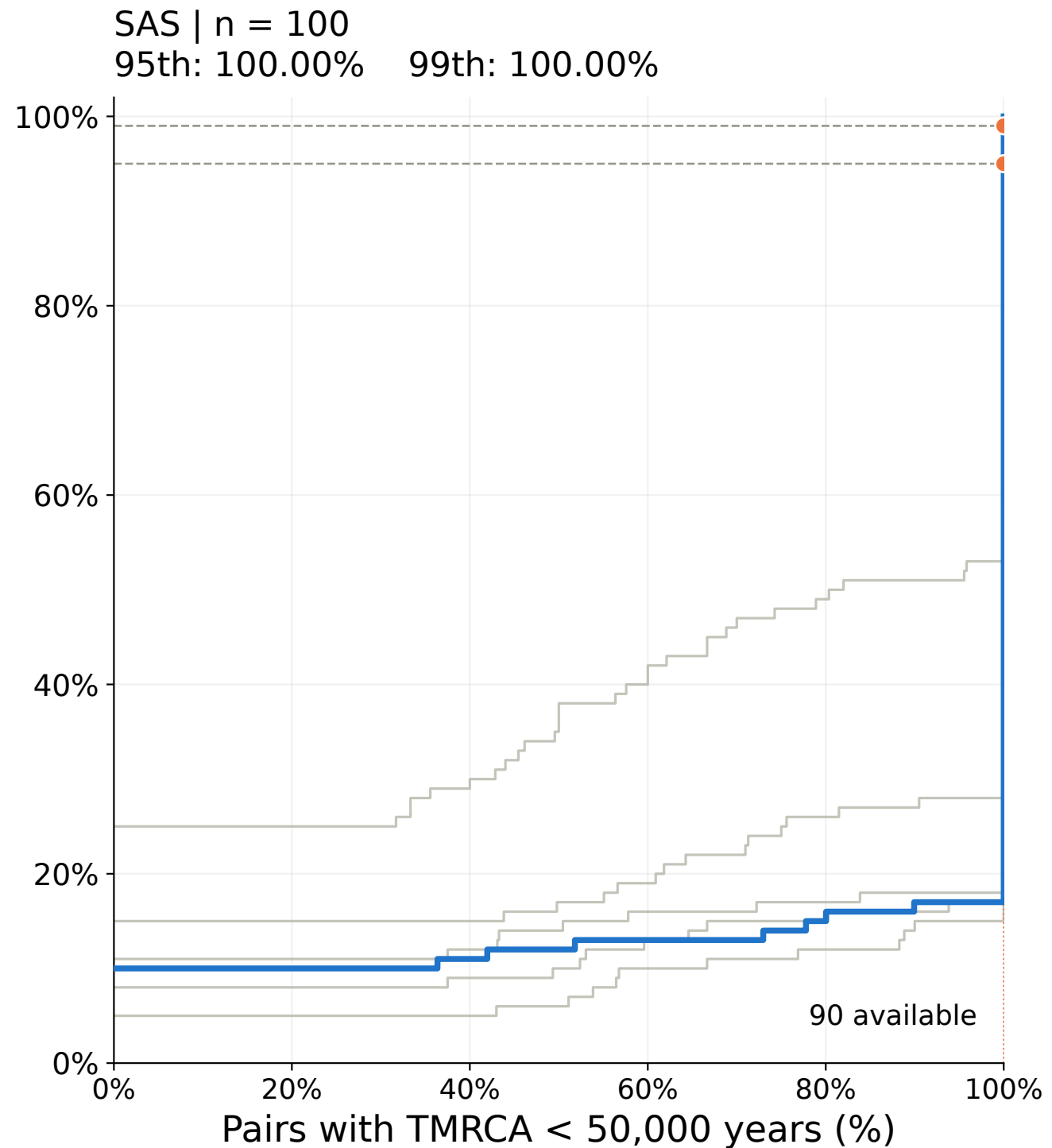
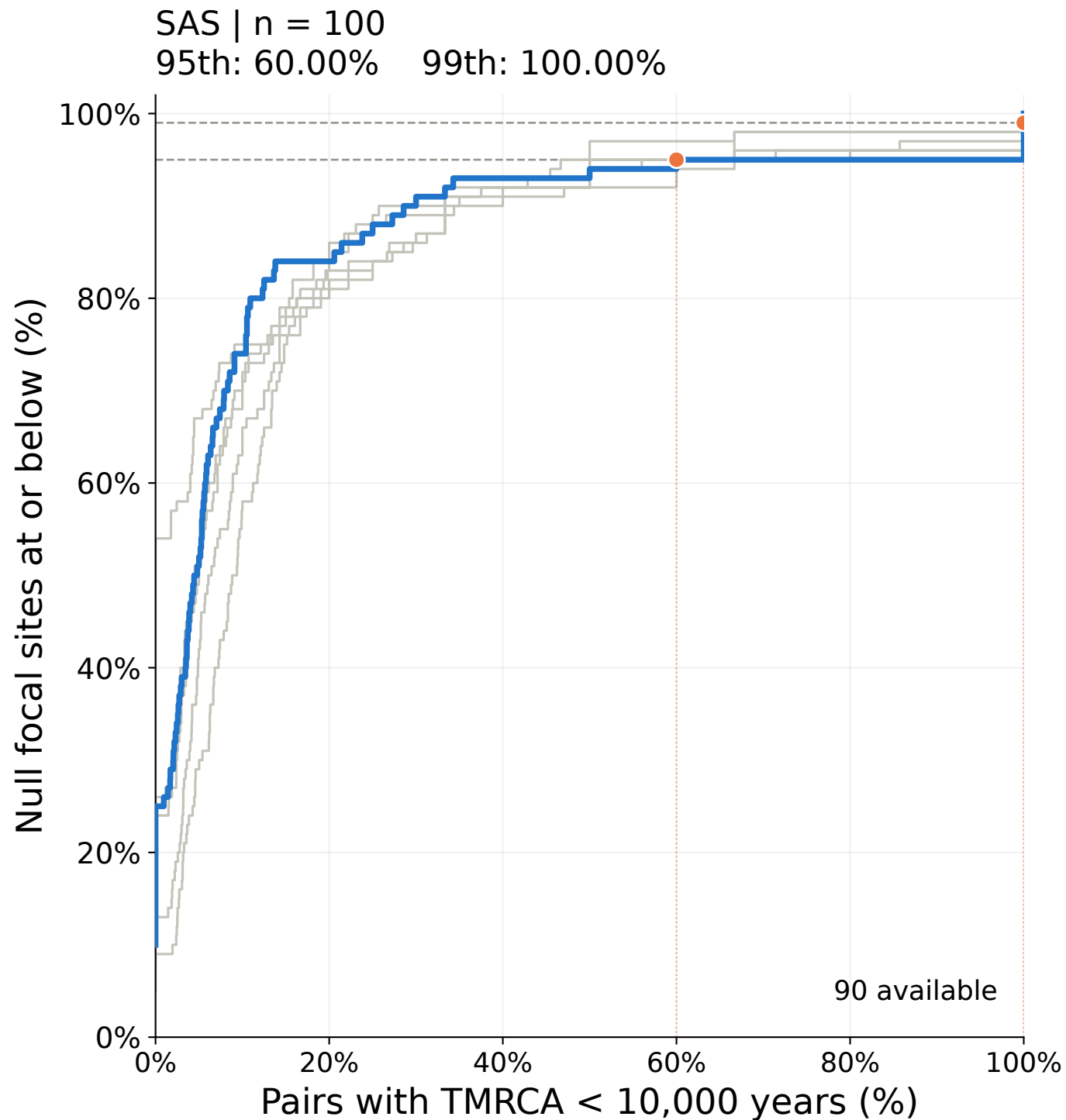
Empirical $p = \text{count}(\text{null} \geq \text{observed}) / n$; ties count; no +1 adjustment
Blue: focal population; gray: others. Null no-calls remain in n , below finite support.

Truth TMRCA | ALT/ALT pairs



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Truth TMRCA | ALT/ALT pairs



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Blue: focal population; gray: others. Null no-calls remain in n , below finite support.